Workflow demonstration

Assessment workflow demo: ALB

WORKFLOW DEMO

2026-09-05

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1 Executive summary

This is a workflow demonstration, not a stock assessment. It uses ALB outputs to demonstrate the model-to-report tools. The diagnostic-model figures use the archived MFCL-compatible KIMSA fit from job 24781. Stepwise, starting-value, self-test, retrospective, sensitivity, ensemble, profile and additional-output figures use illustrative display values to demonstrate report construction. They are not new assessment estimates.

Replace the illustrative study bundles with fitted results, write the assessment interpretation in the section files, and update the report cover before submission.

2 Introduction

3 Data compilation

The diagnostic model retains the ALB observation timing and two-region structure. Catch, abundance-index, length-composition and conditional age-at-length observations are available in the interactive diagnostic supplement.

Table 2 describes the fishery roles and observation periods. Figure 6 shows data availability. Index observations and predictions follow the model likelihood's geometric-mean centering convention.

4 Model description

The archived ALB fit uses the MFCL-compatible age-region state, age-based selectivity and five-pass hybrid F, without tagging. Its original F upper bound of 3 is retained for reproduction; new model specifications default to 6.

Length compositions use square_fita with fishery divisors and a raw-record sample-size cap of 1000. The composition settings table records the fitted likelihood configuration.

Figure 13, Figure 14 and Figure 15 describe the biological inputs and estimates; Figure 40 shows fishery selectivity. Additional-format panels illustrate output types that are not estimated by this tag-free ALB reference model.

5 Model development

Figure 16 illustrates how model-development results are assembled. Each study retains its model definitions and change descriptions. Replace the example model collections with the actual development sequence before interpreting the differences.

6 Results

Figure 39 shows annual reference-model trajectories. Stocks use time-weighted annual means; recruitment and catches use annual totals. Depletion is the ratio of annual spawning-potential summaries. Index points and predictions use annual geometric means. The interactive viewer retains the original time resolution.

Length-residual panels retain the original fitted records and their compressed-tail support. Conditional age-at-length fits are pooled by region using raw sample counts, preserving the model's prediction scale.

Index fits and residuals (Figure 77, Figure 18), length-composition residuals and conditional age-at-length fits (Figure 26) are generated from the same result bundle used by the interactive viewer.

The following studies demonstrate the intended diagnostic presentation: starting-value trials (Figure 27), simulation-estimation recovery (Figure 30), retrospective analysis (Figure 32), objective profiles (Figure 33) and sensitivity comparisons. Their displayed values are illustrative.

7 Reporting examples

The ensemble, management-quantity and projection panels demonstrate the reporting formats. They do not establish ALB stock status. Supply actual reference points, model retention decisions, uncertainty draws and projection results before writing management conclusions.

8 Discussion

9 Tables

Table 1: Illustrative output. Assessment resources and output provenance.

Resource	Purpose
Numeric display fixture	Demonstrate reporting inputs

Table 2: Fishery definitions and observation units.

Fishery	Region	Role	First year	Last year	Unit
1	1	Catch	1954	2022	fish
2	1	Catch	1954	2022	fish
3	1	Catch	1954	2022	fish
4	1	Catch	1954	2022	fish
5	1	Catch	1992	2022	fish
6	1	Catch	1982	2022	fish
7	1	Catch	1987	2022	fish
8	1	Catch	1954	2022	fish
9	1	Catch	1984	2022	fish
10	1	Catch	1985	2022	fish
11	1	Catch	1982	2022	t
12	1	Catch	1987	2022	t
13	1	Catch	1983	1990	t
14	2	Catch	1957	2022	fish
15	2	Catch	1961	2021	fish
16	2	Catch	1963	2021	fish
17	2	Catch	1986	2006	t
18	1	Index	1954	2022	relative index
19	1	Index	1992	2022	relative index
20	2	Index	1954	2022	relative index

Table 3: Illustrative output. Dirichlet–multinomial composition settings.

Fishery	Likelihood	Group	Theta
Fishery A	dirichlet_multinomial	1	20

Table 4: Illustrative output. Groups sharing composition-weight parameters.

Group	Fishery
1	Fishery A

Table 5: Illustrative output. Tag-release and reporting groups.

Programme	Region	Mixing
Programme A	Region 1	Two supplied model periods

Table 6: Illustrative output. Parameters on or near their declared bounds.

Parameter	Estimate	Lower	Upper
Steepness	0.7	0.2	0.99

Table 7: Illustrative output. Parameter correlations from the supplied covariance estimate.

Parameter	Partner	Correlation
log R0	Index q	-0.65

Table 8: Management quantities and supplied uncertainty summaries.

Quantity	Period	Estimate	Lower	Upper
$SB/SB_{F=0}$	2016–2020	0.45	0.35	0.55

Table 9: Spawning-depletion comparisons for the declared reference periods.

Period	Reference	Ratio
2016–2020	2012–2015	0.95

Table 10: Stock-status probabilities for the declared reference points.

Condition	Probability
$F/F_{MSY} > 1$	0.05

Table 11: Model reference points and their definitions.

Quantity	Estimate	Unit	Definition
SB_{MSY}	1e+06	t	Spawning potential at MSY

Table 12: Comparison of supplied uncertainty summaries.

Source	Quantity	Lower	Upper
Supplied illustrative quantiles	$SB/SB_{F=0}$	0.35	0.55

Table 13: Projected quantities by scenario and reporting period.

Scenario	Period	Quantity	Median
Reference catch	2031–2040	$SB/SB_{F=0}$	0.45

Table 14: Model configurations.

Model	Parent	Change
alb	NA	No change description supplied

Table 15: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
alb	ALB reference	passed	3	0	0

Table 16: Length-composition likelihood and weighting settings by fishery.

Fishery	Likelihood	Divisor	Record cap	Epsilon scale	Flag161
3	square_fita	51	1000	1	0
4	square_fita	164	1000	1	0
6	square_fita	89	1000	1	0
7	square_fita	29	1000	1	0
8	square_fita	372	1000	1	0
9	square_fita	212	1000	1	0
10	square_fita	125	1000	1	0
11	square_fita	27	1000	1	0
12	square_fita	37	1000	1	0
13	square_fita	358	1000	1	0
14	square_fita	123	1000	1	0
15	square_fita	241	1000	1	0
16	square_fita	450	1000	1	0
17	square_fita	1	1000	1	0
18	square_fita	746	1000	1	0
19	square_fita	55	1000	1	0

Fishery	Likelihood	Divisor	Record cap	Epsilon scale	Flag161
20	square_fita	61	1000	1	0

Table 17: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
catch	F1_R1	1954	2022	274
catch	F10_R1	1985	2022	147
conditional_age	F10_R1	2010	2010	17
length	F10_R1	1989	2016	74
catch	F11_R1	1982	2022	344
conditional_age	F11_R1	2006	2010	63
length	F11_R1	1996	2022	143
catch	F12_R1	1987	2022	161
length	F12_R1	1987	1998	27
catch	F13_R1	1983	1990	17
length	F13_R1	1988	1989	2
catch	F14_R2	1957	2022	262
length	F14_R2	1966	2022	119
catch	F15_R2	1961	2021	229
length	F15_R2	1968	2021	65
catch	F16_R2	1963	2021	195
length	F16_R2	1968	2021	17
catch	F17_R2	1986	2006	22
length	F17_R2	1992	1992	1
length	F18_R1	1965	2022	57
index	F18_R1_G18	1954	2022	69
length	F19_R1	1997	2022	25
index	F19_R1_G19	1992	2022	31
catch	F2_R1	1954	2022	274
length	F20_R2	1966	2018	39
index	F20_R2_G20	1954	2022	69
catch	F3_R1	1954	2022	275
conditional_age	F3_R1	2009	2009	4
length	F3_R1	1965	2022	178
catch	F4_R1	1954	2022	274
length	F4_R1	1965	2022	145
catch	F5_R1	1992	2022	121
catch	F6_R1	1982	2022	161
conditional_age	F6_R1	2009	2010	81
length	F6_R1	1992	2022	116

Kind	Series	First	Last	N
catch	F7_R1	1987	2022	135
conditional_age	F7_R1	2009	2010	30
length	F7_R1	1993	2013	25
catch	F8_R1	1954	2022	271
conditional_age	F8_R1	2010	2010	15
length	F8_R1	1965	2022	89
catch	F9_R1	1984	2022	126
conditional_age	F9_R1	2007	2013	45
length	F9_R1	1999	2020	17

Table 18: Index fit on the log scale; the first decade starts at each index's first observation.

Index	N	Log RMSE	First-decade RMSE
18	69	0.180	0.269
19	31	0.349	0.502
20	69	0.209	0.127

Table 19: Model configurations.

Model	Parent	Change
initial		Initial configuration
data-update	initial	Update observations
biology-update	data-update	Update biological assumptions
reference	biology-update	Select diagnostic configuration

Table 20: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
initial	Initial	passed	3	0	0
data-update	Data update	passed	3	0	0
biology-update	Biology update	passed	3	0	0
reference	Reference	passed	3	0	0

Table 21: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31

Kind	Series	First	Last	N
index	F20_R2_G20	1954	2022	69

Table 22: Model configurations.

Model	Parent	Change
reference	NA	No change description supplied
trial-1	NA	No change description supplied
trial-2	NA	No change description supplied
trial-3	NA	No change description supplied
trial-4	NA	No change description supplied
trial-5	NA	No change description supplied
trial-6	NA	No change description supplied
trial-7	NA	No change description supplied
trial-8	NA	No change description supplied

Table 23: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
reference	Reference	passed	3	0	0
trial-1	Trial 1	passed	3	0	0
trial-2	Trial 2	passed	3	0	0
trial-3	Trial 3	passed	3	0	0
trial-4	Trial 4	passed	3	0	0
trial-5	Trial 5	passed	3	0	0
trial-6	Trial 6	passed	3	0	0
trial-7	Trial 7	passed	3	0	0
trial-8	Trial 8	failed	2	1	0

Table 24: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

Table 25: Model configurations.

Model	Parent	Change
truth	NA	No change description supplied
replicate-1	NA	No change description supplied
replicate-2	NA	No change description supplied
replicate-3	NA	No change description supplied
replicate-4	NA	No change description supplied
replicate-5	NA	No change description supplied

Table 26: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
truth	Generating truth	passed	3	0	0
replicate-1	Refit 1	passed	3	0	0
replicate-2	Refit 2	passed	3	0	0
replicate-3	Refit 3	passed	3	0	0
replicate-4	Refit 4	passed	3	0	0
replicate-5	Refit 5	passed	3	0	0

Table 27: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

Table 28: Model configurations.

Model	Parent	Change
reference	NA	No change description supplied
peel-1	NA	No change description supplied
peel-2	NA	No change description supplied
peel-3	NA	No change description supplied
peel-4	NA	No change description supplied
peel-5	NA	No change description supplied

Table 29: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
reference	Reference	passed	3	0	0
peel-1	Peel 1	passed	3	0	0
peel-2	Peel 2	passed	3	0	0
peel-3	Peel 3	passed	3	0	0
peel-4	Peel 4	passed	3	0	0
peel-5	Peel 5	passed	3	0	0

Table 30: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

Table 31: Model configurations.

Model	Parent	Change
reference	NA	No change description supplied
lower	NA	No change description supplied
upper	NA	No change description supplied

Table 32: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
reference	Reference	passed	3	0	0
lower	Lower	passed	3	0	0
upper	Upper	passed	3	0	0

Table 33: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

Table 34: Model configurations.

Model	Parent	Change
reference	NA	No change description supplied
member-1	NA	No change description supplied
member-2	NA	No change description supplied
member-3	NA	No change description supplied
member-4	NA	No change description supplied
member-5	NA	No change description supplied
member-6	NA	No change description supplied

Table 35: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
reference	Reference	passed	3	0	0
member-1	Member 1	passed	3	0	0
member-2	Member 2	passed	3	0	0
member-3	Member 3	passed	3	0	0
member-4	Member 4	passed	3	0	0
member-5	Member 5	passed	3	0	0
member-6	Member 6	passed	3	0	0

Table 36: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

Table 37: Model configurations.

Model	Parent	Change
profiles	NA	No change description supplied

Table 38: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
profiles	Illustrative objective profiles	unknown	0	0	0

Table 39: Numbers of fitted observations and composition samples by series.

Kind	Series	N

Table 40: Model configurations.

Model	Parent	Change
formats	NA	No change description supplied

Table 41: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
formats	Illustrative reporting inputs	unknown	0	0	1

Table 42: Length-composition likelihood and weighting settings by fishery.

Fishery	Likelihood	Group	Theta
Fishery A	dirichlet_multinomial	1	20

Table 43: Illustrative output. Fishery definitions and observation units.

Fishery	Unit	Region
Longline	fish	Region 1
Purse seine	fish	Region 1
Other	fish	Region 1

Table 44: Numbers of fitted observations and composition samples by series.

Kind	Series	N

10 Figures

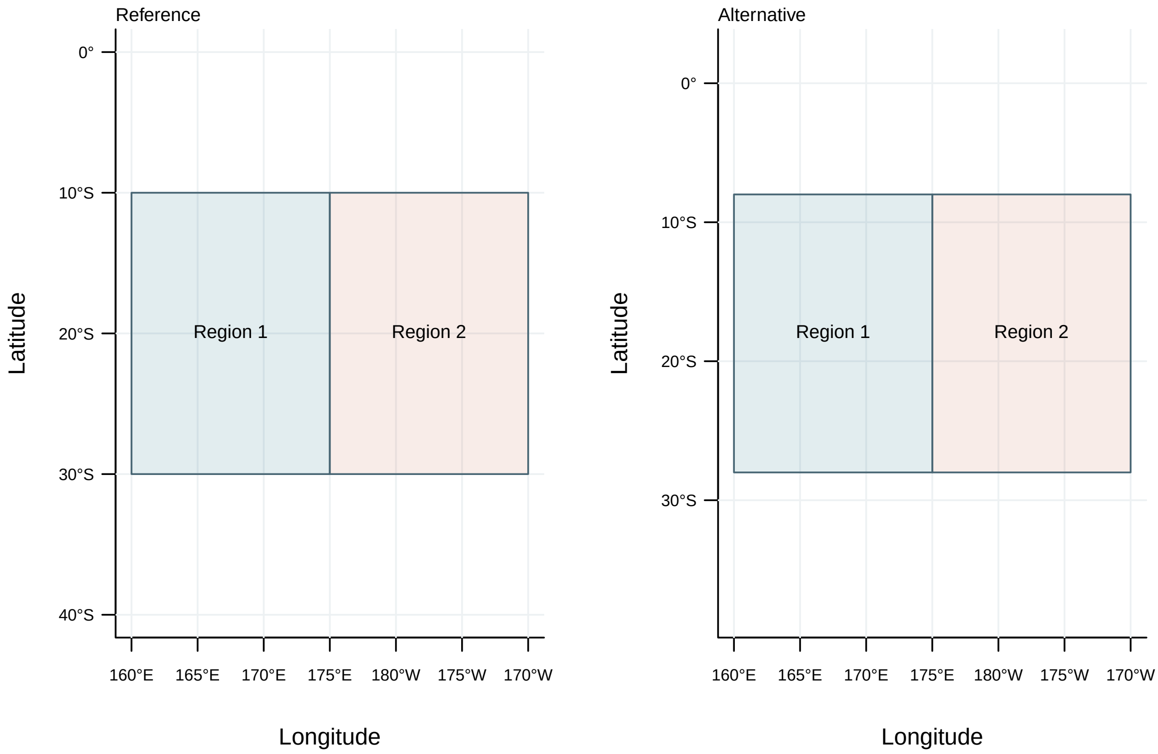

Figure 1: Illustrative output. Model regions.

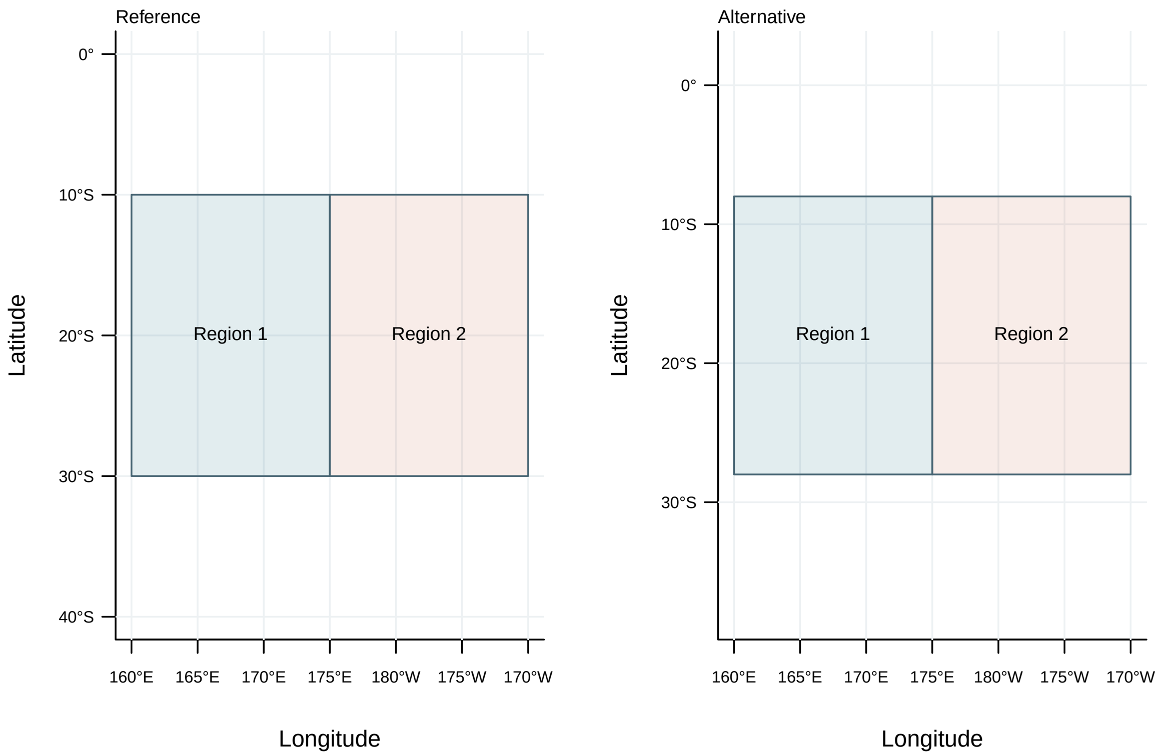

Figure 2: Illustrative output. Comparison of model spatial structures.

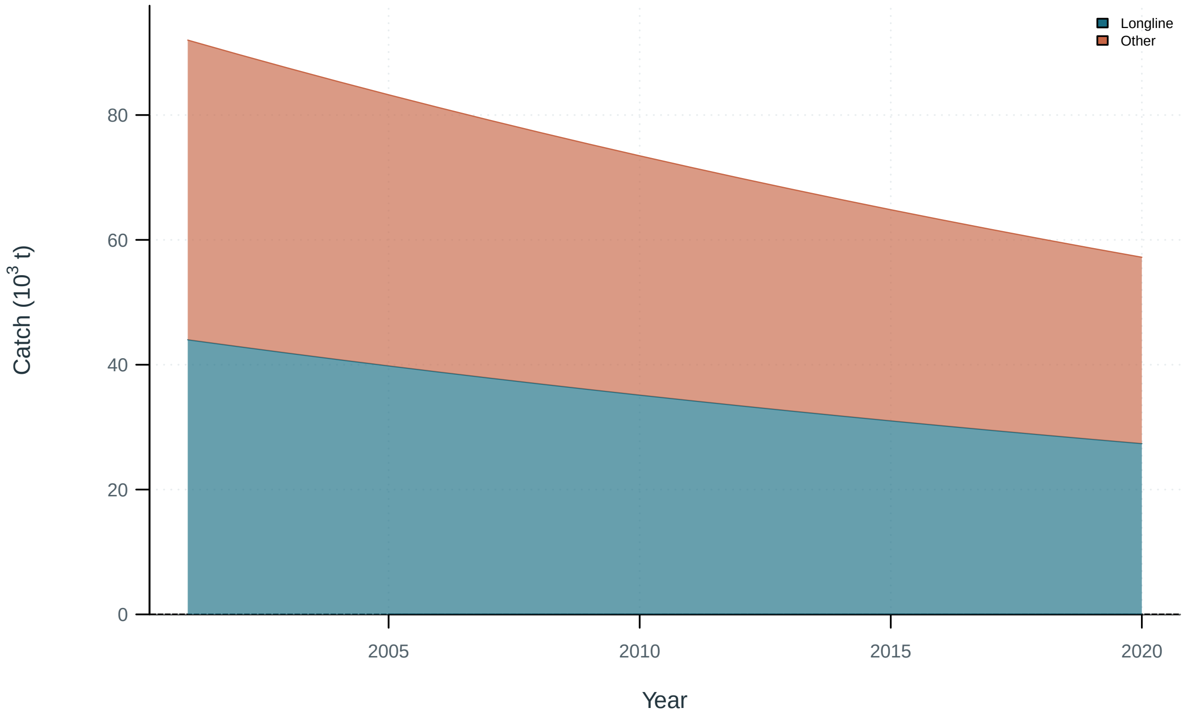

Figure 3: Illustrative output. Catch by fishing group.

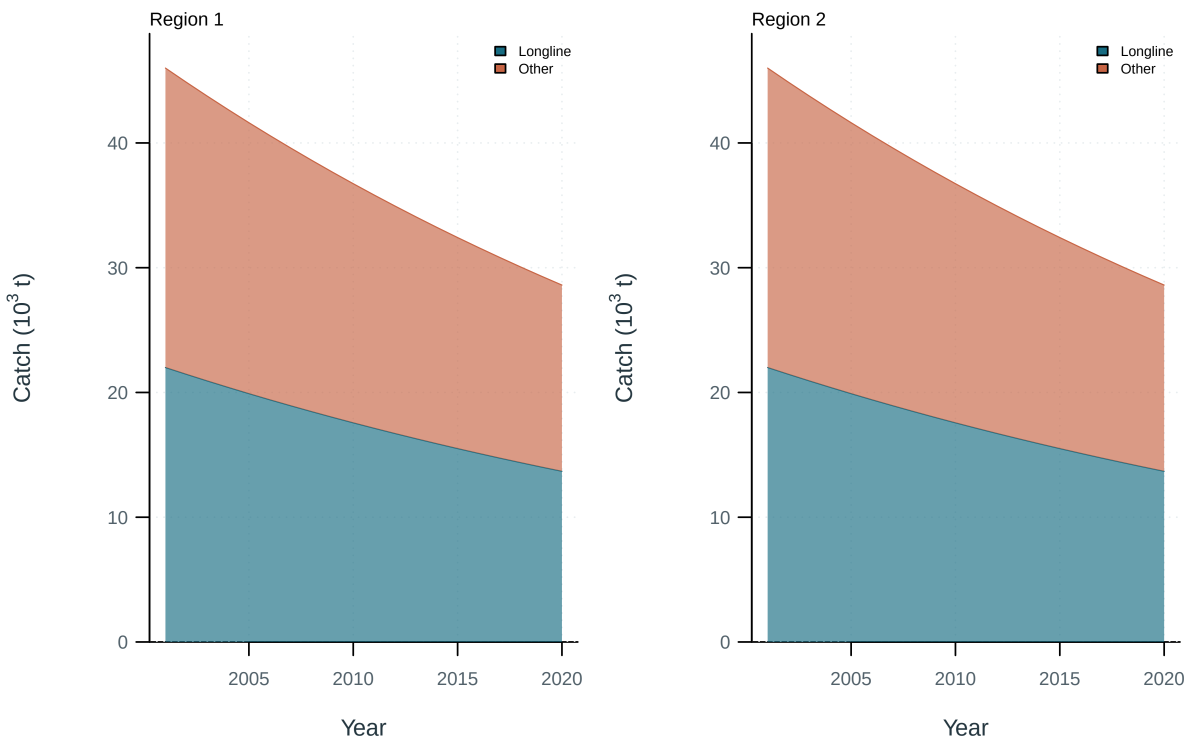

Figure 4: Illustrative output. Catch by fishing group and region.

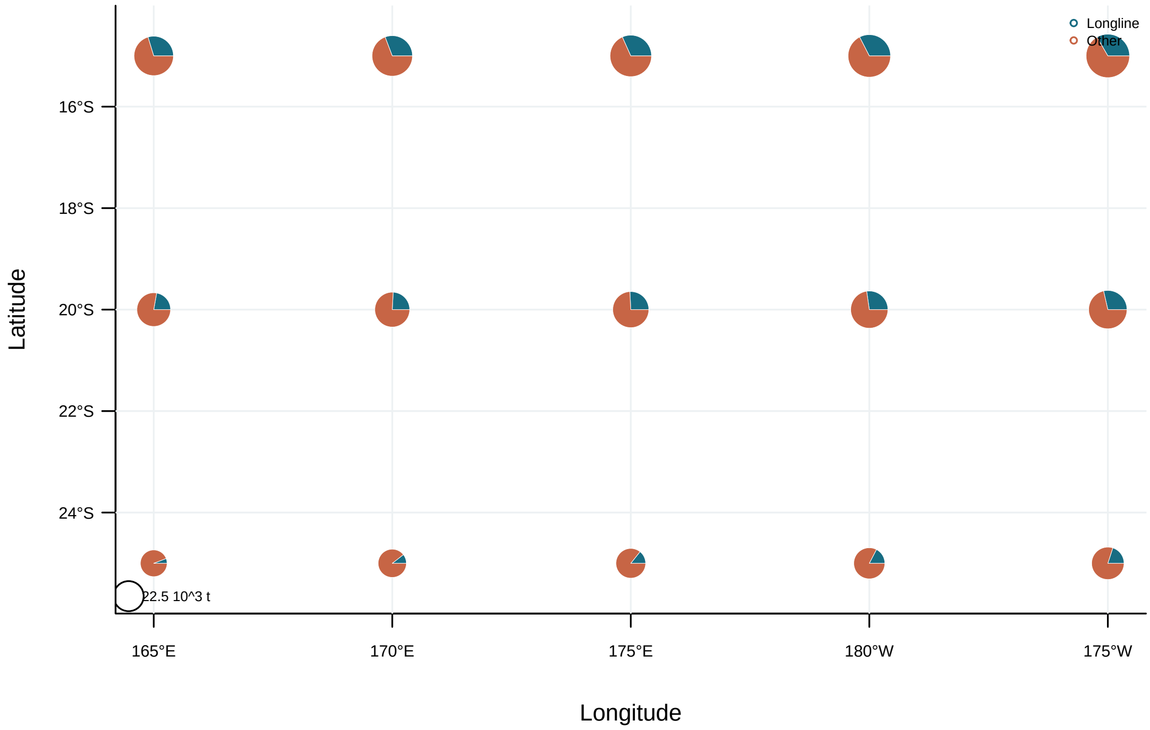

Figure 5: Illustrative output. Spatial catch distribution by fishing group.

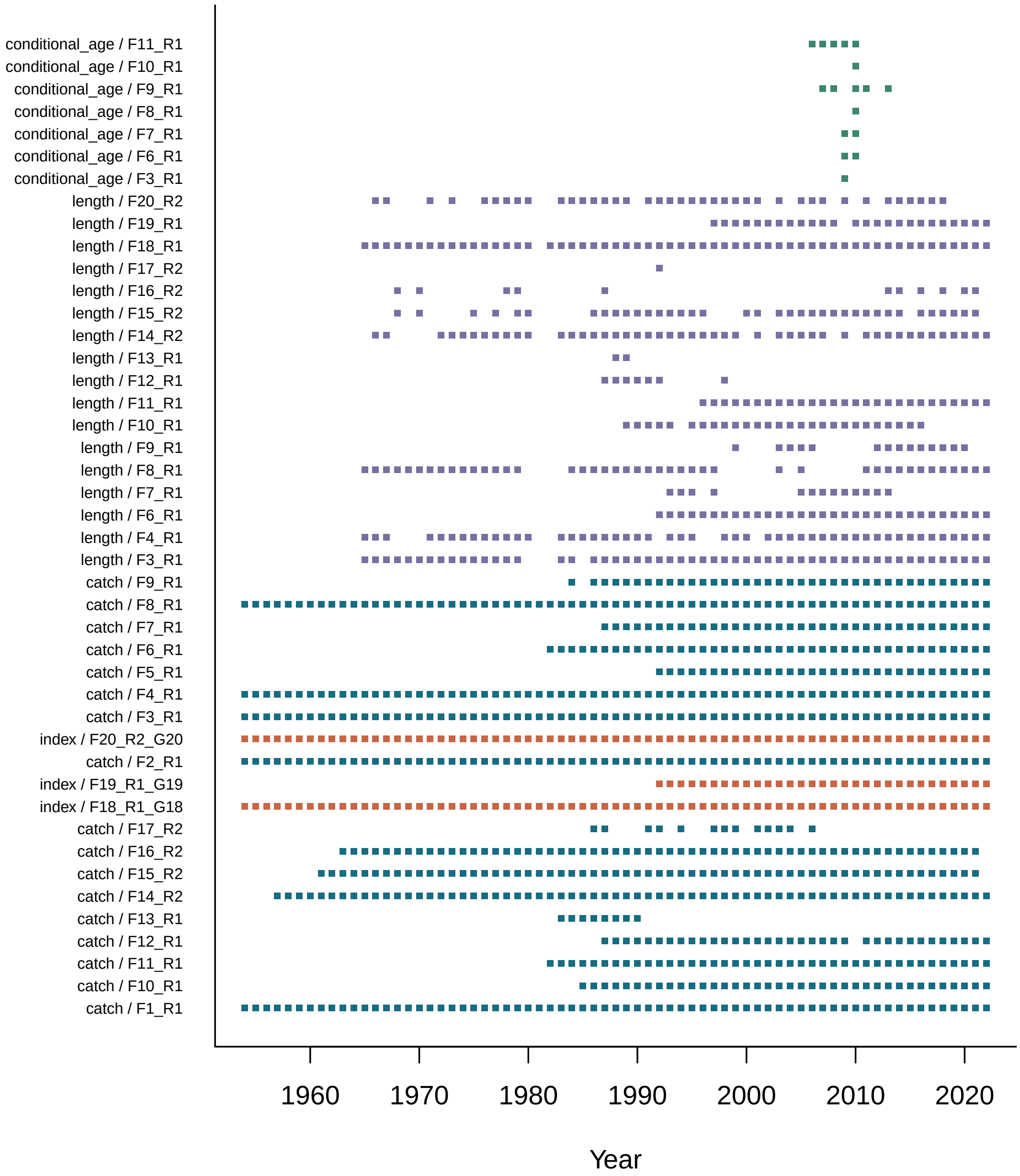

Figure 6: Availability of catch, abundance-index and composition observations by series and year.

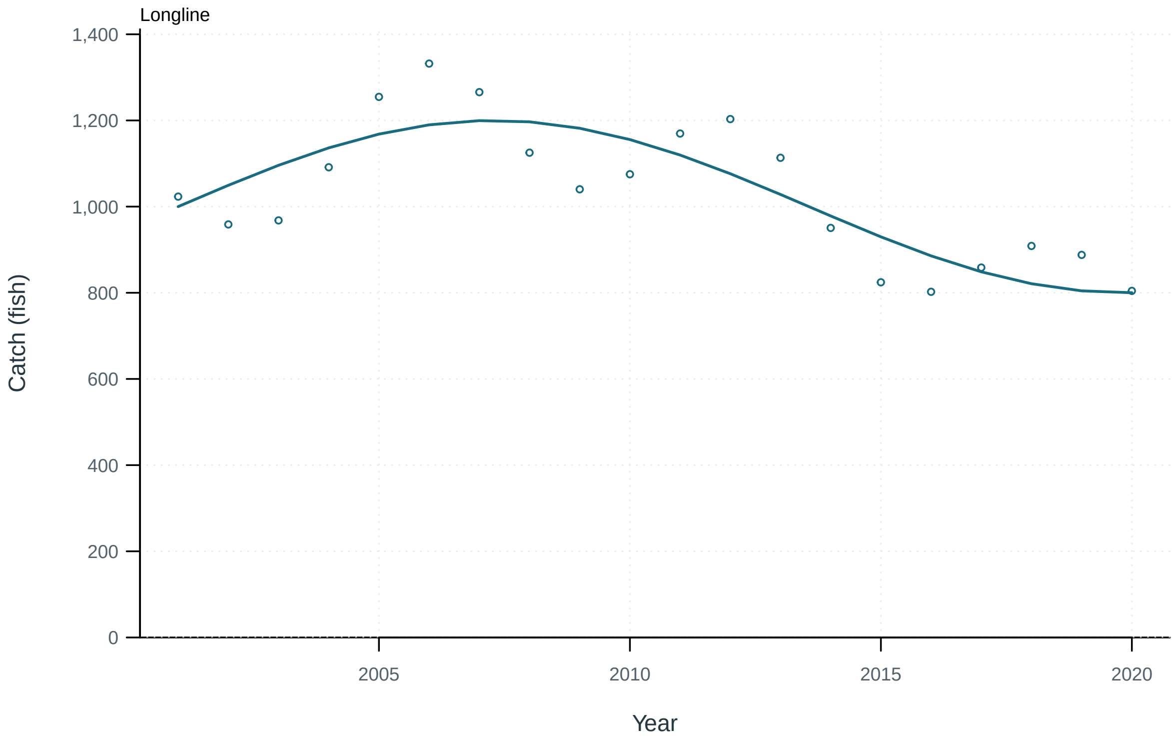

Figure 7: Illustrative output. Observed and predicted catch by fishery.

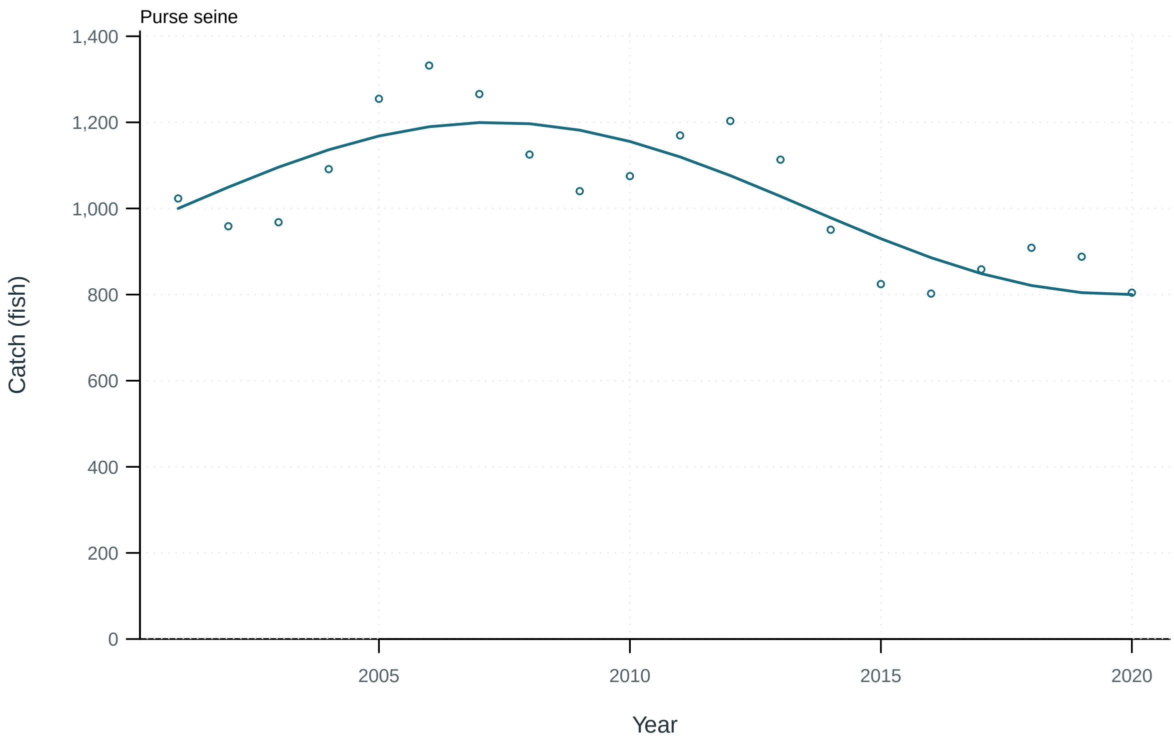

Figure 8: Illustrative output. Observed and predicted catch by fishery.

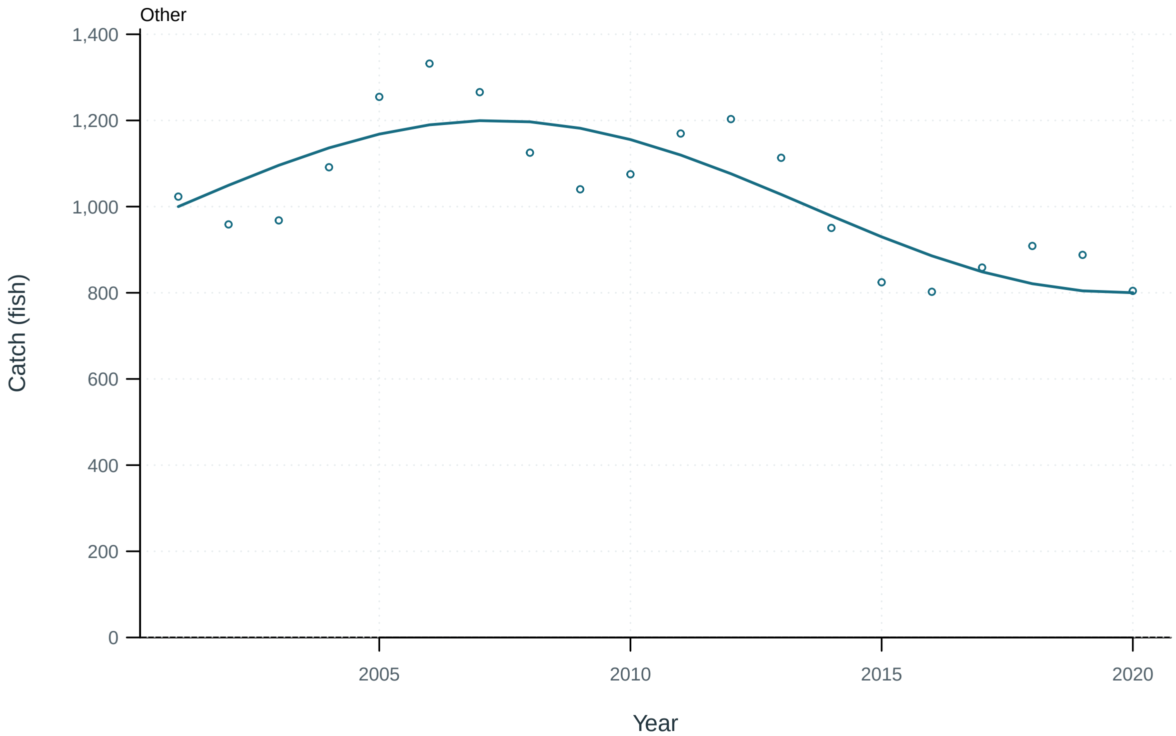

Figure 9: Illustrative output. Observed and predicted catch by fishery.

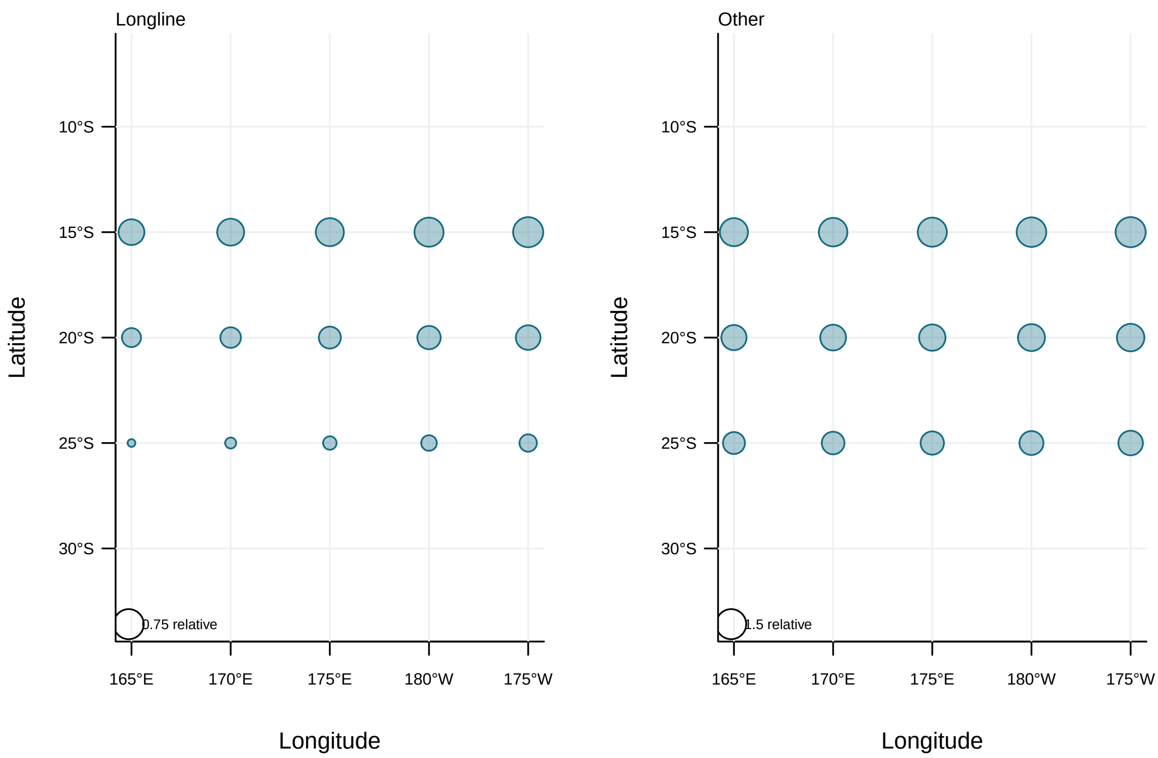

Figure 10: Illustrative output. Spatial distribution of nominal abundance indices.

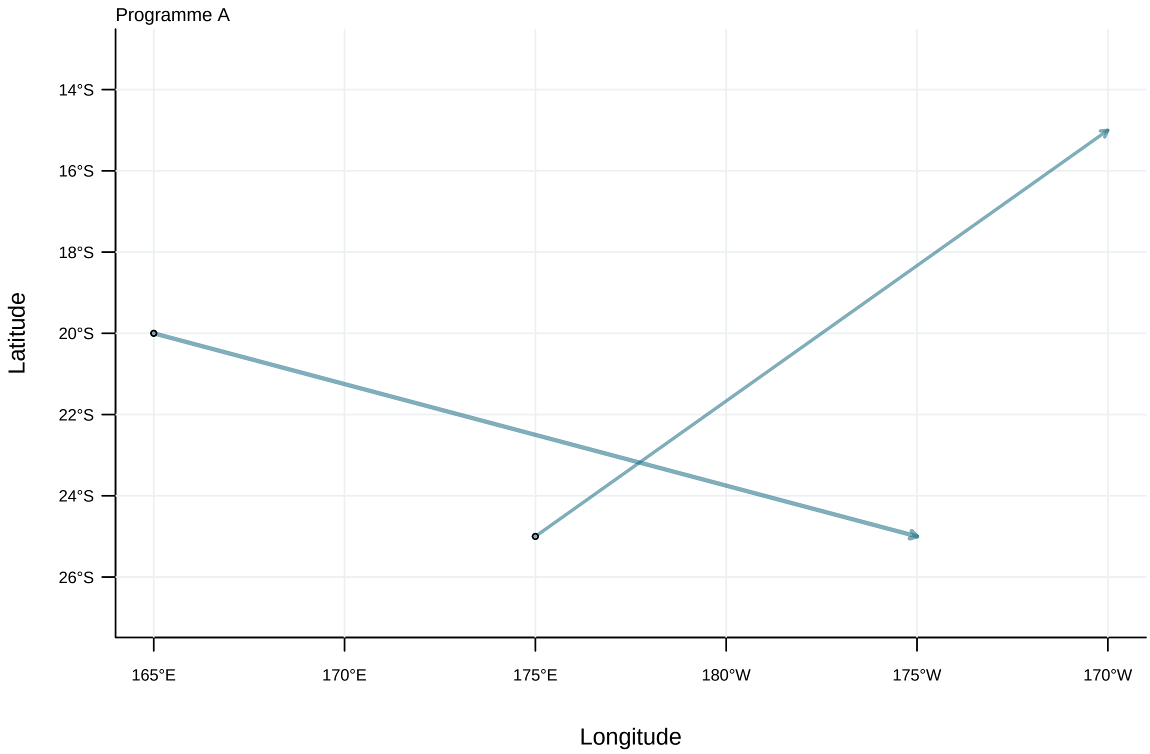

Figure 11: Illustrative output. Tag-release and recapture displacements by programme.

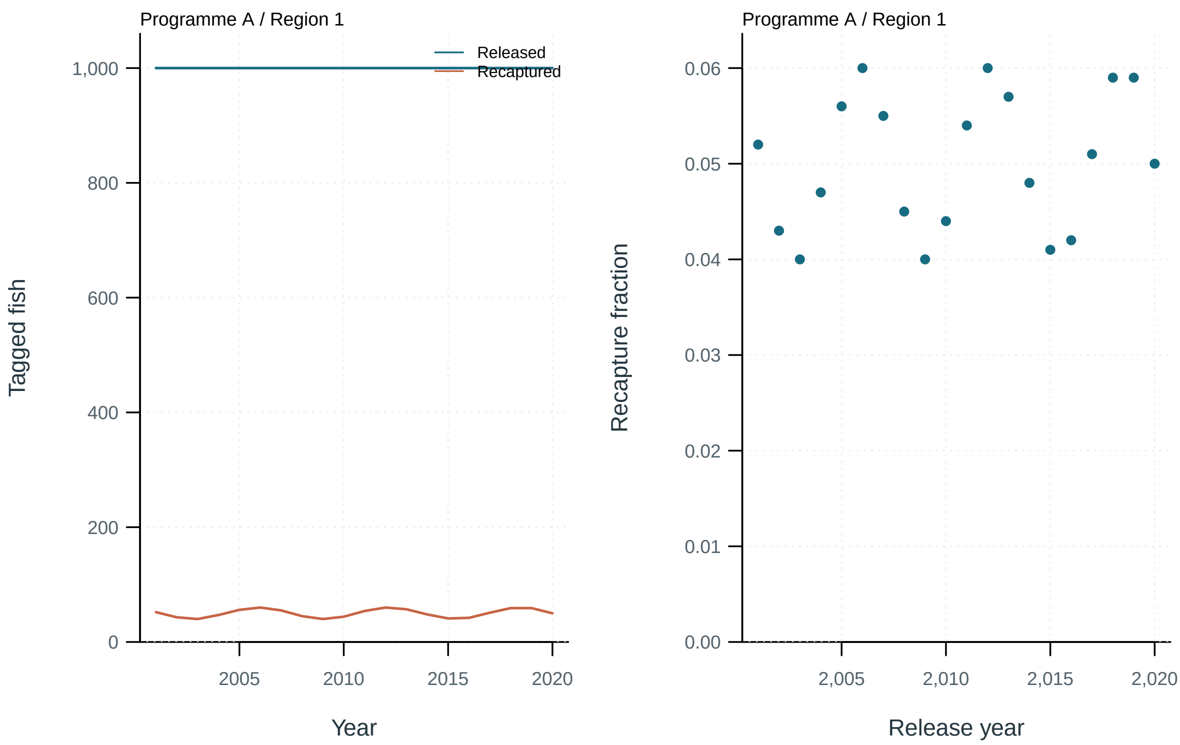

Figure 12: Illustrative output. Tag releases, recaptures and recapture fractions by cohort.

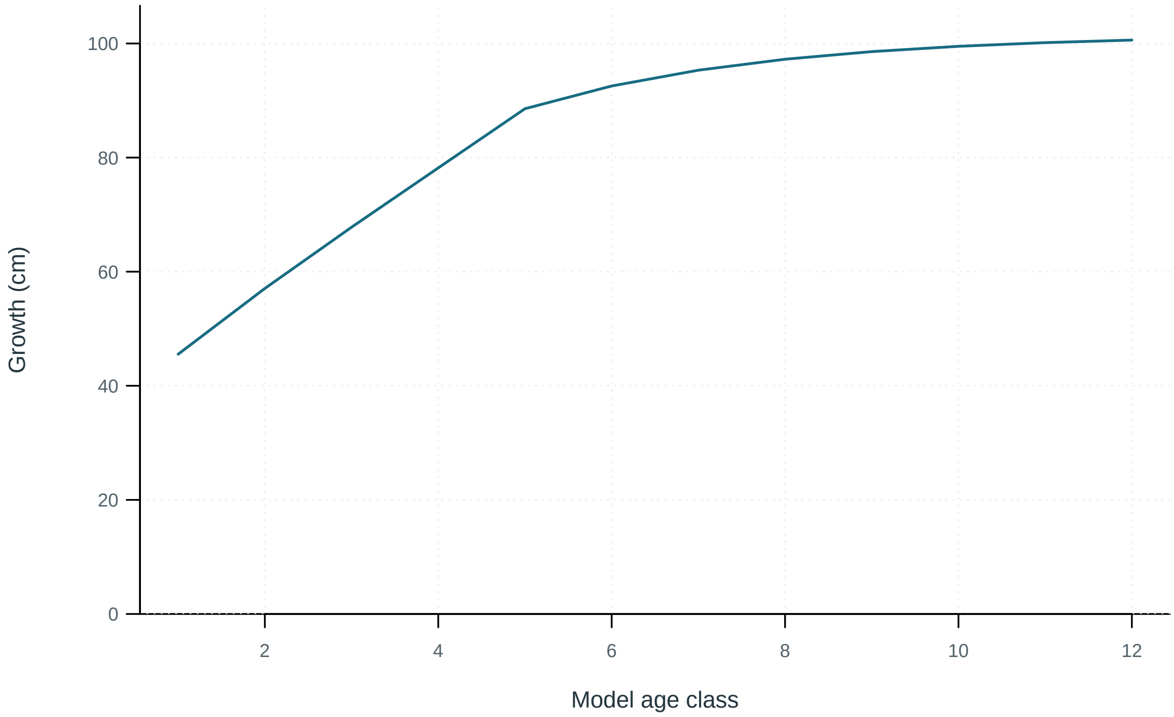

Figure 13: Mean length at the start of each model age class.

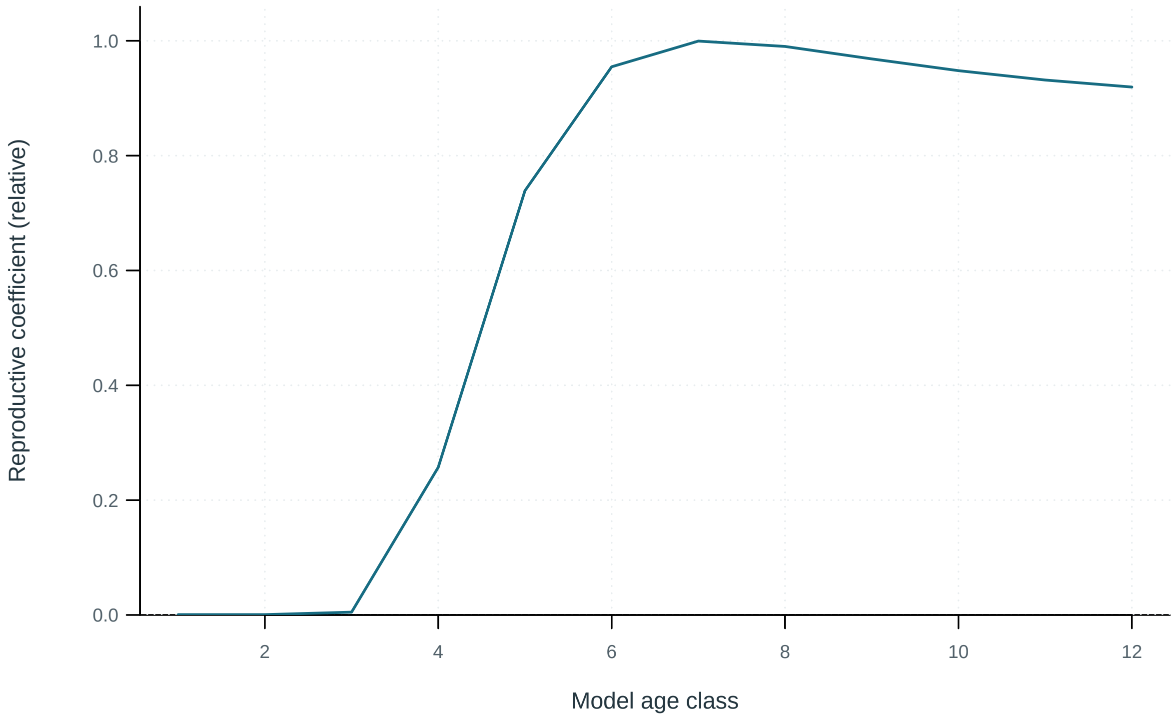

Figure 14: Model reproductive coefficient at age.

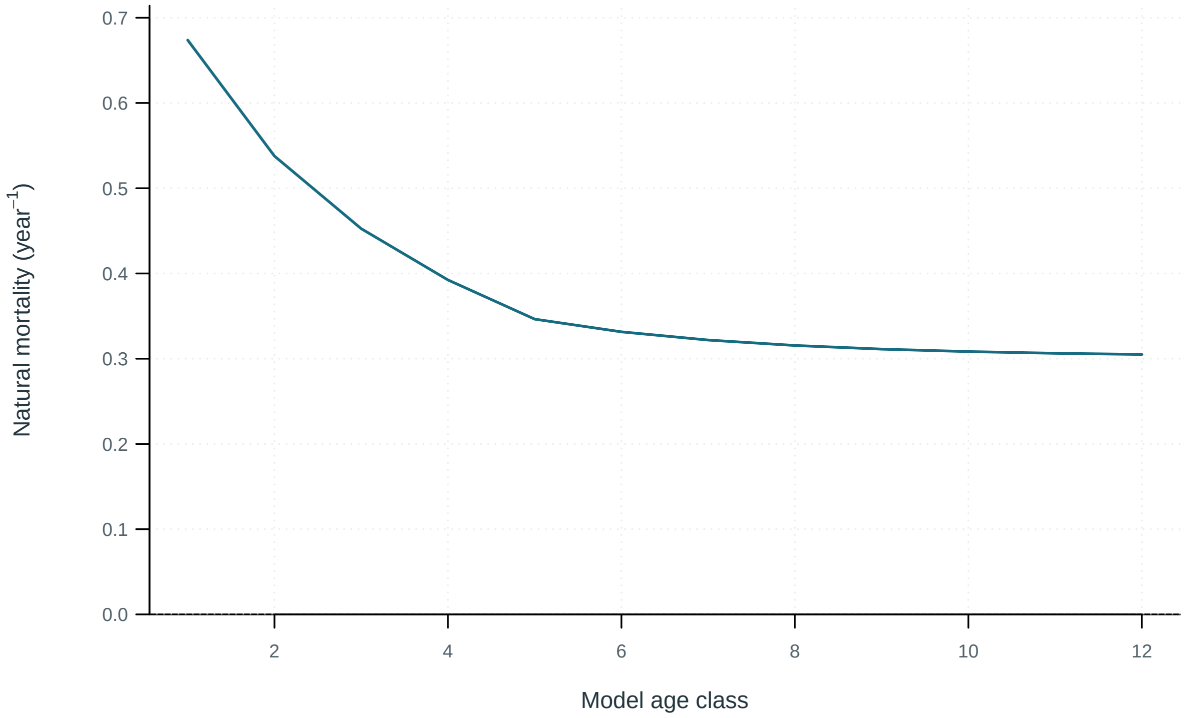

Figure 15: Annual instantaneous natural mortality at age.

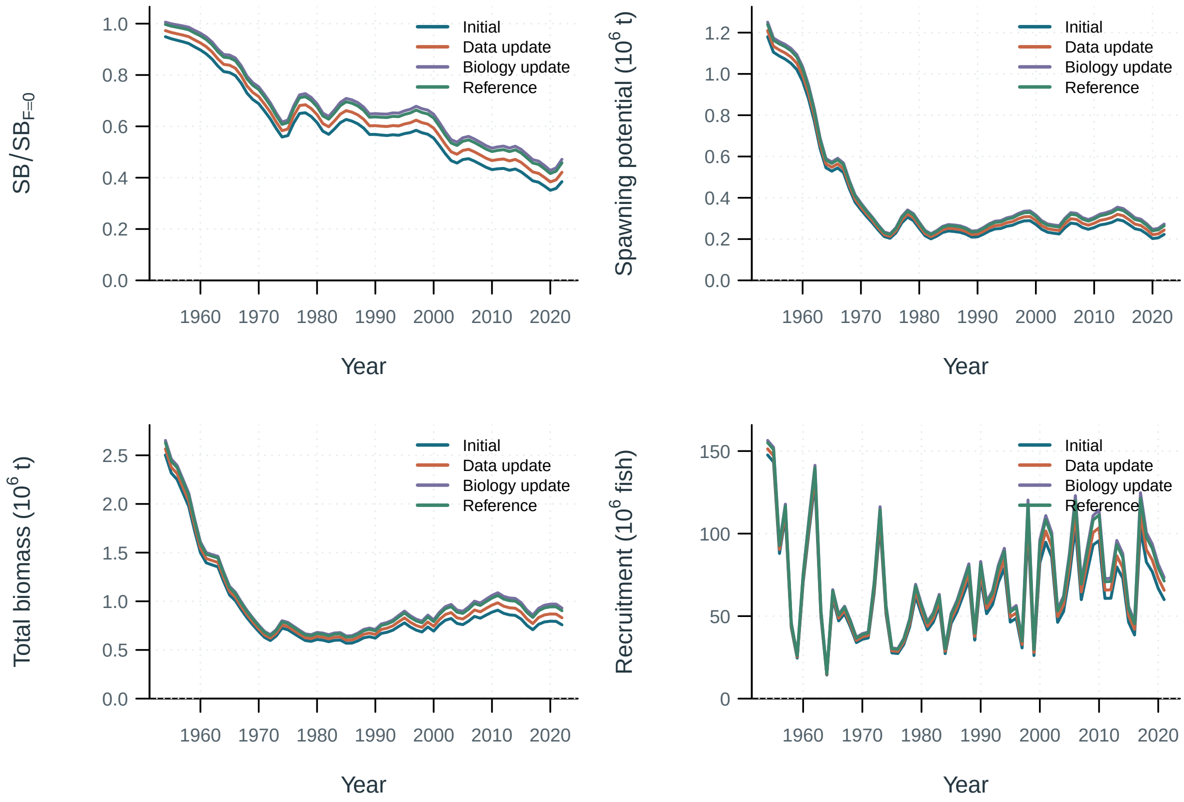

Figure 16: Stepwise model trajectories.

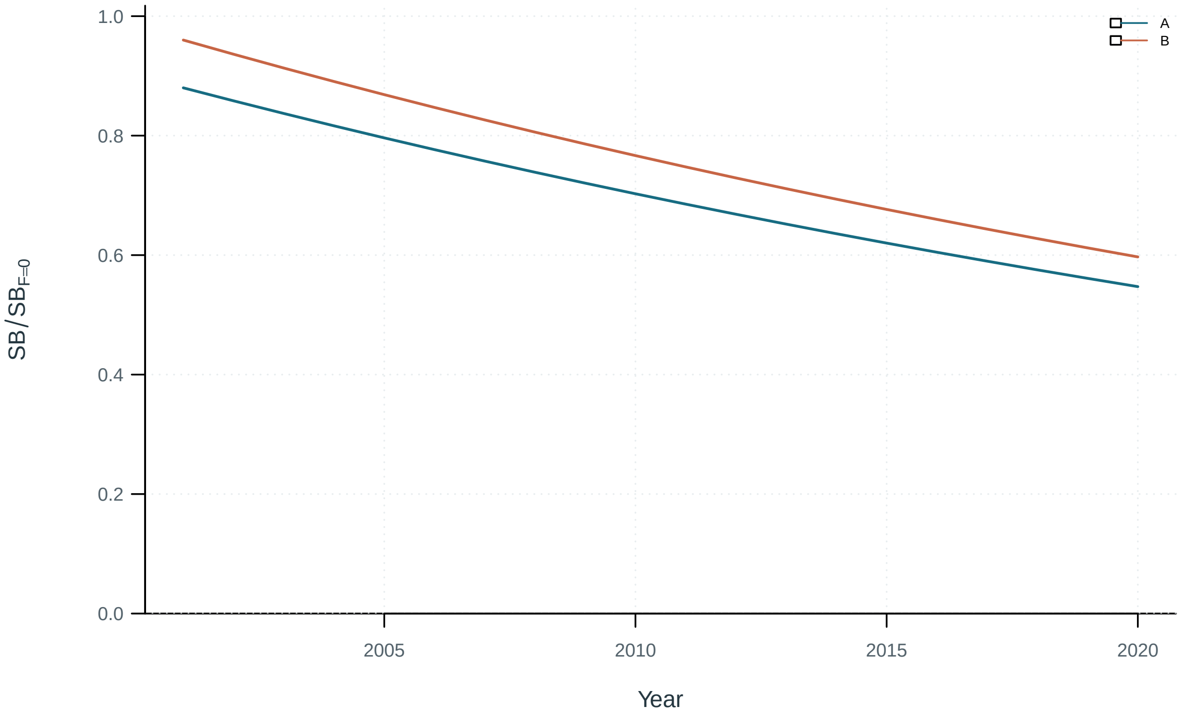

Figure 17: Selected model-development trajectories.

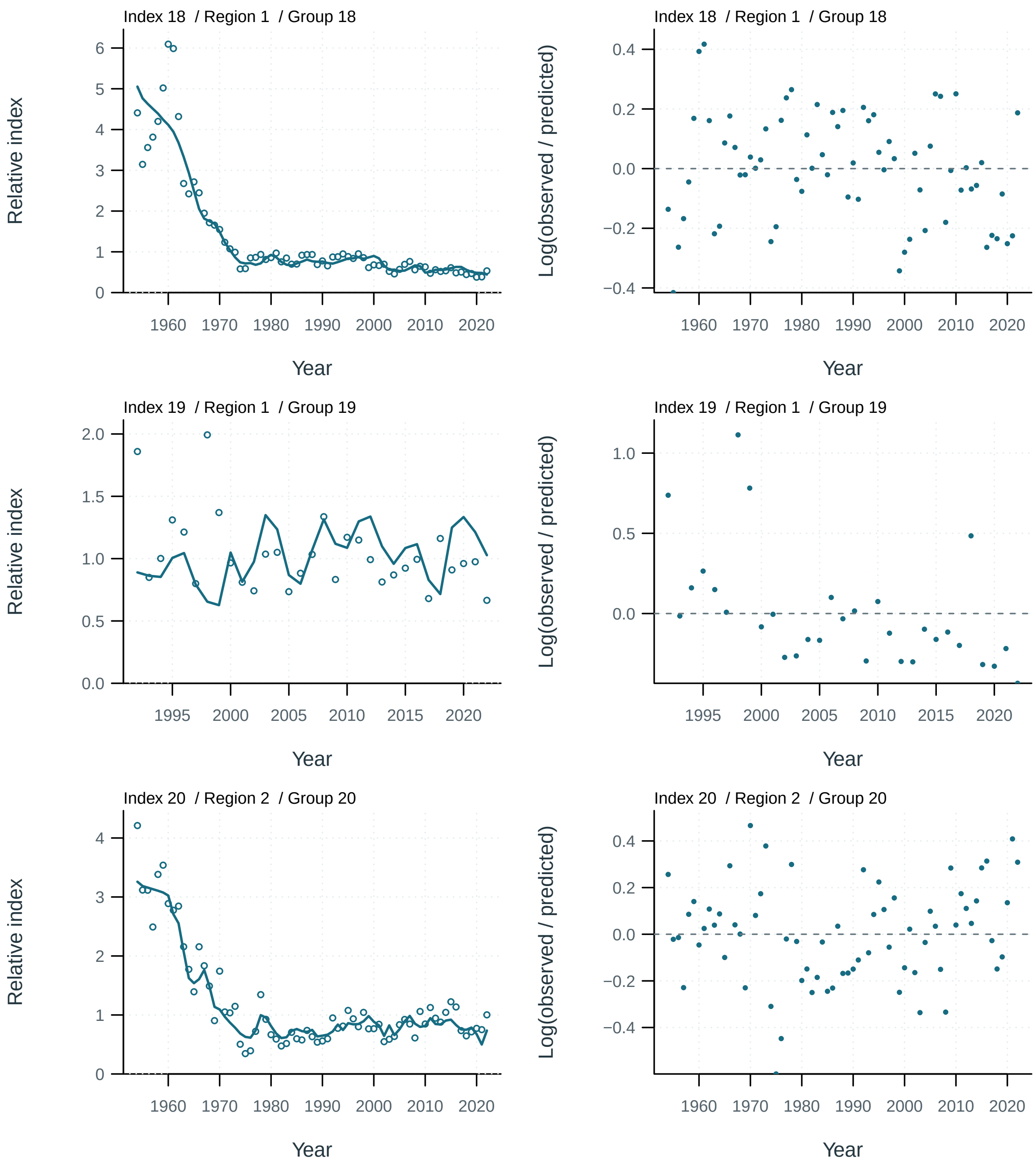

Figure 18: Abundance-index fits and log residuals by series. Residuals are log(observed/predicted).

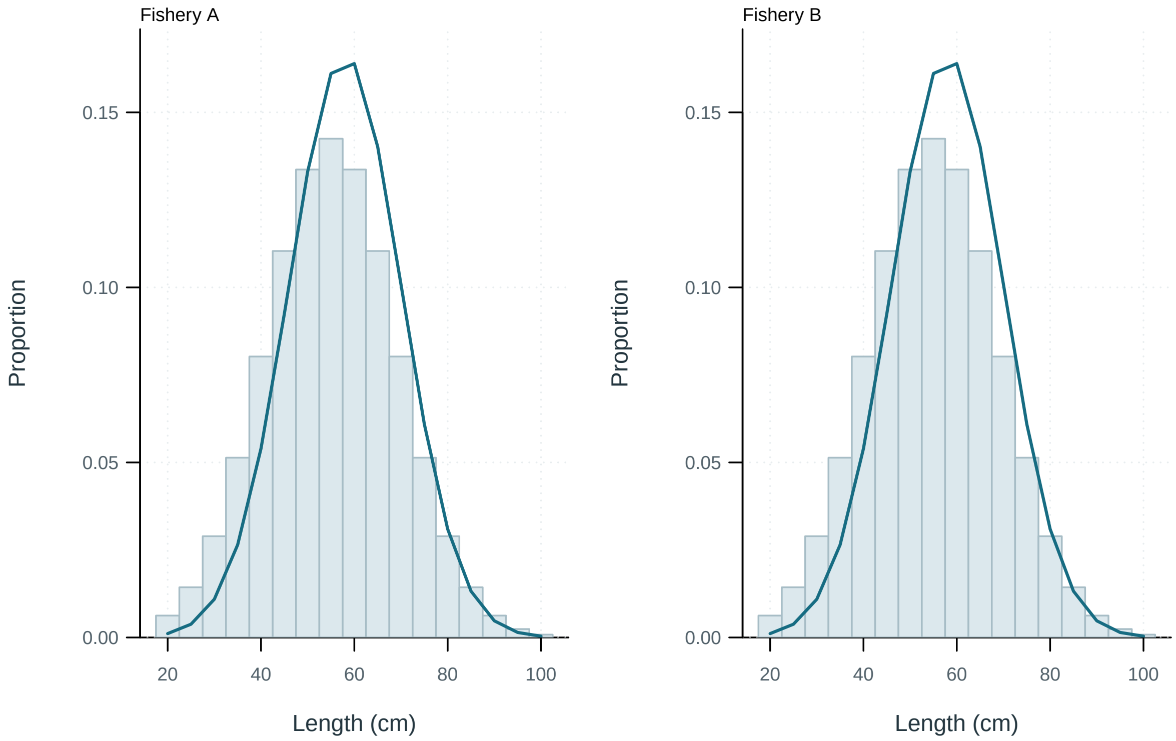

Figure 19: Illustrative output. Pooled observed and predicted length distributions.

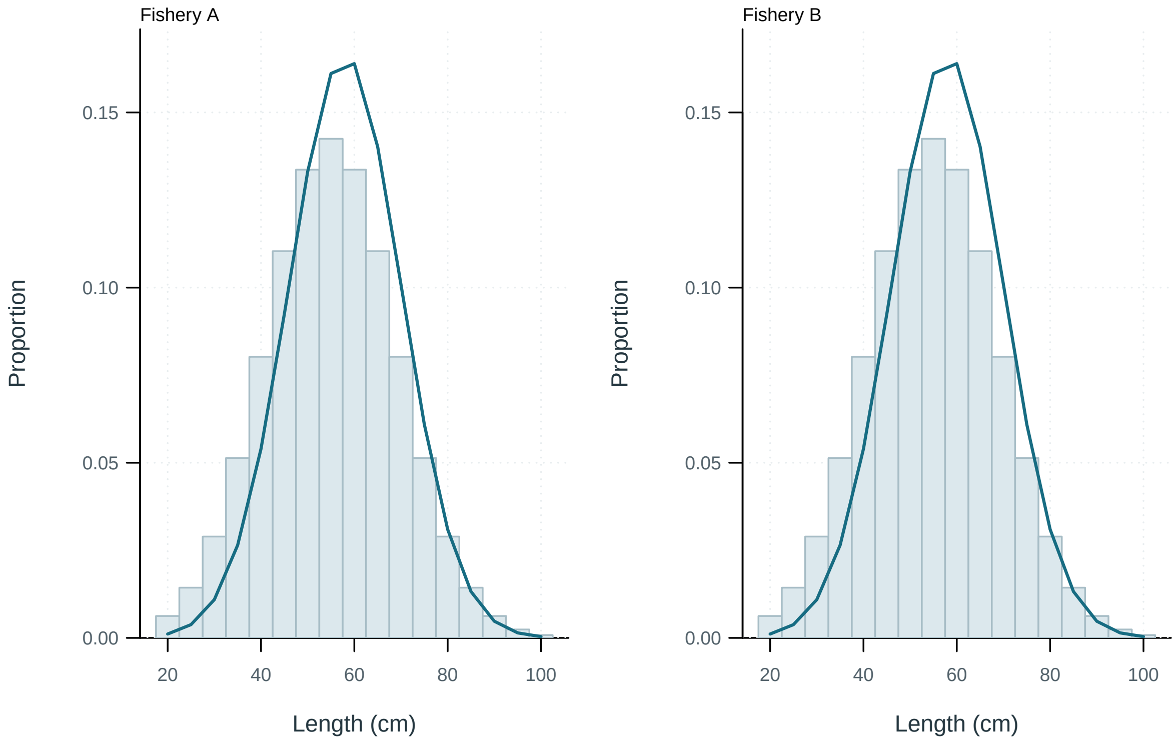

Figure 20: Illustrative output. Pooled observed and predicted length distributions.

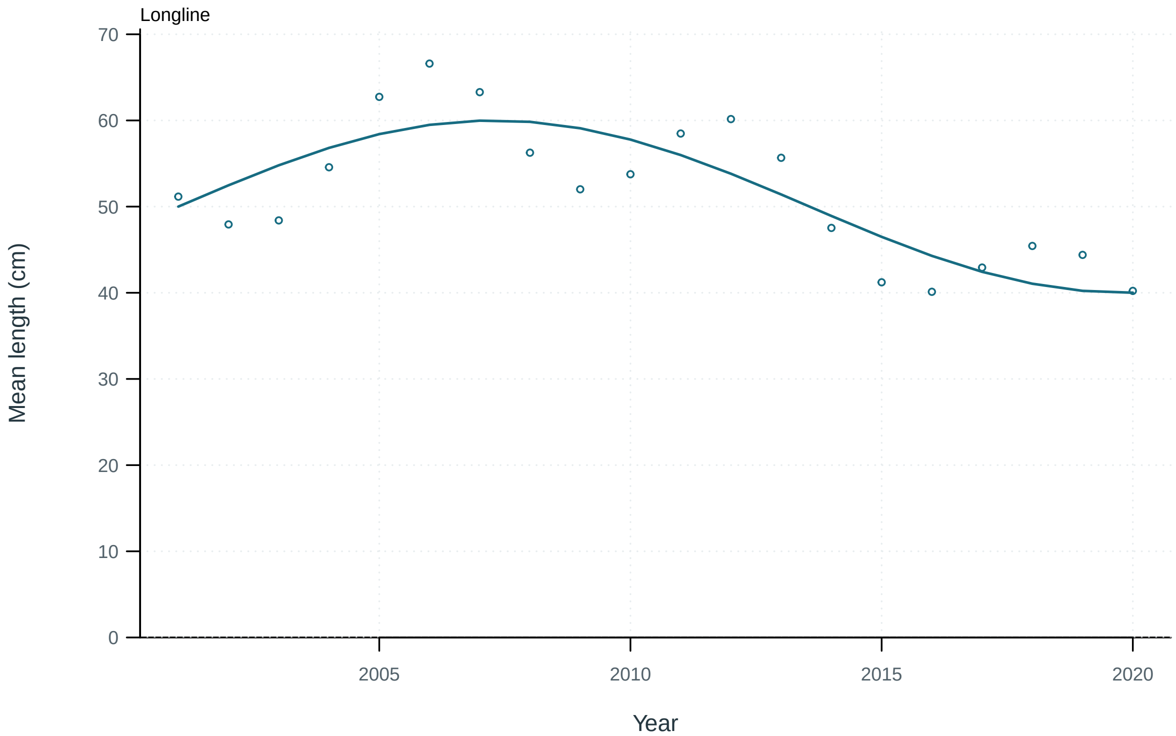

Figure 21: Illustrative output. Observed and predicted length summaries through time.

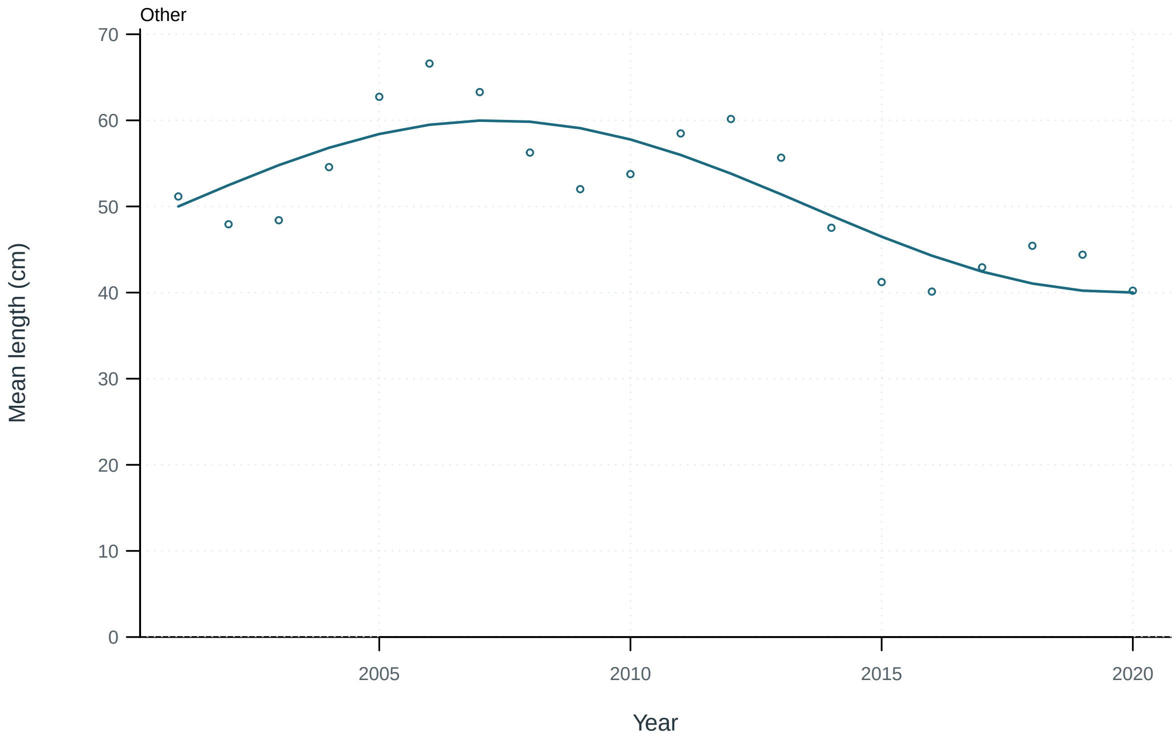

Figure 22: Illustrative output. Observed and predicted length summaries through time.

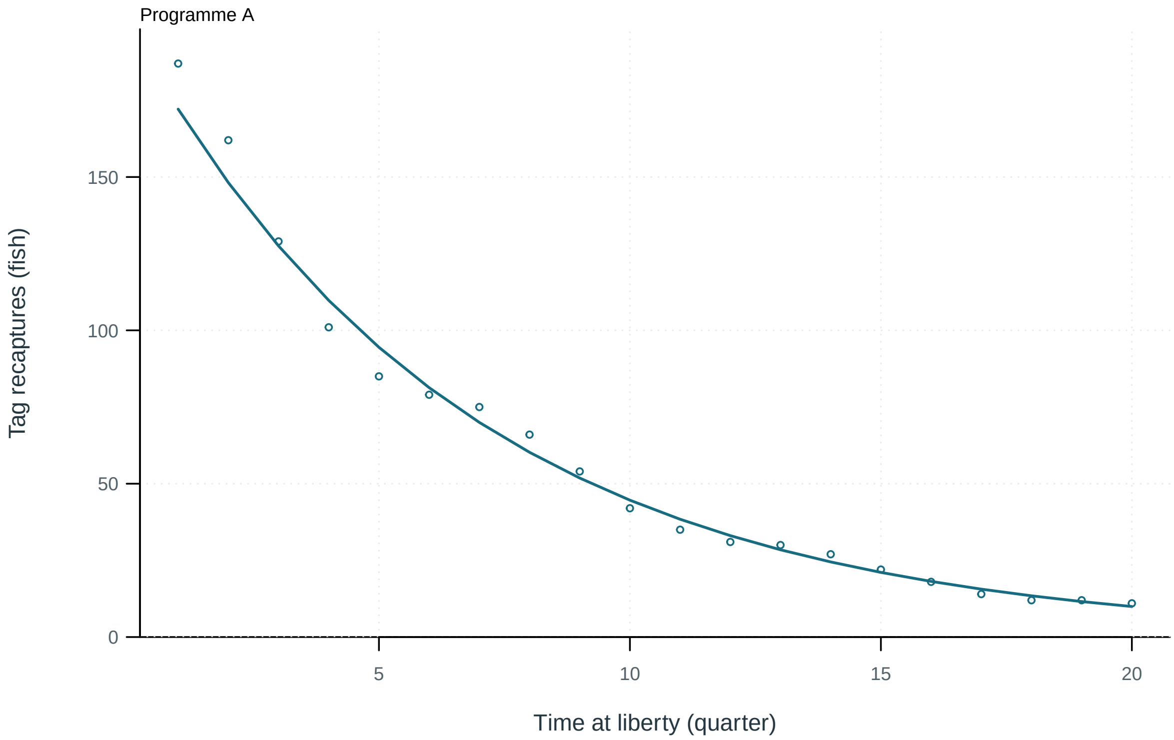

Figure 23: Illustrative output. Observed and predicted tag recaptures by time at liberty.

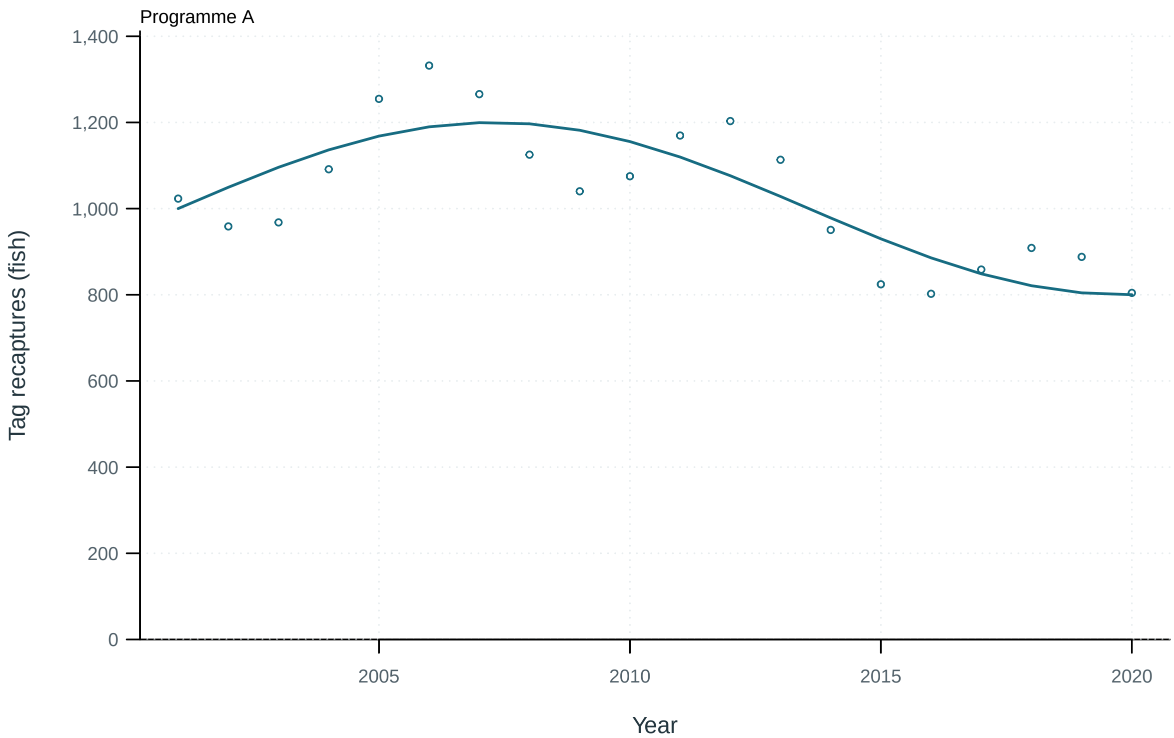

Figure 24: Illustrative output. Observed and predicted tag recaptures through time.

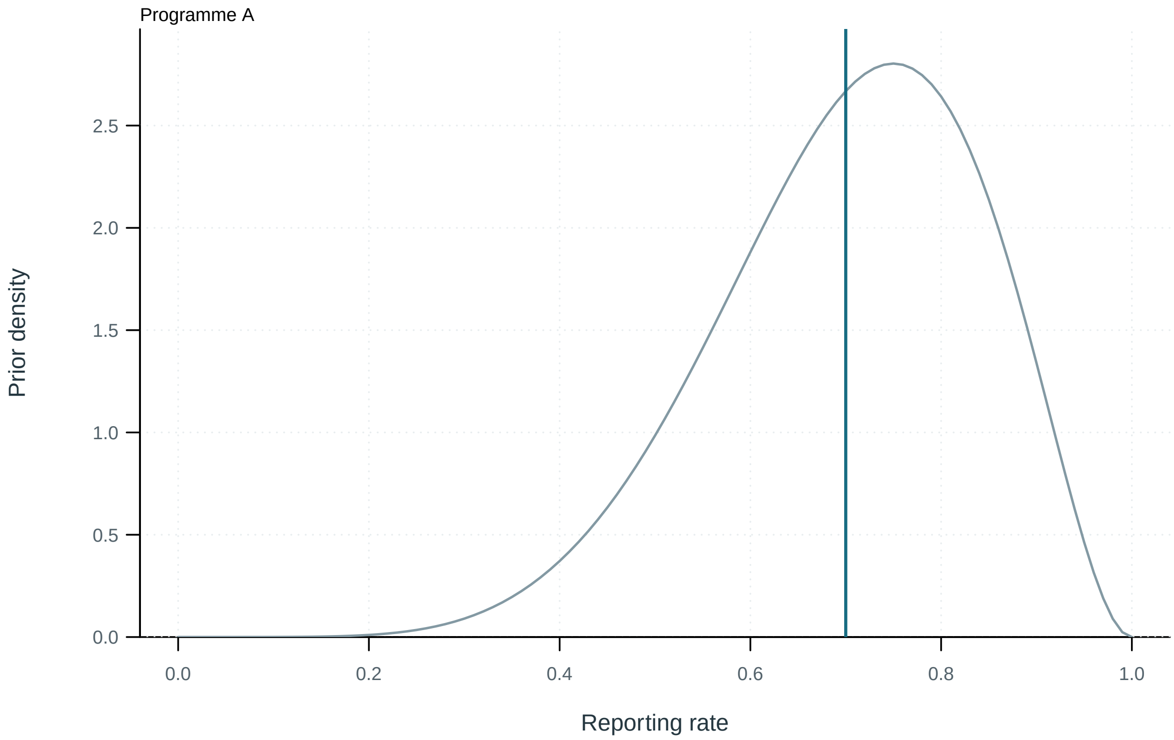

Figure 25: Illustrative output. Estimated tag-reporting rates and supplied priors.

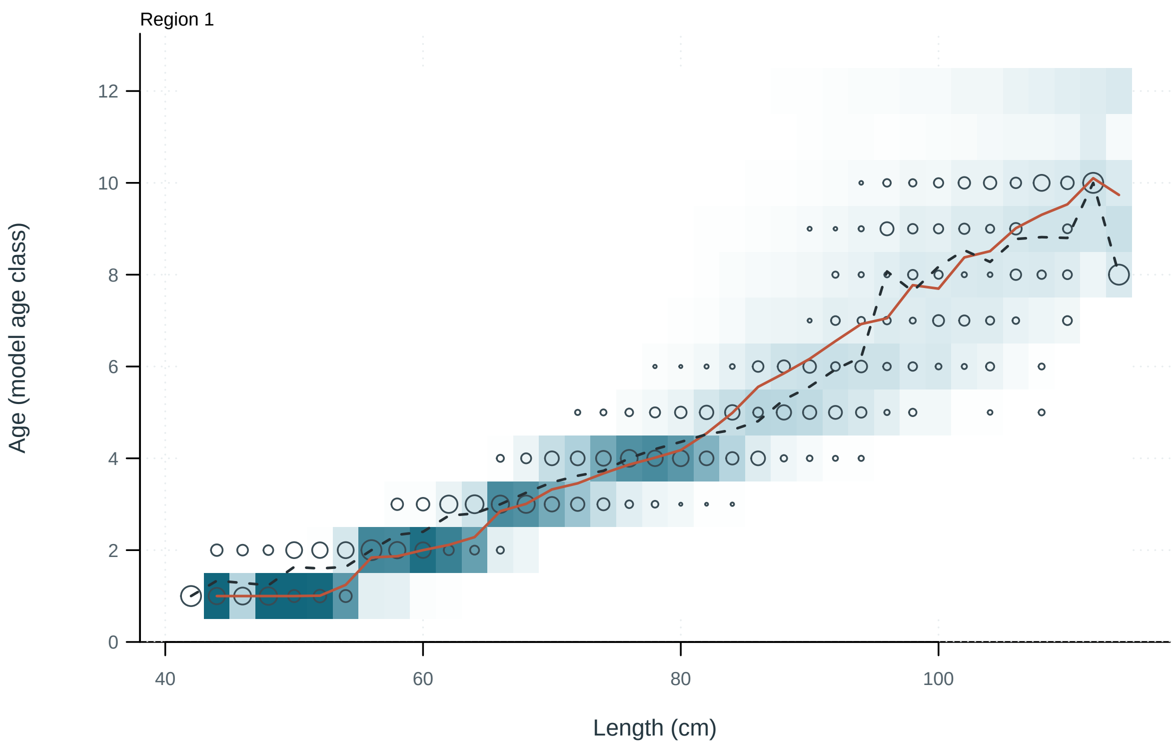

Figure 26: Observed and predicted age distributions conditional on length. Cells: predictions; circles: observations; solid and dashed lines: predicted and observed mean age.

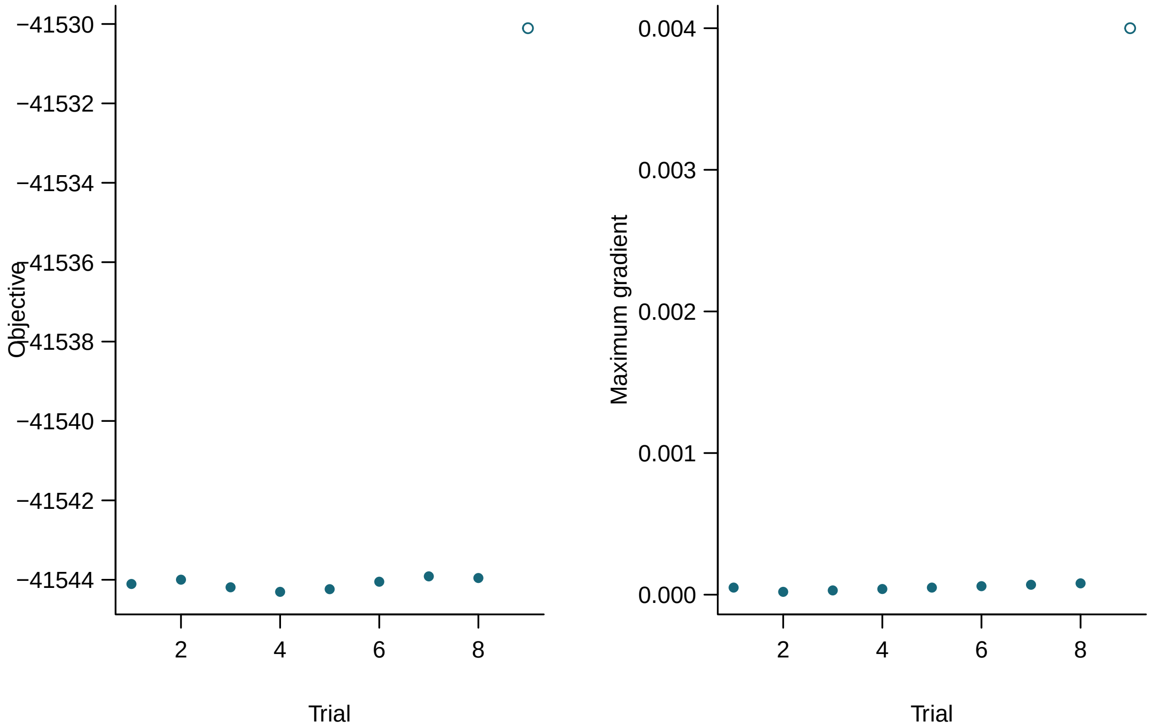

Figure 27: Final objectives and maximum gradients for independent starting values. Filled points meet all recorded numerical checks.

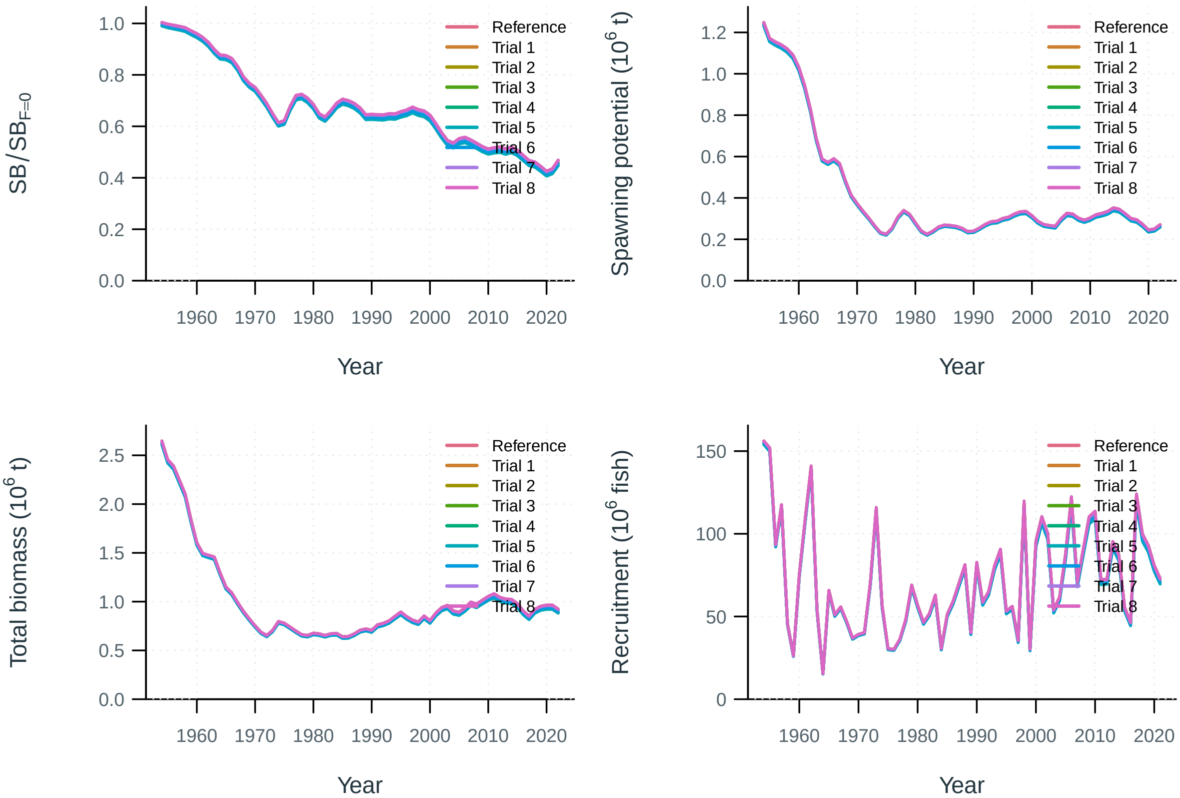

Figure 28: Jitter model trajectories.

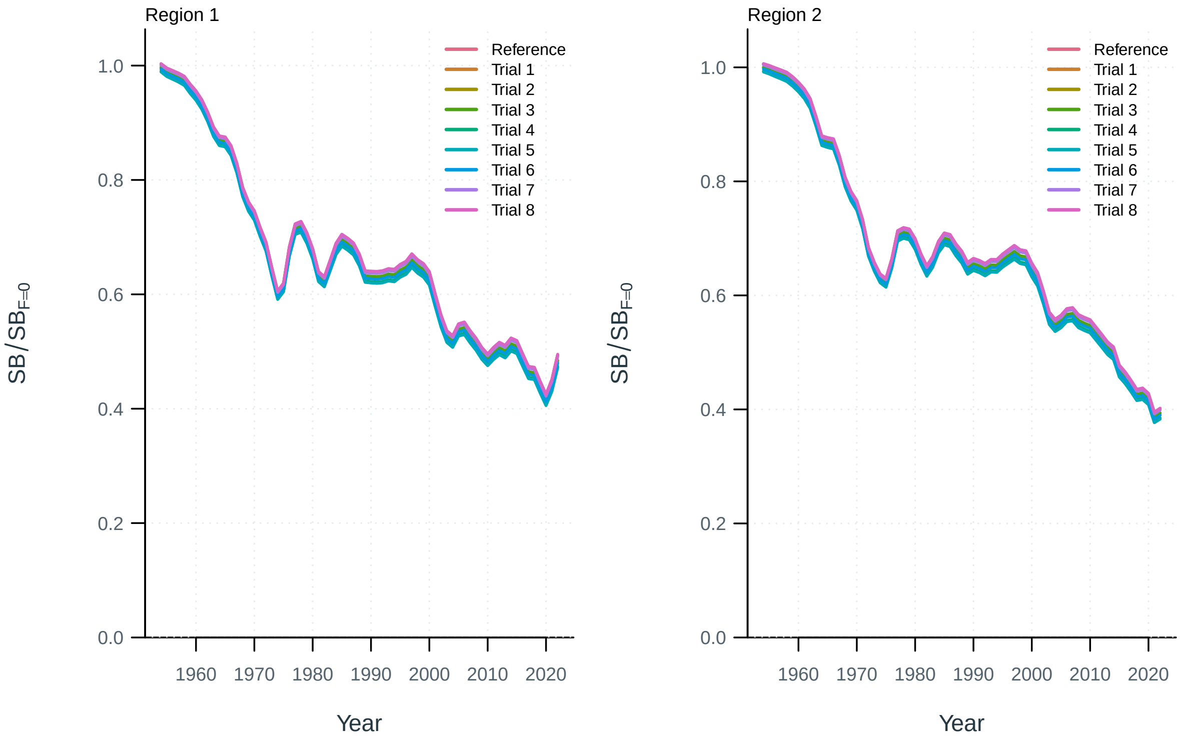

Figure 29: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

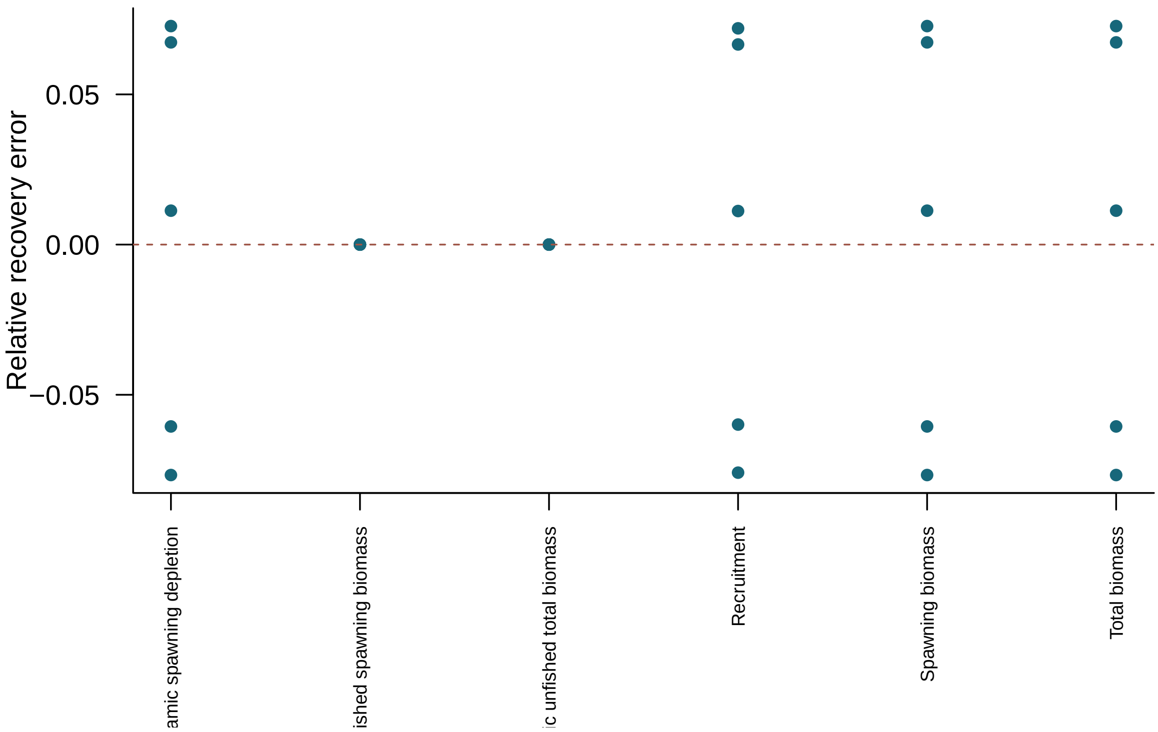

Figure 30: Relative recovery error in terminal quantities for converged simulated-data refits.

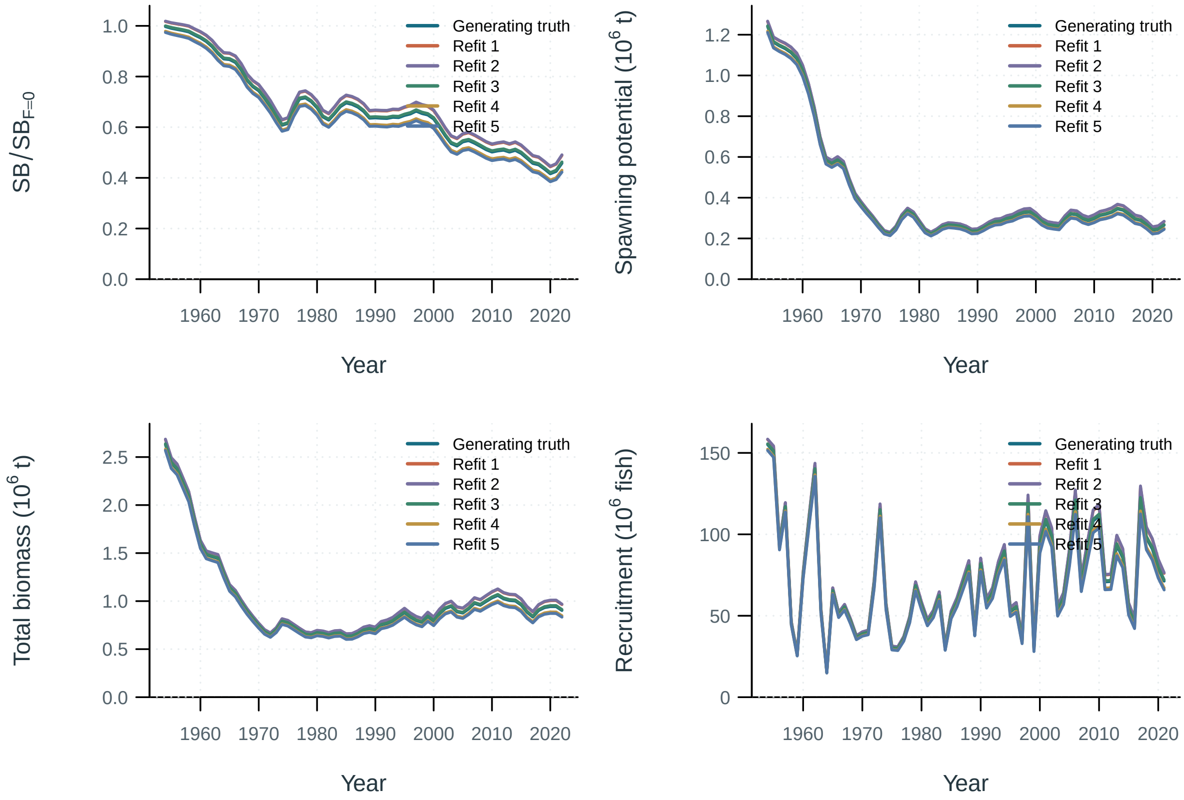

Figure 31: Selftest model trajectories.

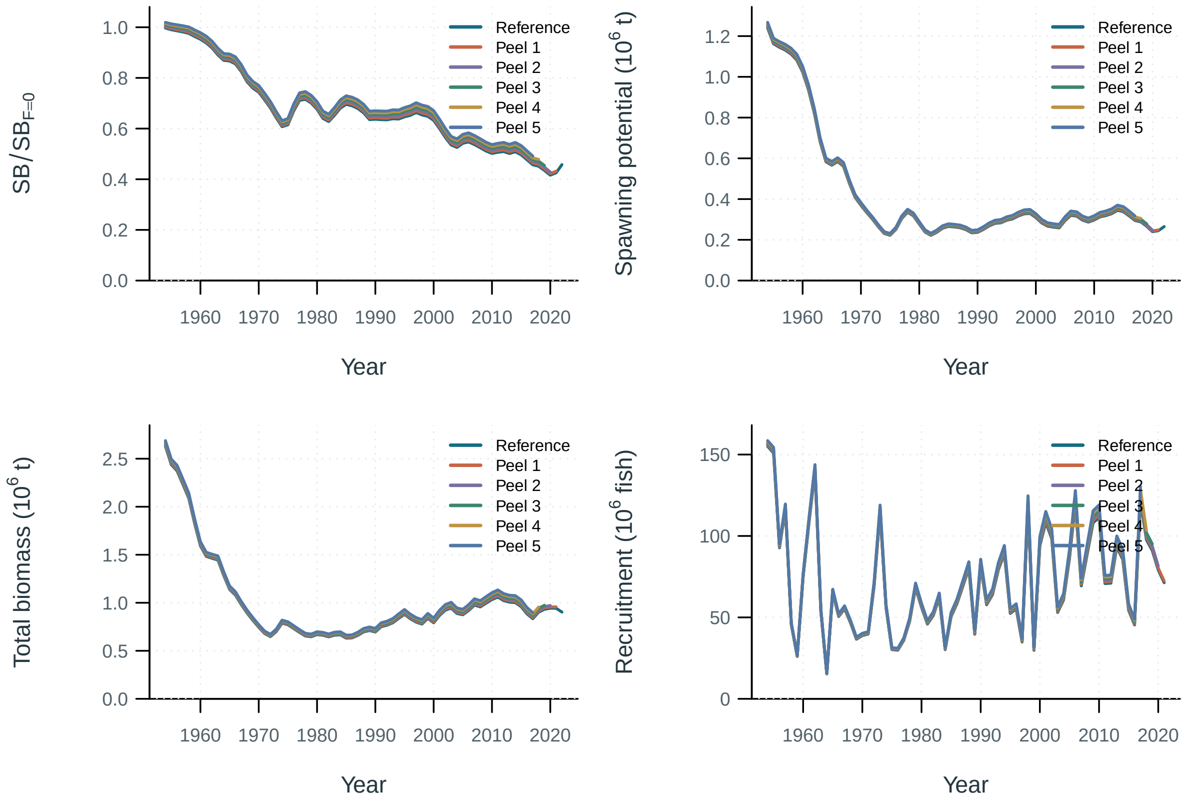

Figure 32: Retrospective model trajectories.

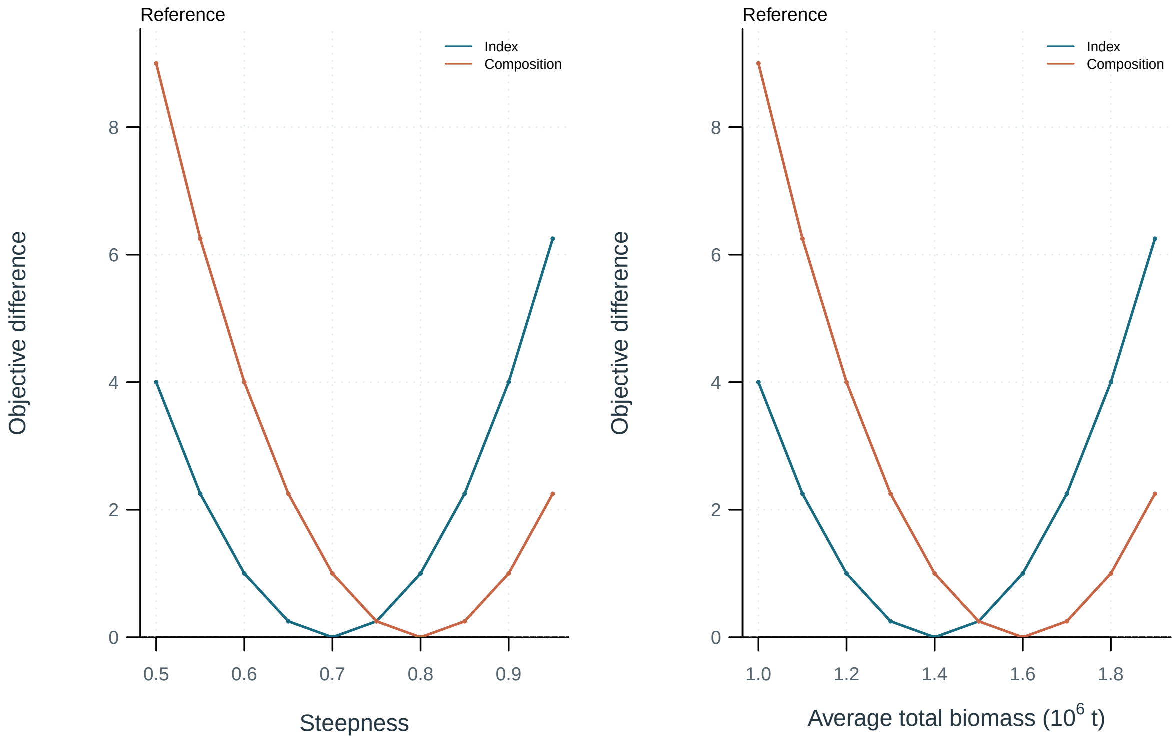

Figure 33: Objective profiles on the supplied parameter or quantity scale. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

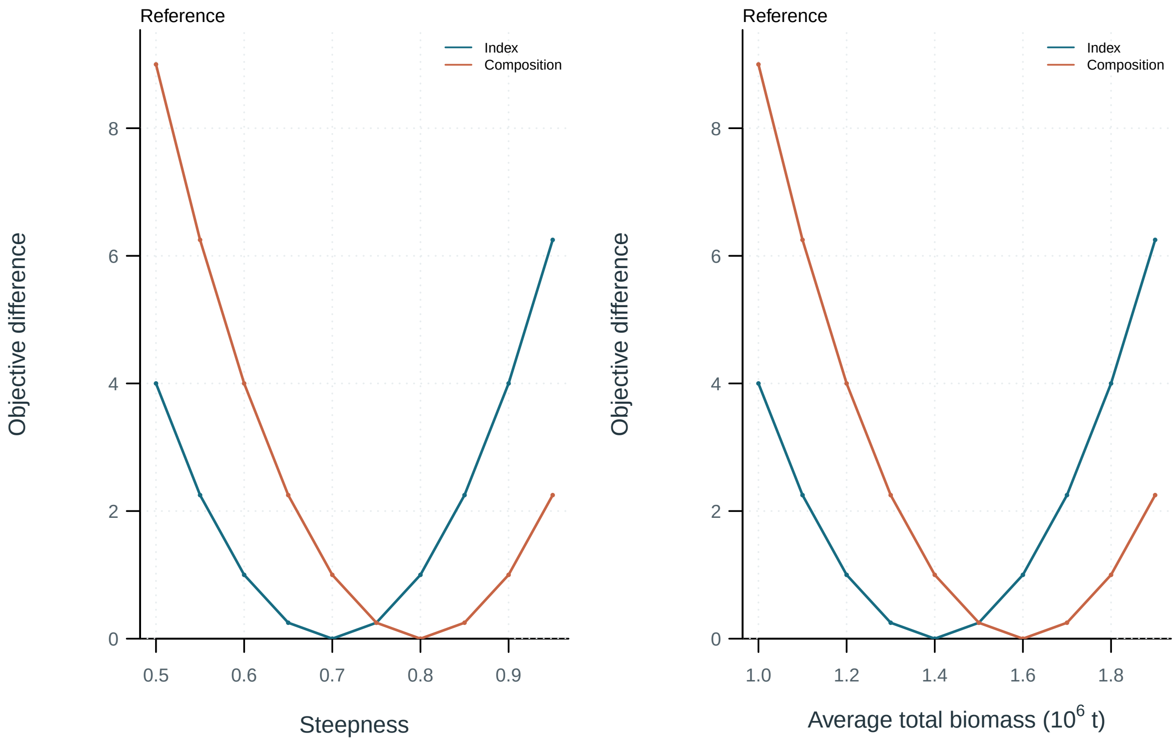

Figure 34: Objective profiles on the supplied parameter or quantity scale. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

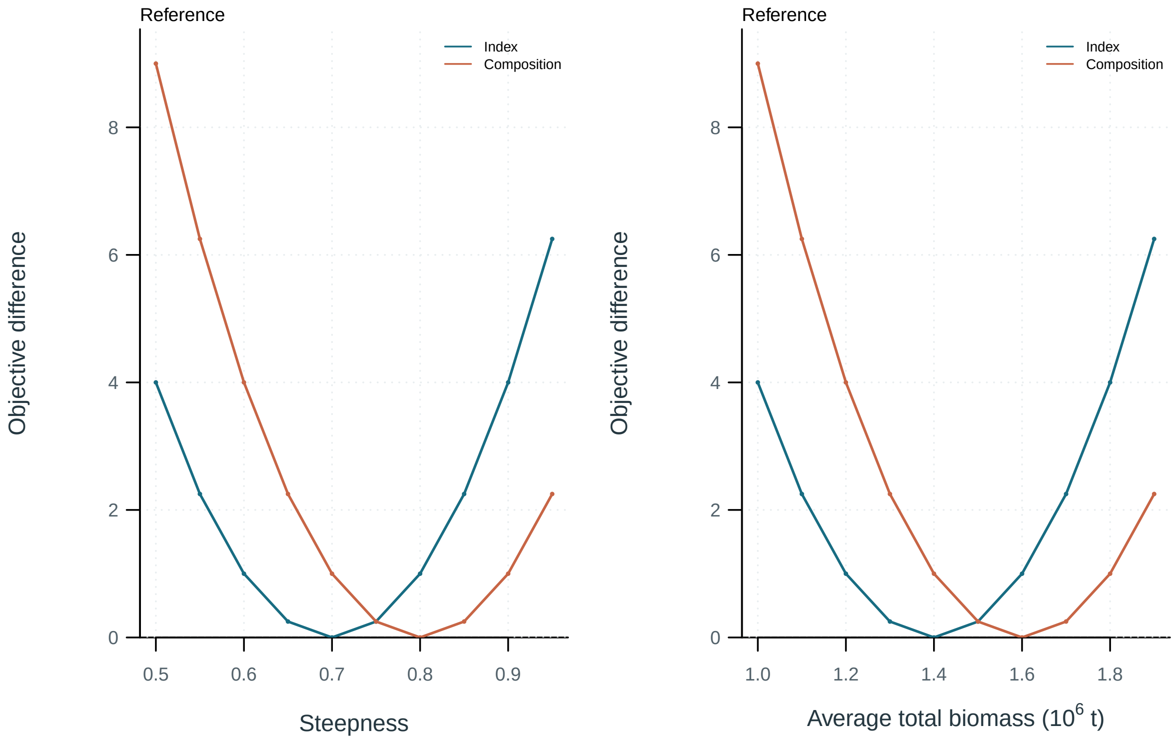

Figure 35: Objective profiles on the supplied parameter or quantity scale. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

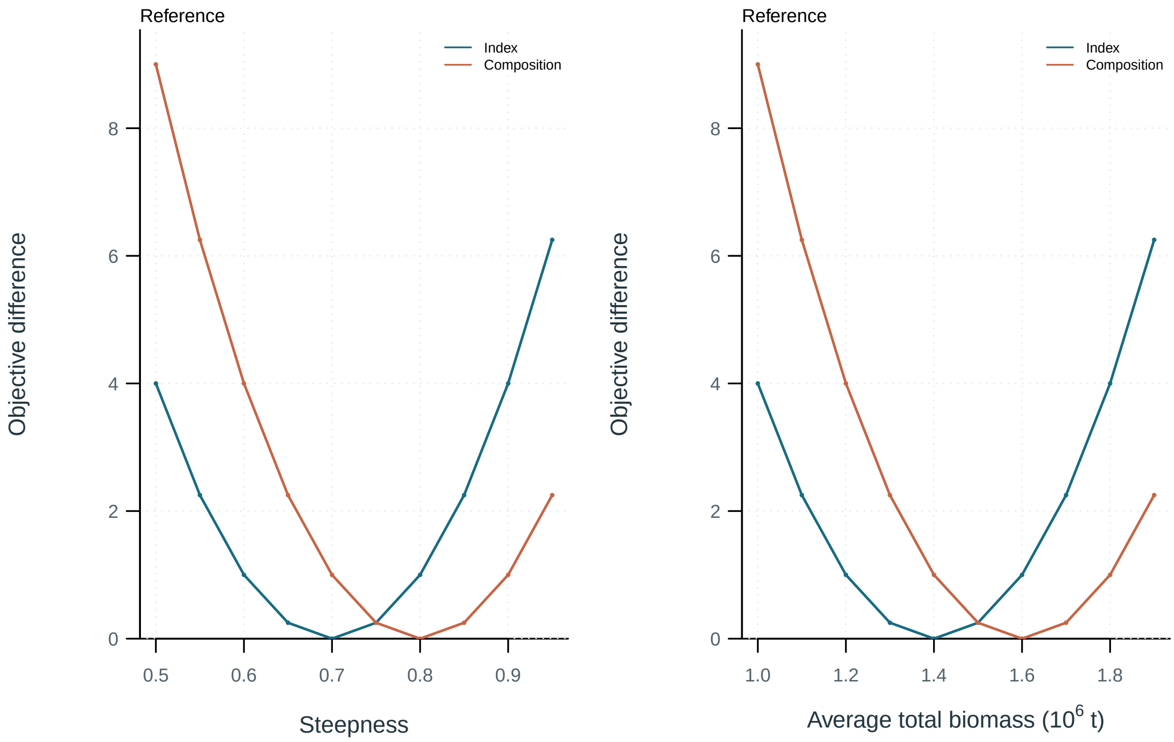

Figure 36: Objective profiles on the supplied parameter or quantity scale. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

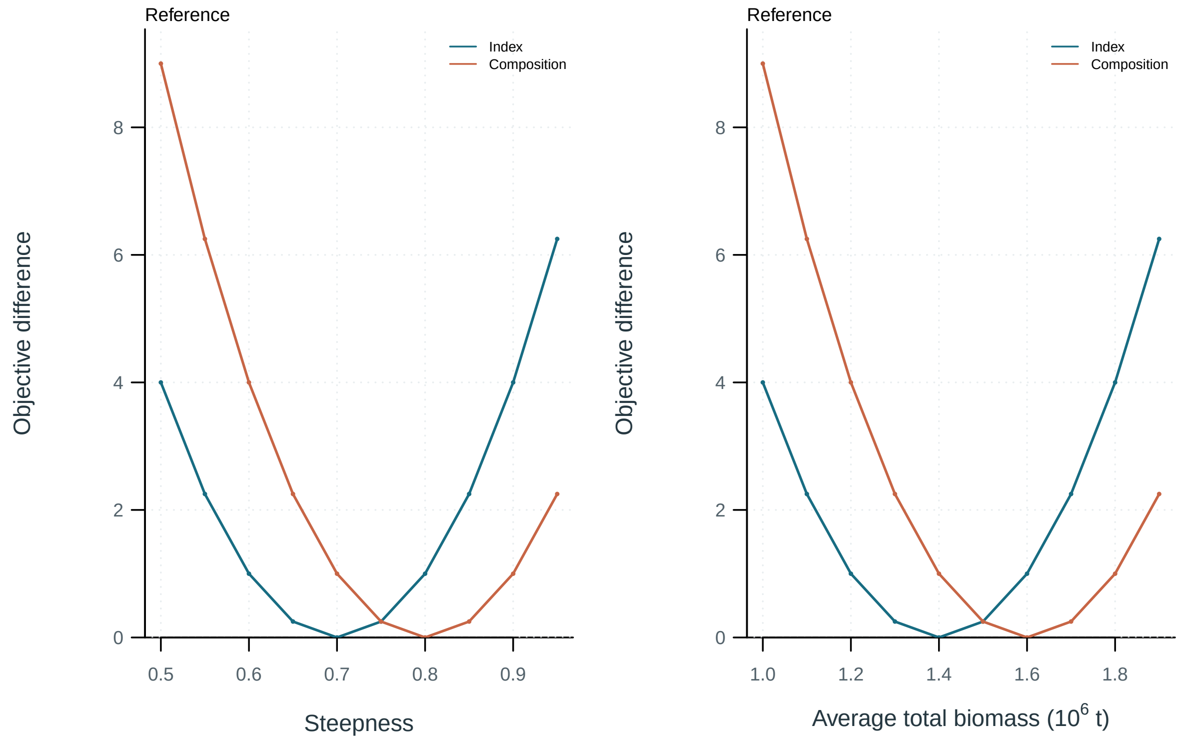

Figure 37: Objective profiles on the supplied parameter or quantity scale. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

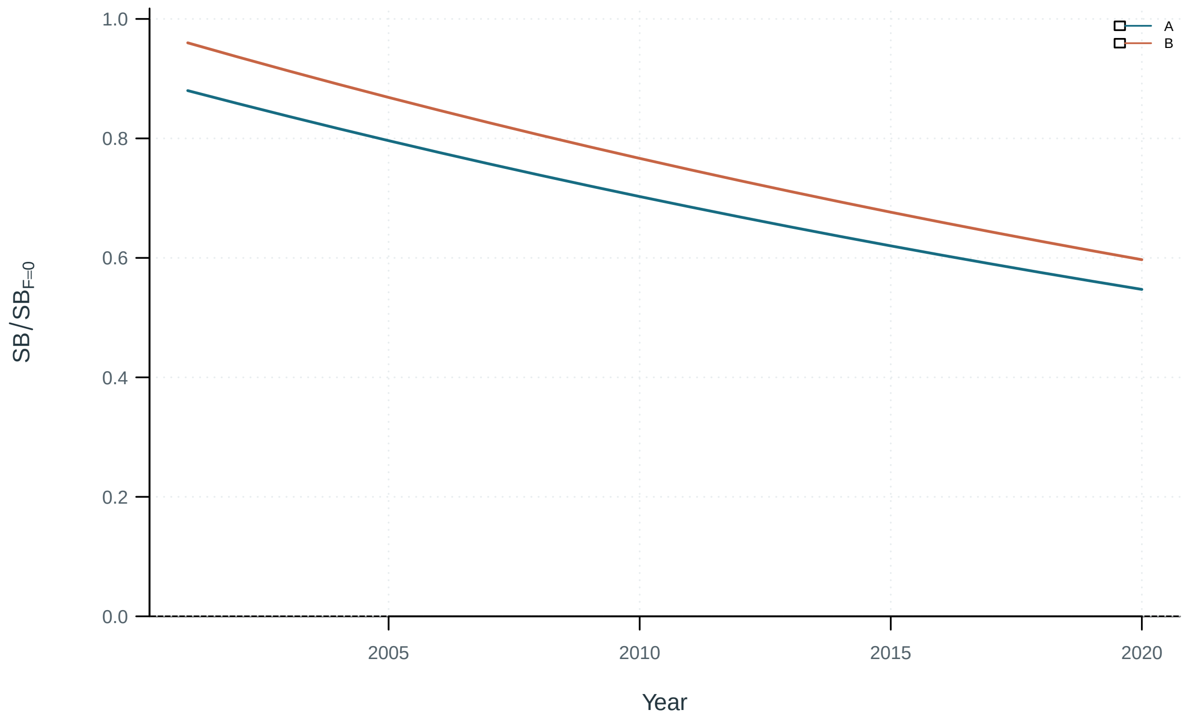

Figure 38: Illustrative output. Diagnostic and production-model trajectories.

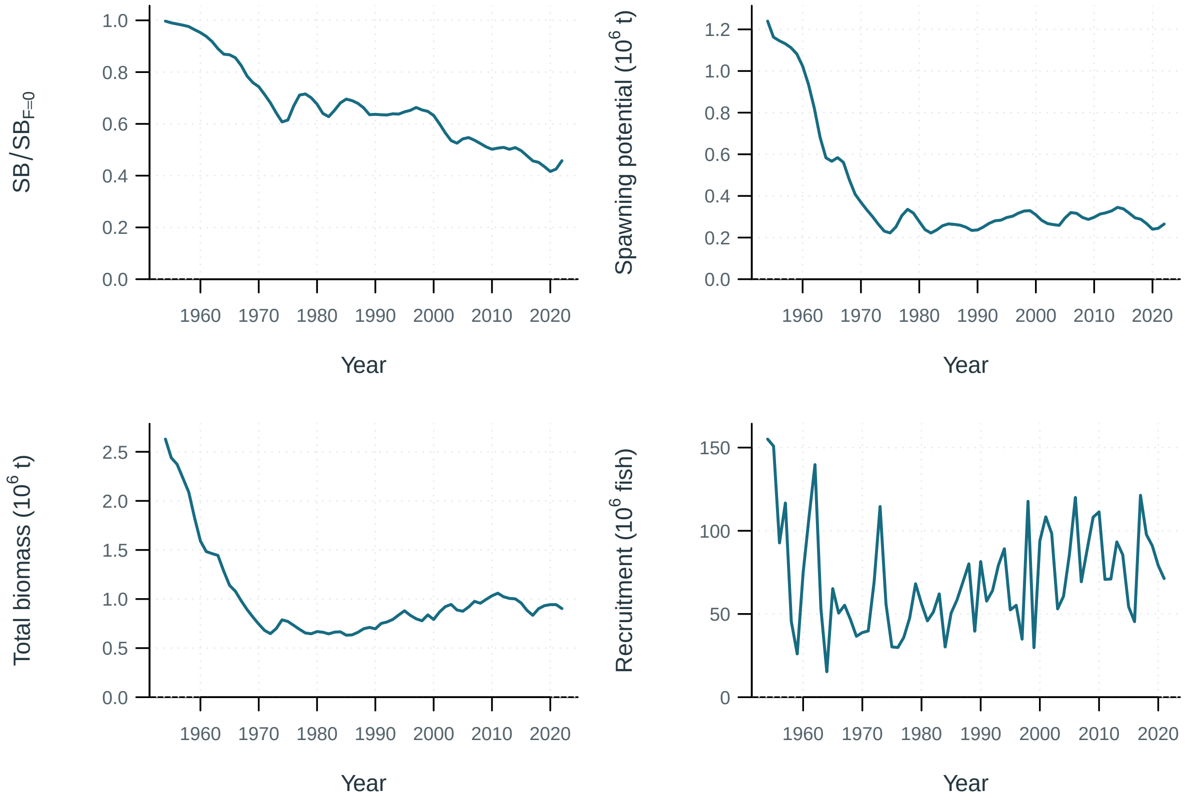

Figure 39: Estimated population trajectories.

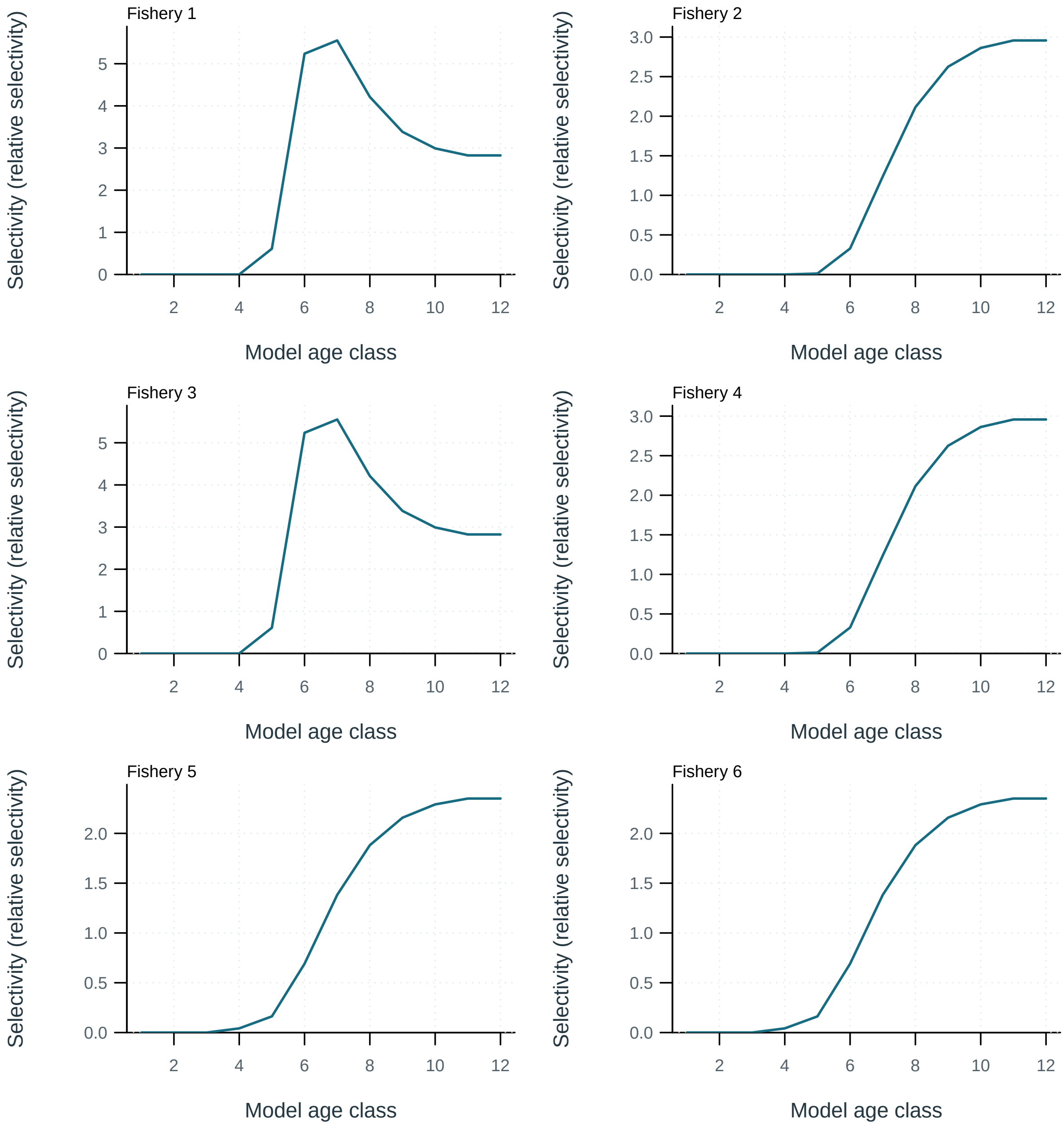

Figure 40: Fishery selectivity at age.

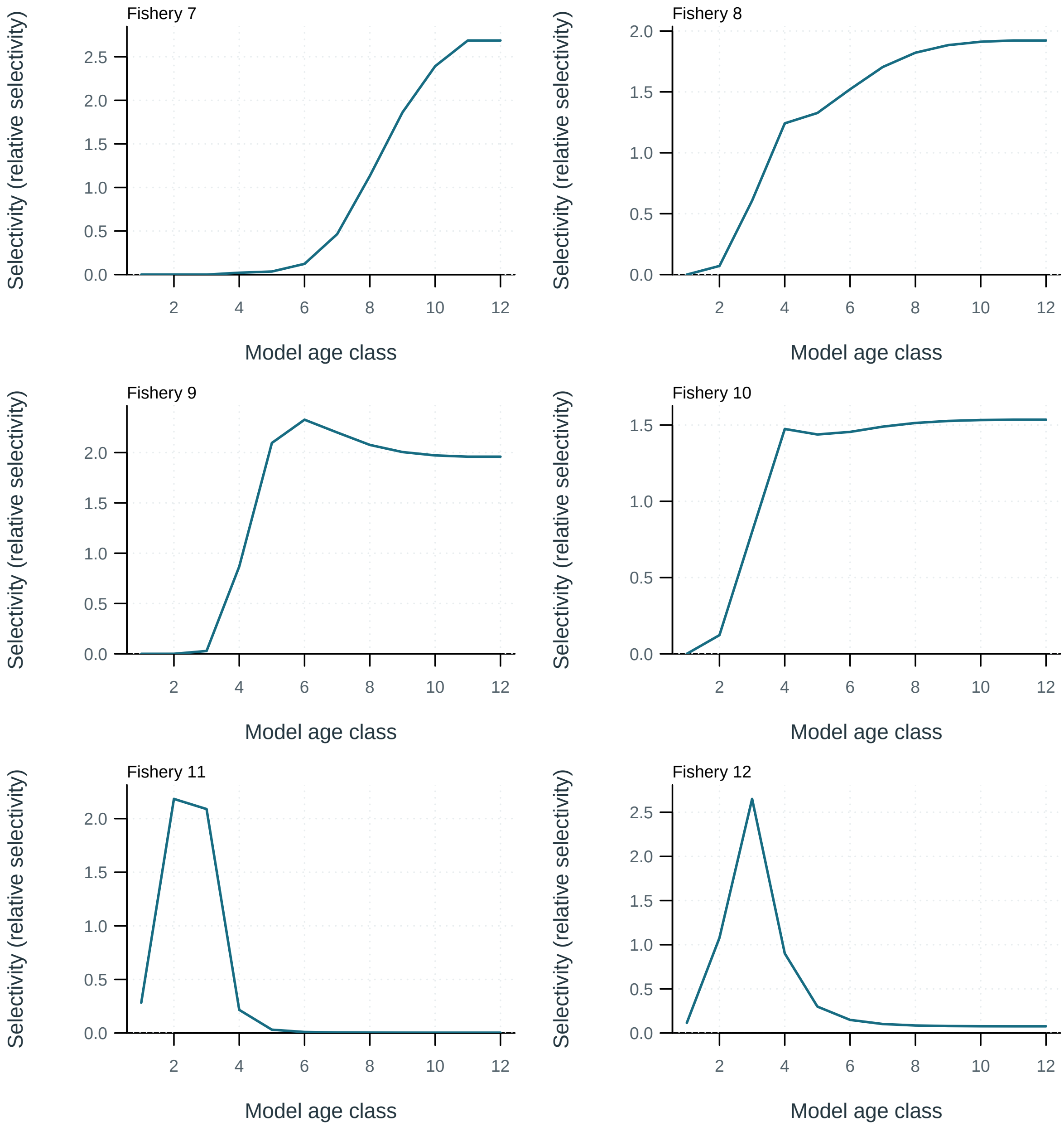

Figure 41: Fishery selectivity at age.

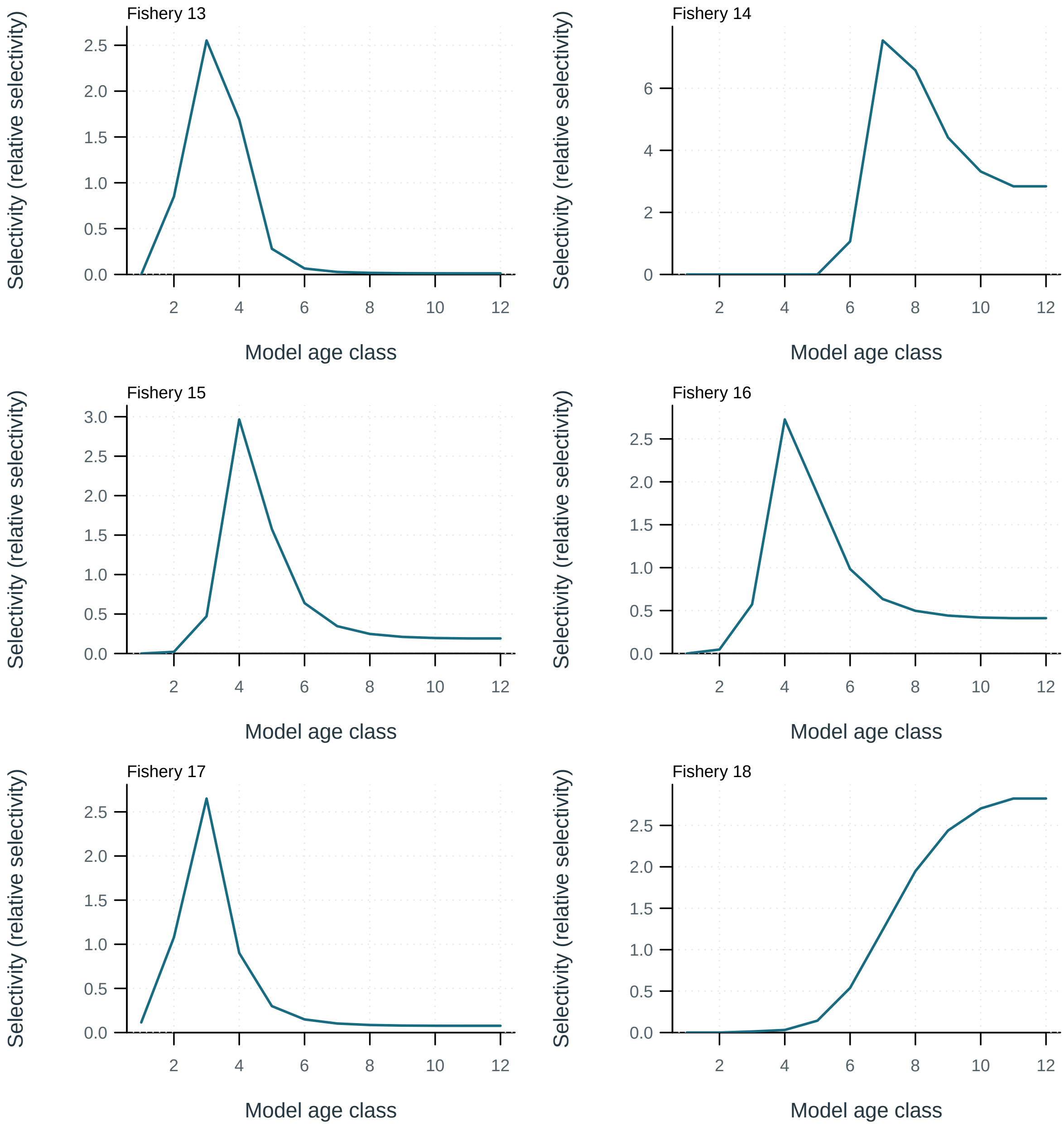

Figure 42: Fishery selectivity at age.

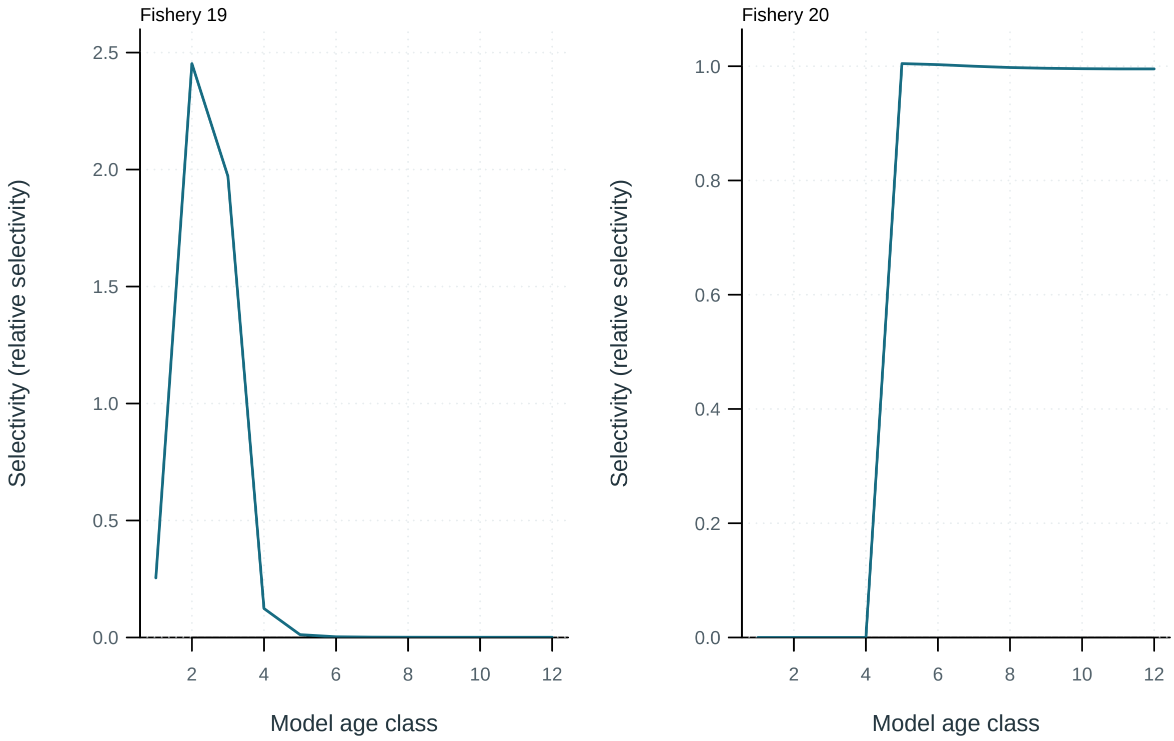

Figure 43: Fishery selectivity at age.

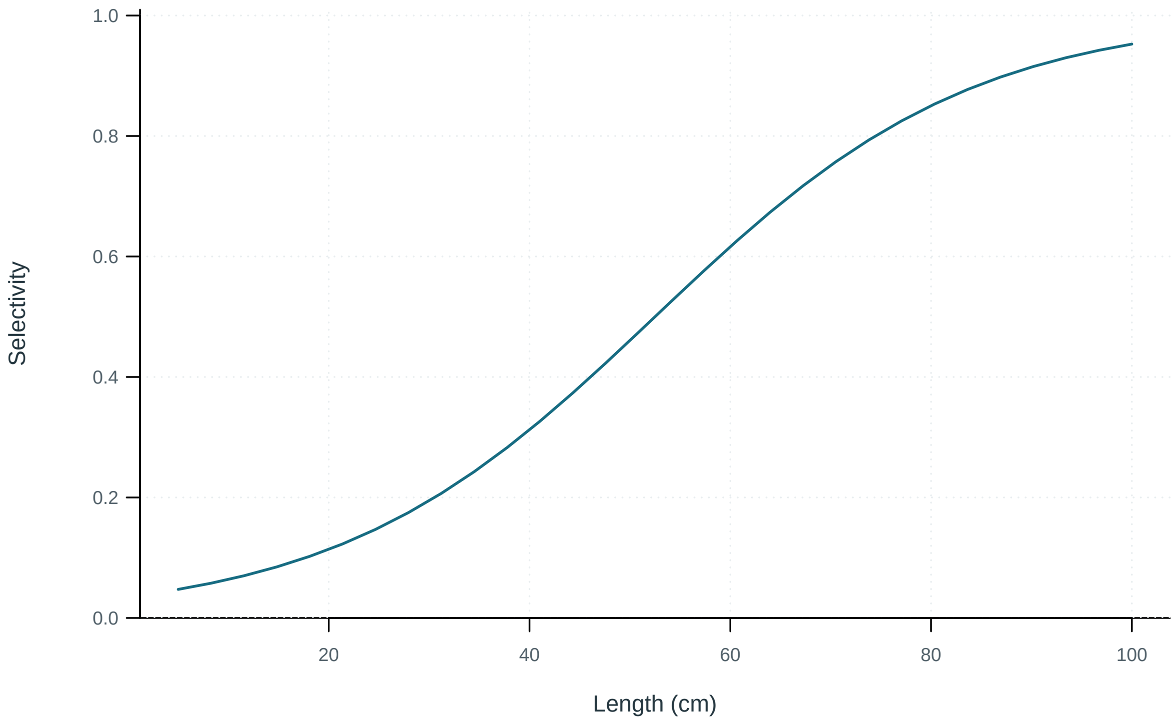

Figure 44: Illustrative output. Selectivity at length by fishery.

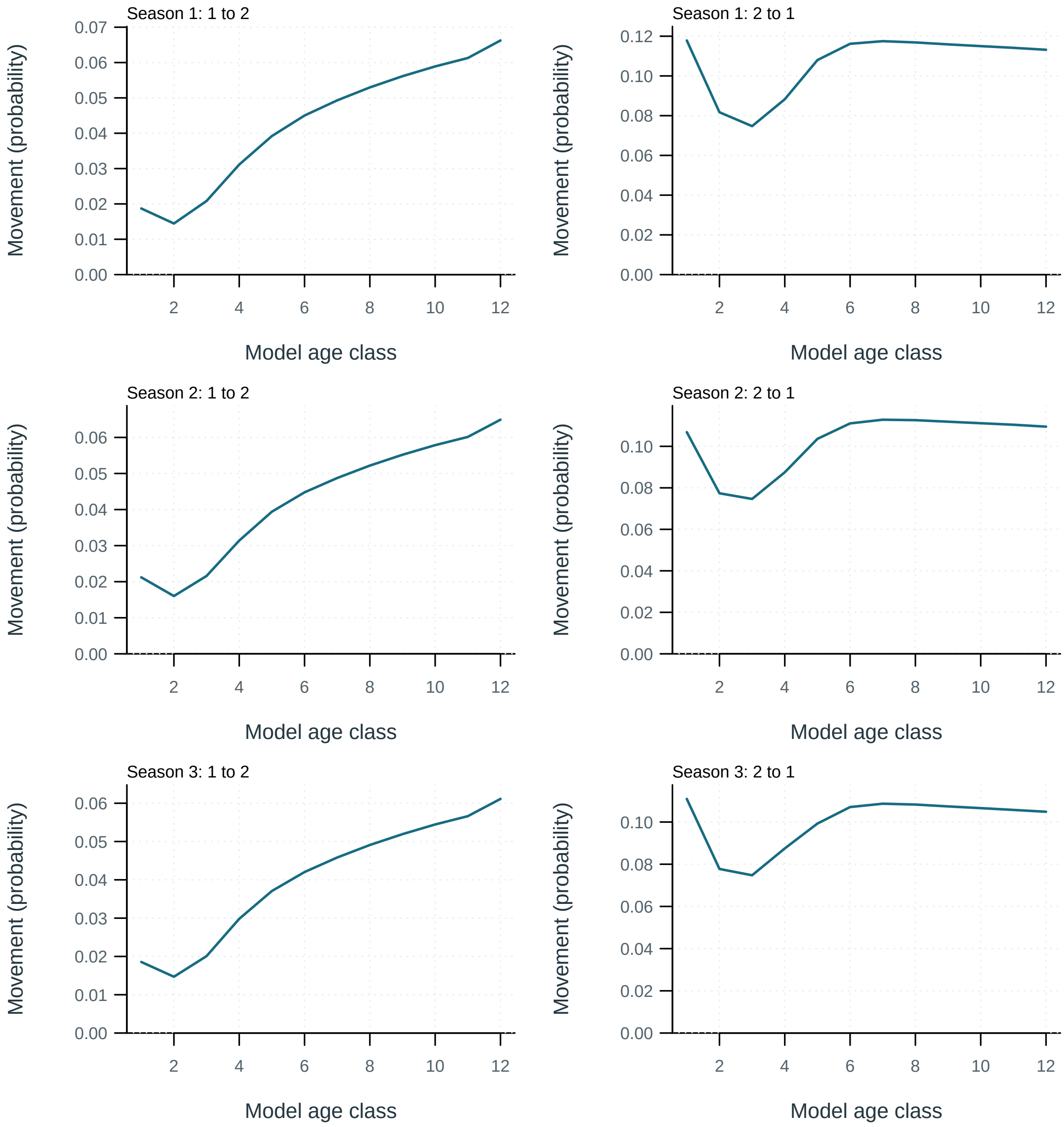

Figure 45: Probability of movement from source to destination region per movement event.

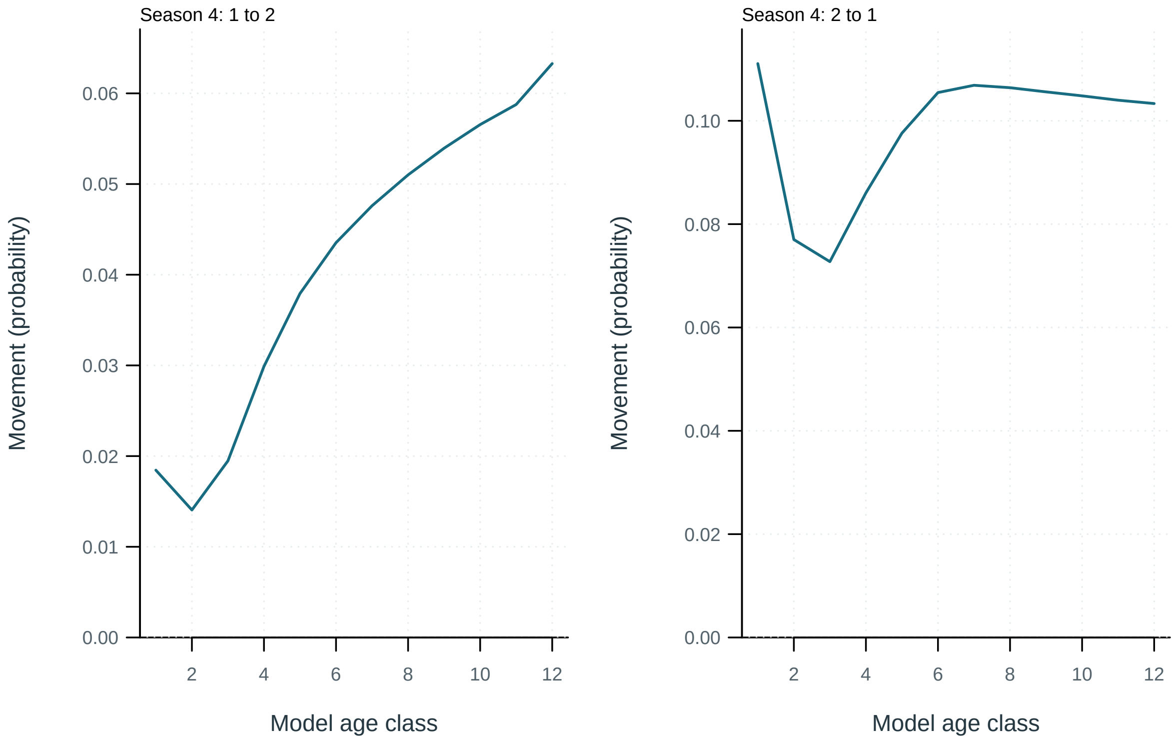

Figure 46: Probability of movement from source to destination region per movement event.

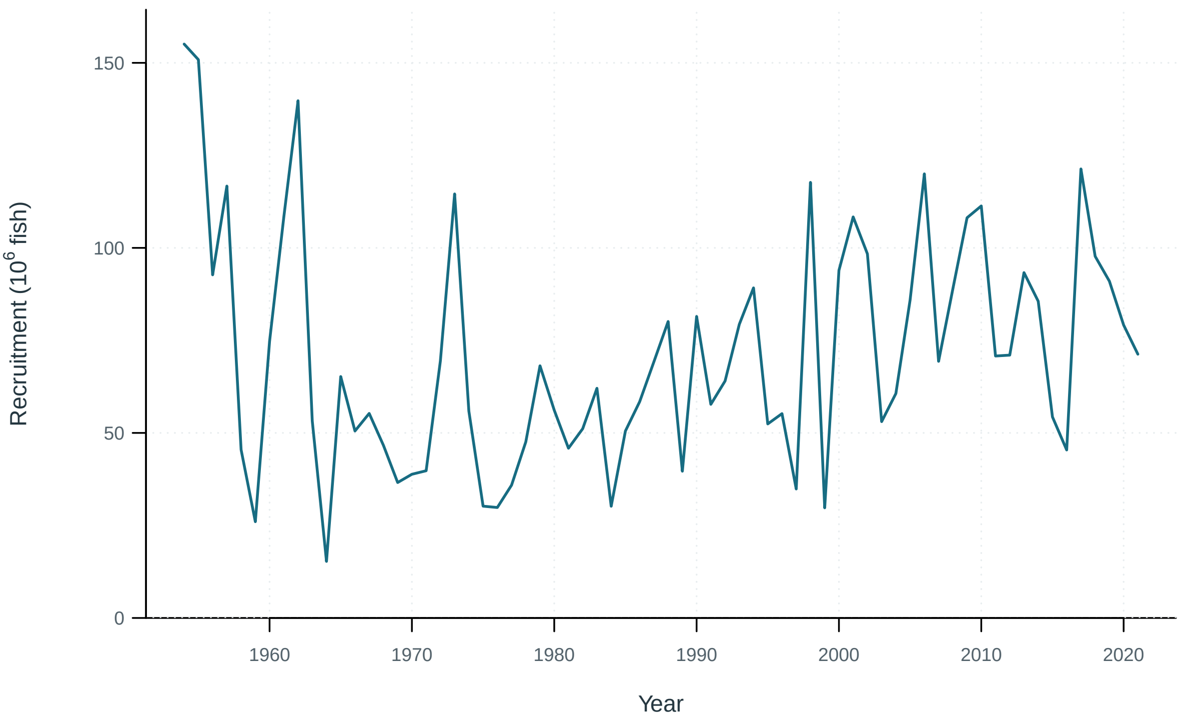

Figure 47: Recruitment by model region.

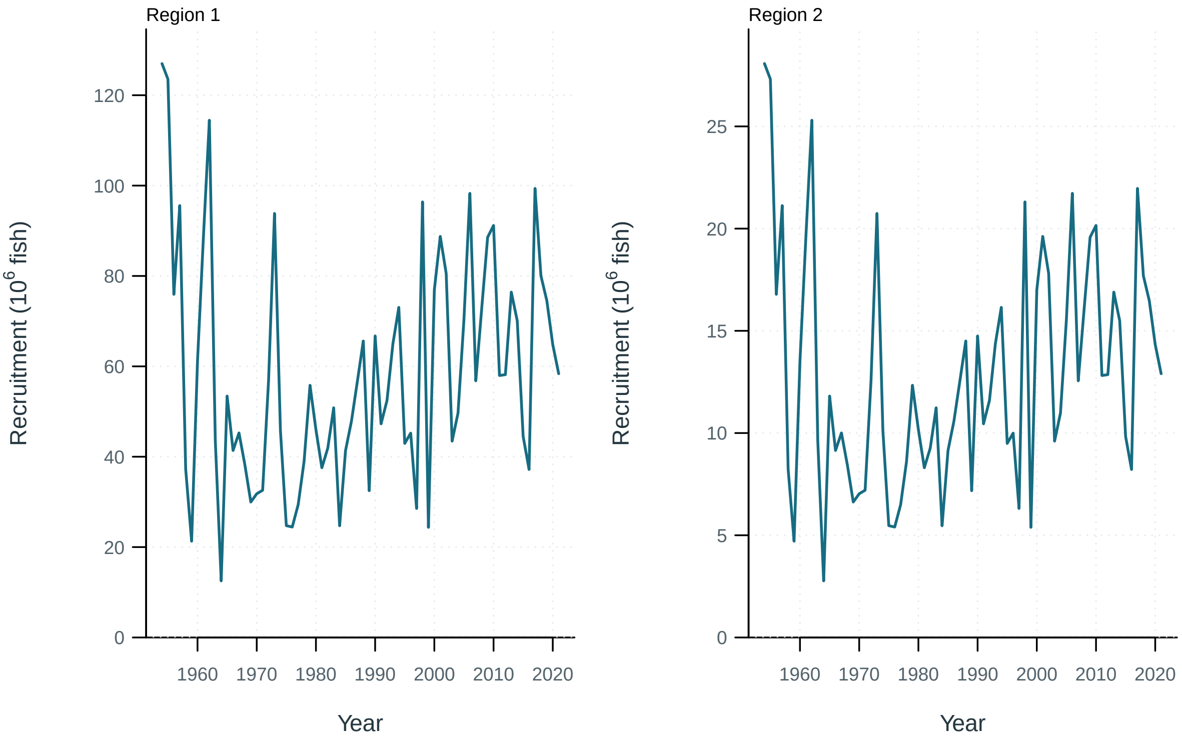

Figure 48: Recruitment by model region.

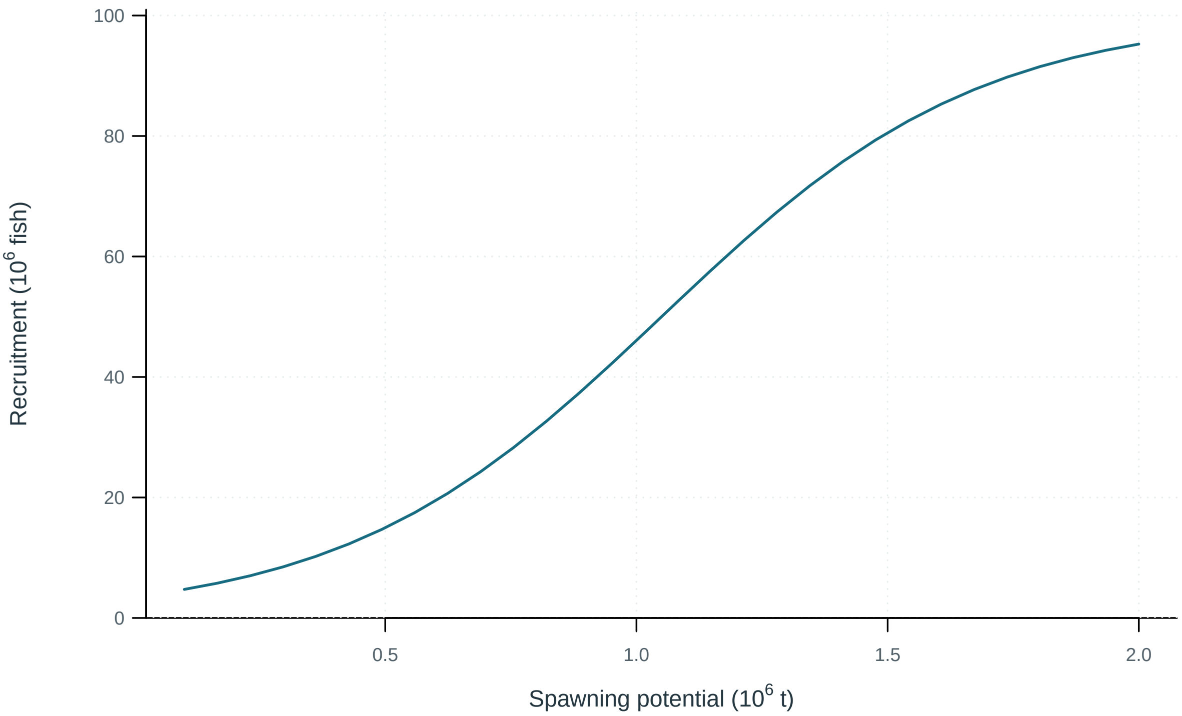

Figure 49: Illustrative output. Stock–recruitment observations and fitted relationship.

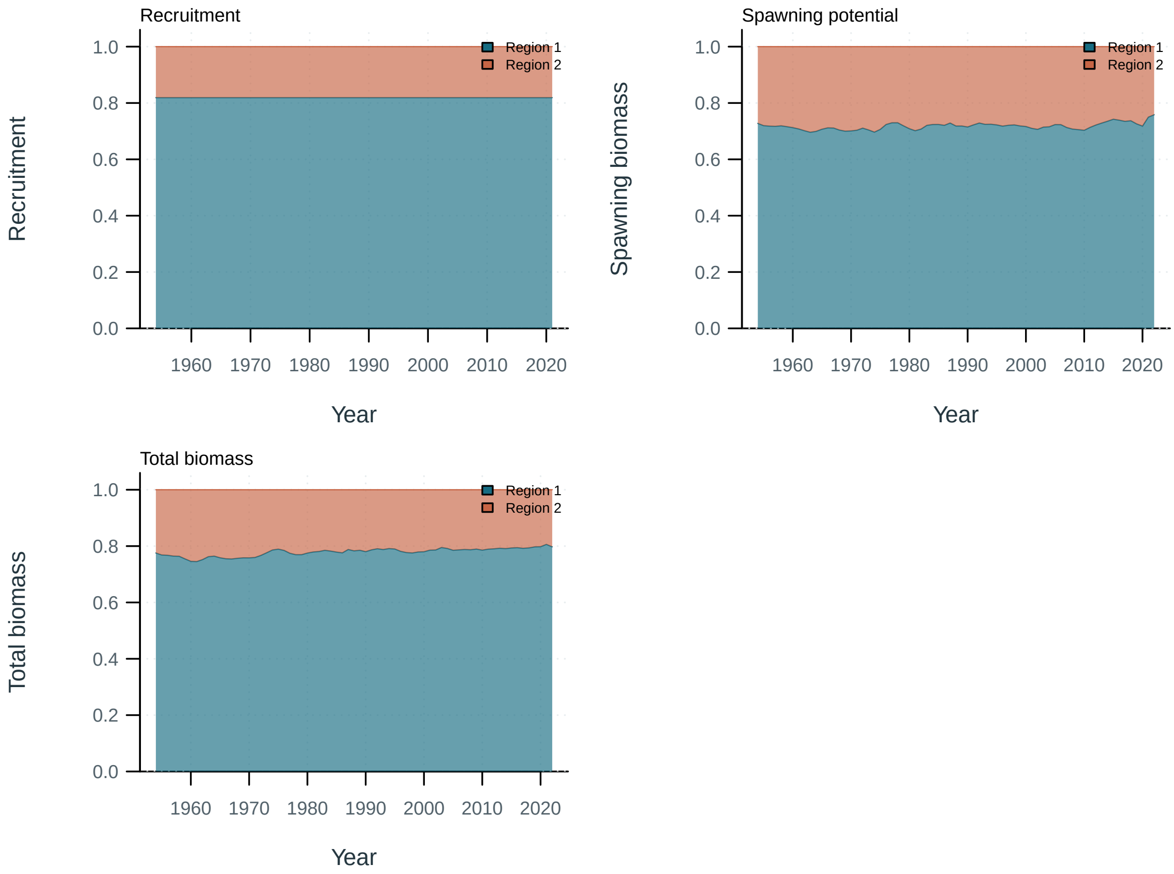

Figure 50: Regional shares of population quantities.

Figure 51: Total biomass under estimated fishing and the dynamic no-fishing counterfactual.

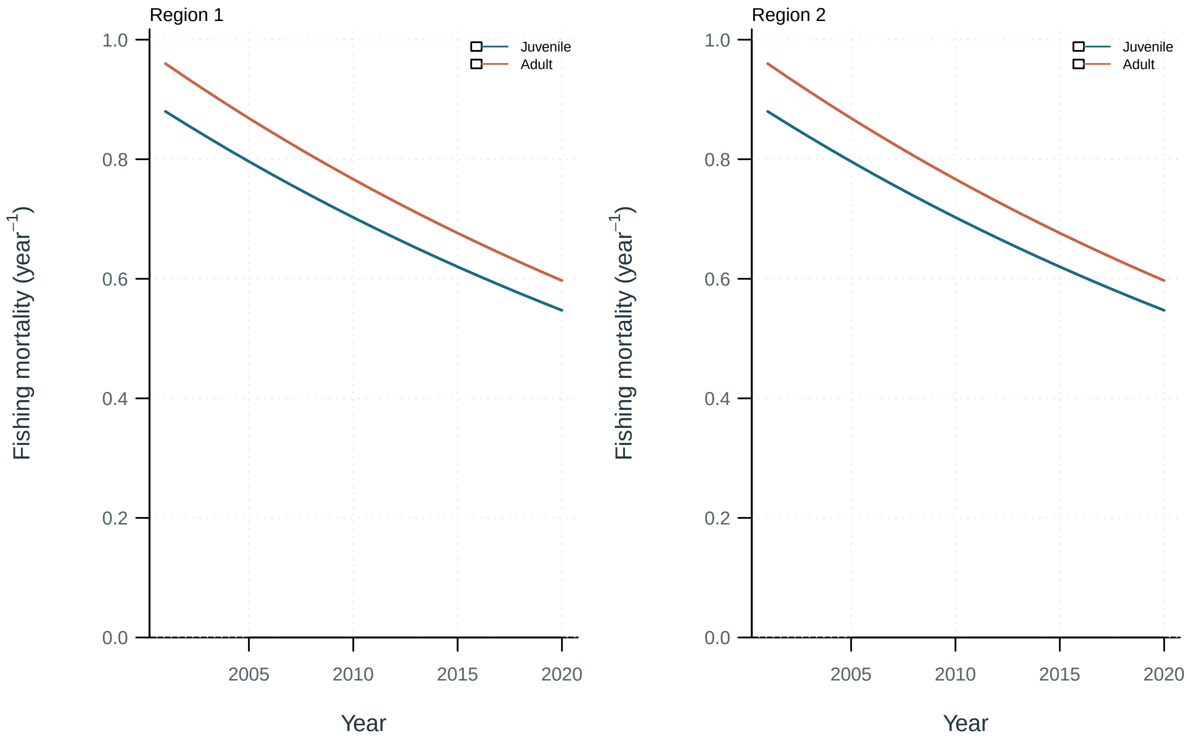

Figure 52: Illustrative output. Fishing mortality by life stage and region.

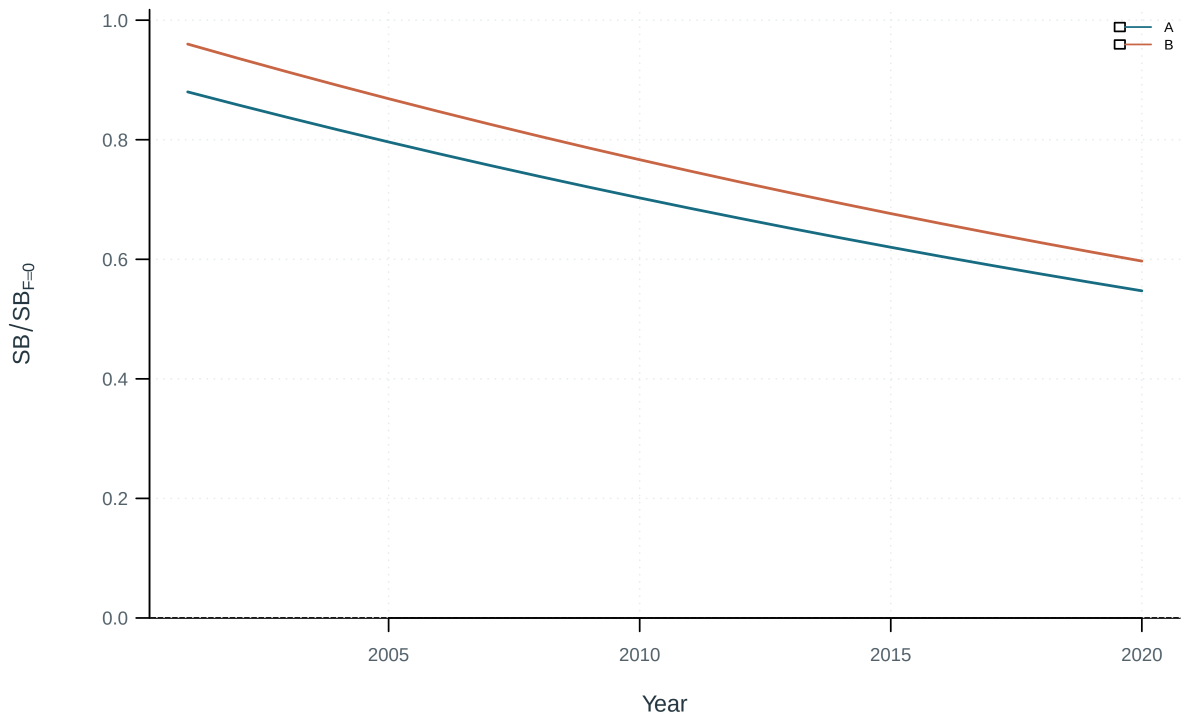

Figure 53: Population trajectories across the stated sensitivity models.

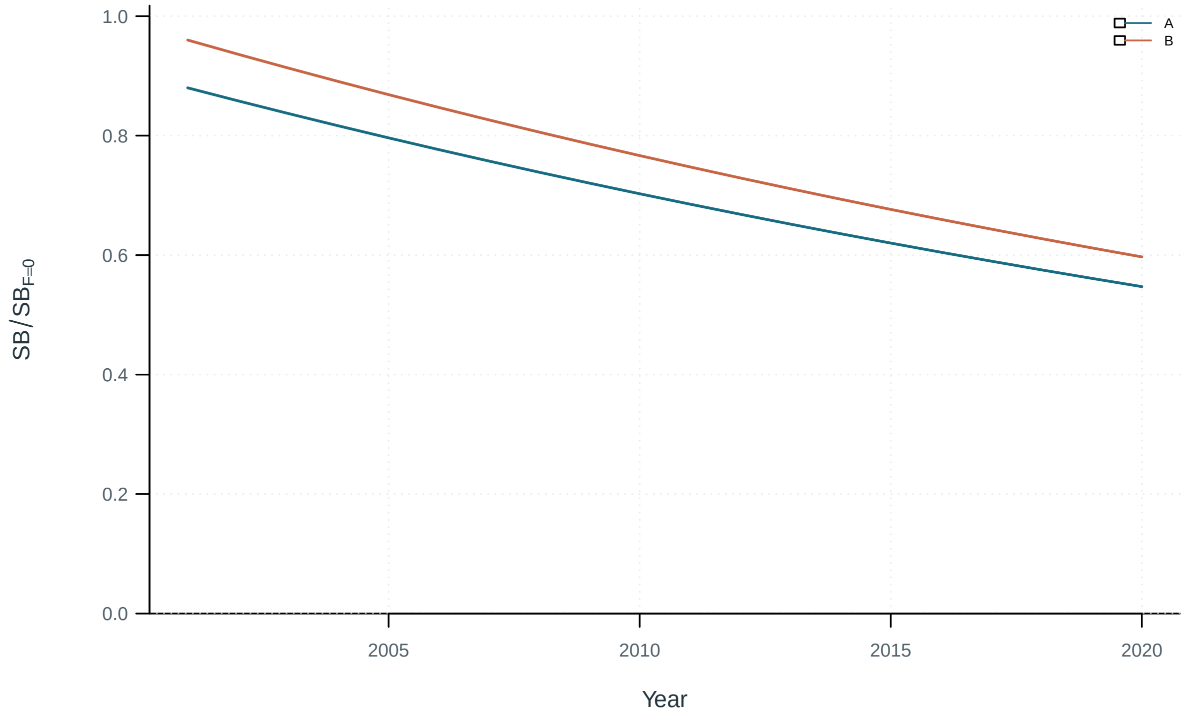

Figure 54: Population trajectories across the stated sensitivity models.

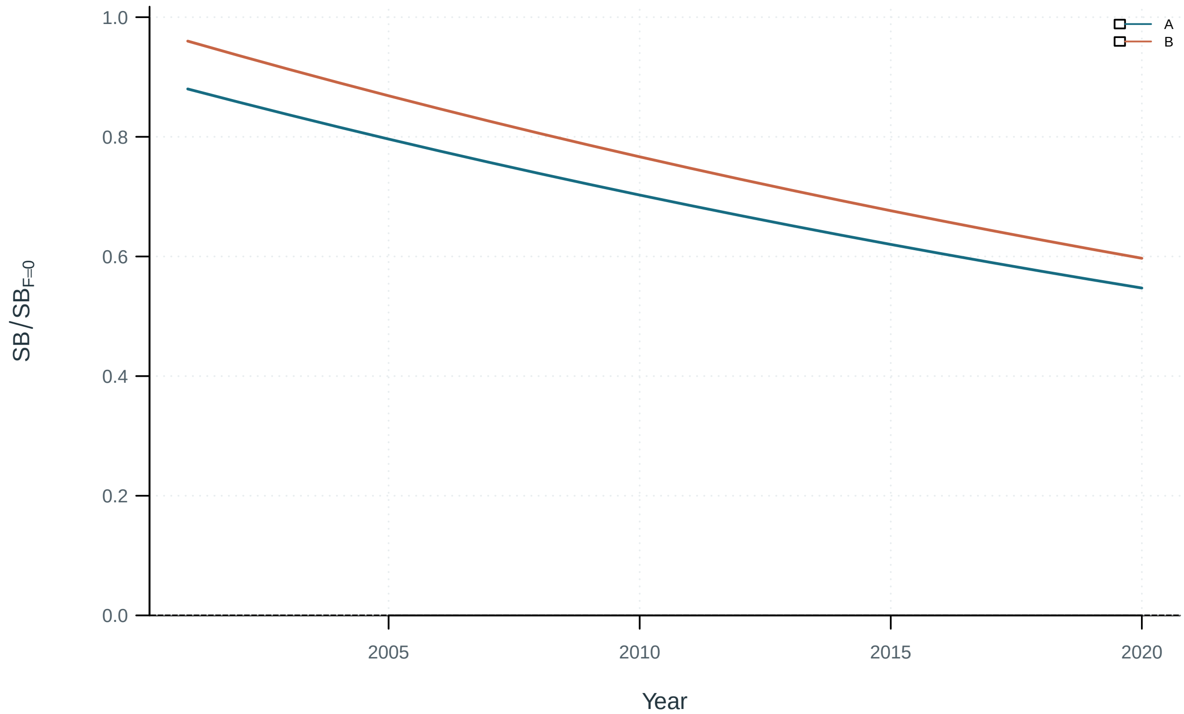

Figure 55: Population trajectories across the stated sensitivity models.

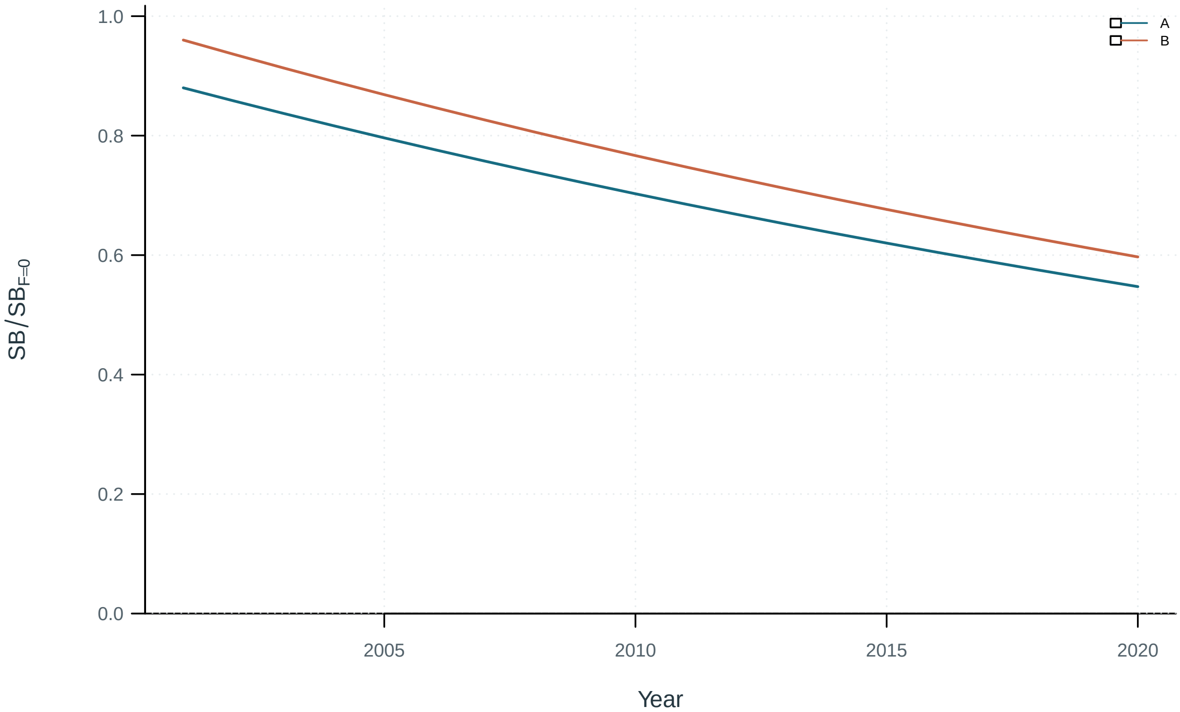

Figure 56: Population trajectories across the stated sensitivity models.

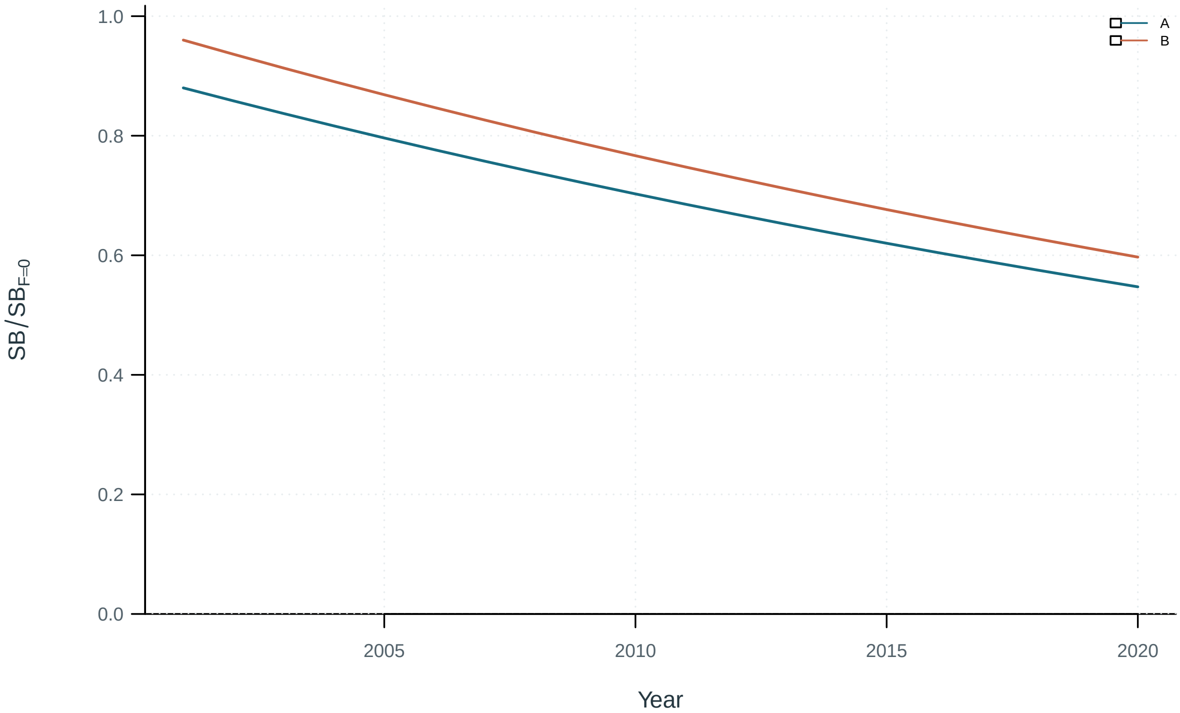

Figure 57: Population trajectories across the stated sensitivity models.

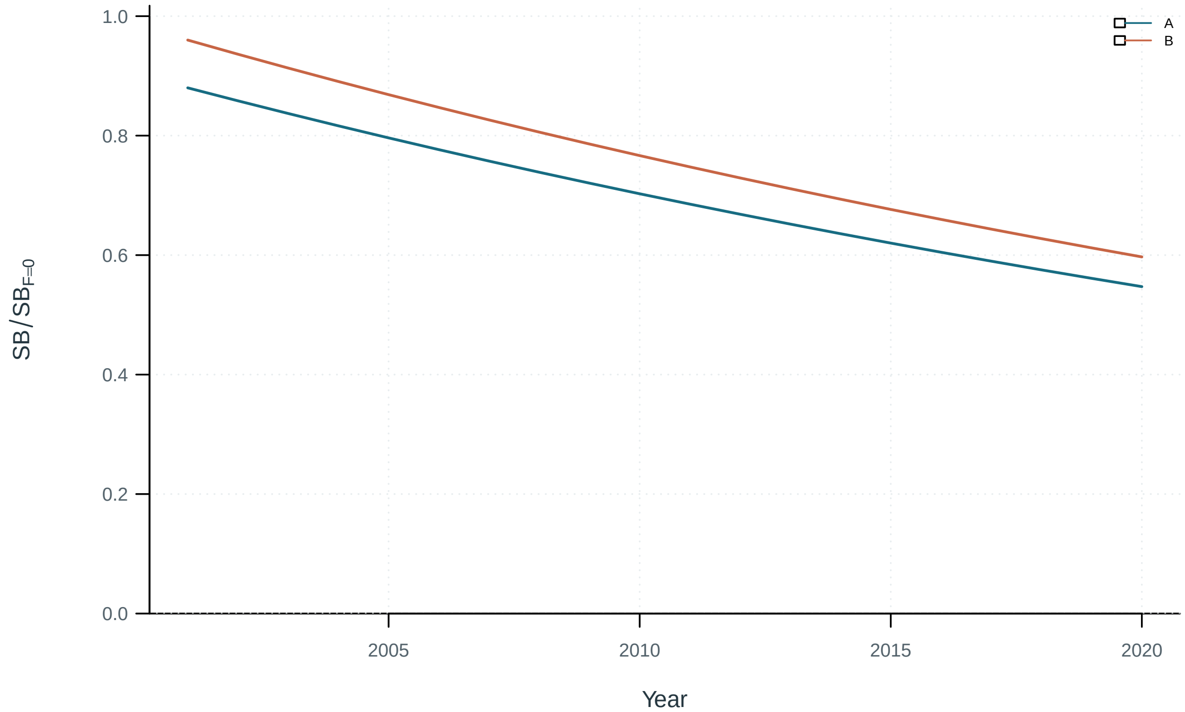

Figure 58: Population trajectories across the stated sensitivity models.

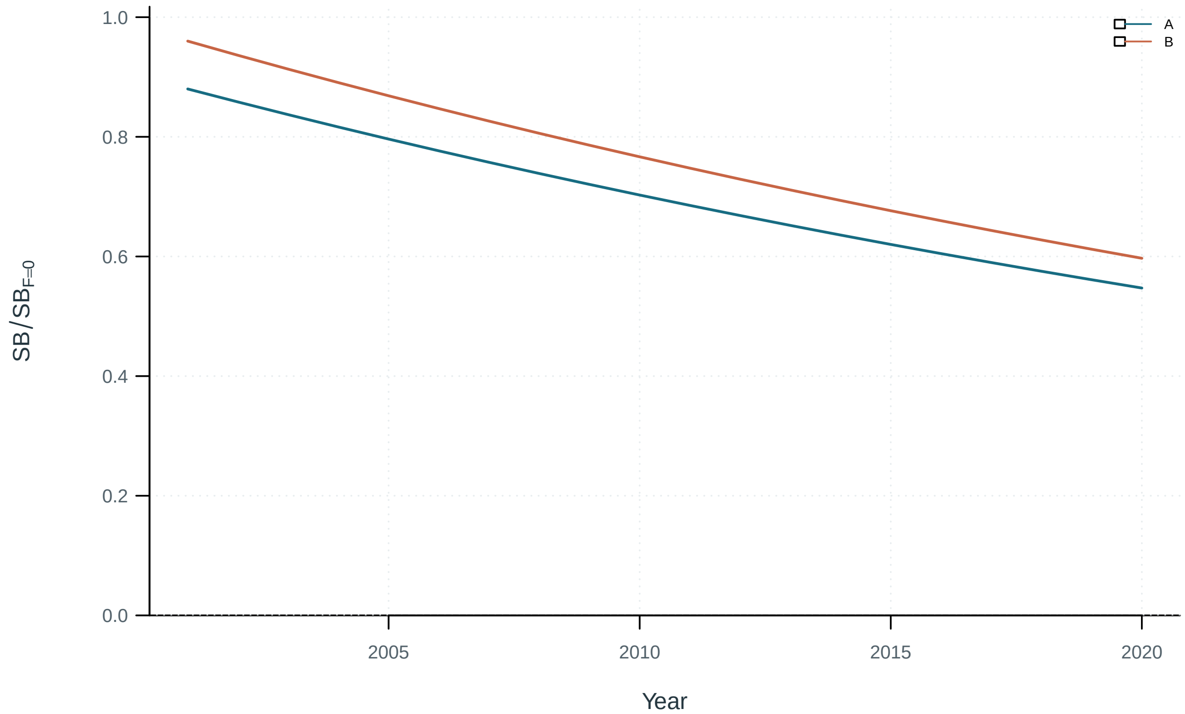

Figure 59: Population trajectories across the stated sensitivity models.

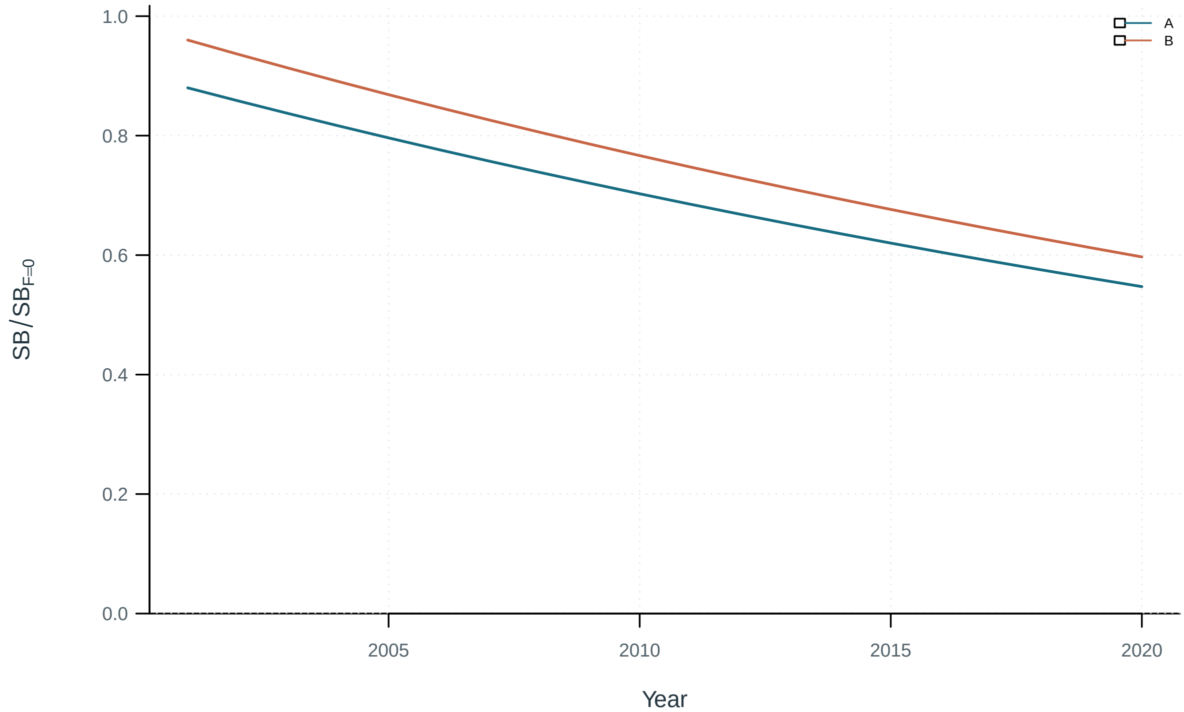

Figure 60: Population trajectories across the stated sensitivity models.

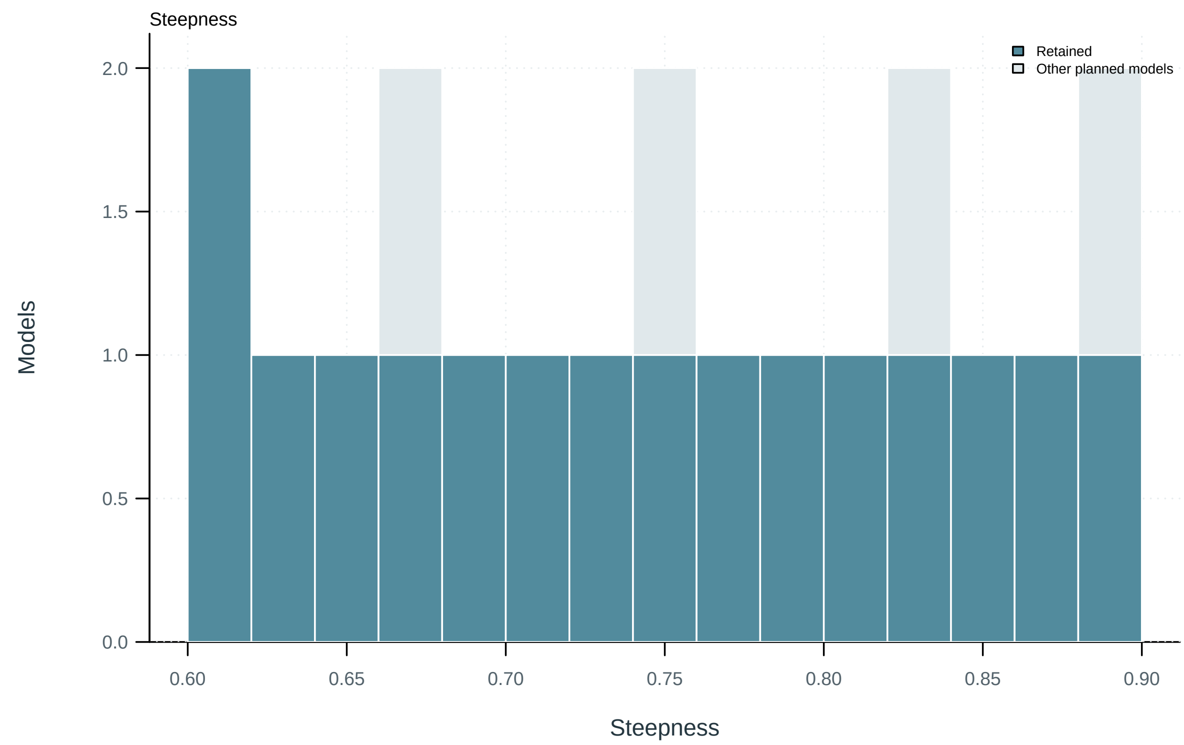

Figure 61: Planned and retained model inputs.

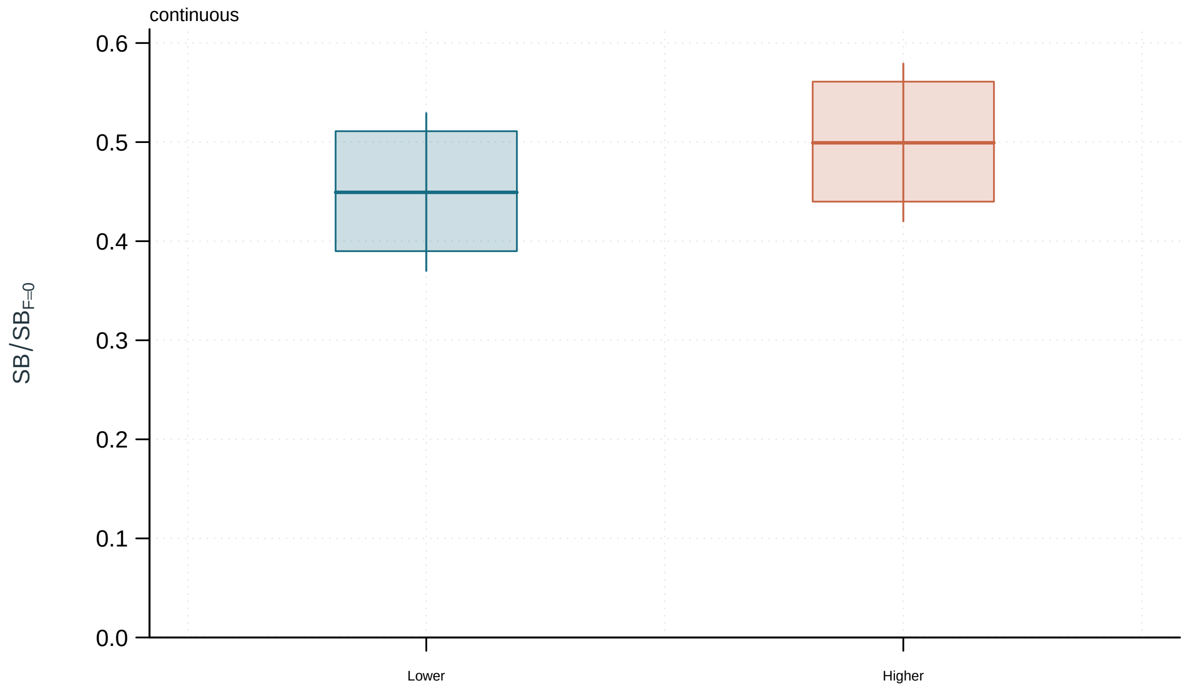

Figure 62: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

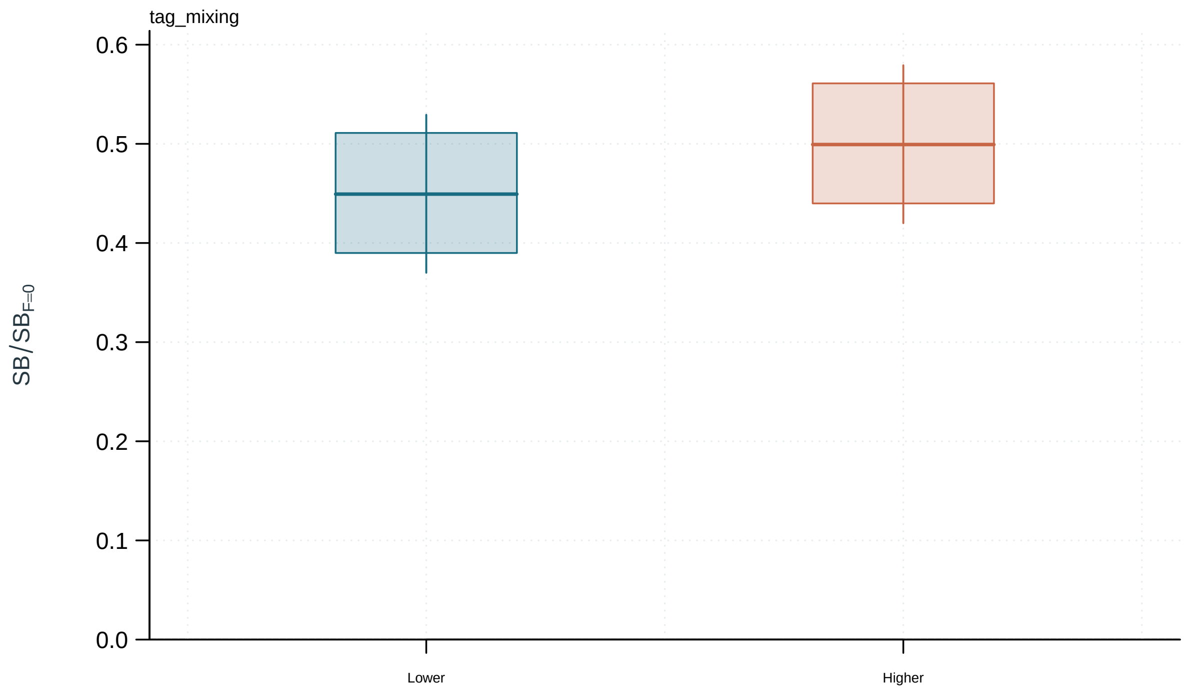

Figure 63: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

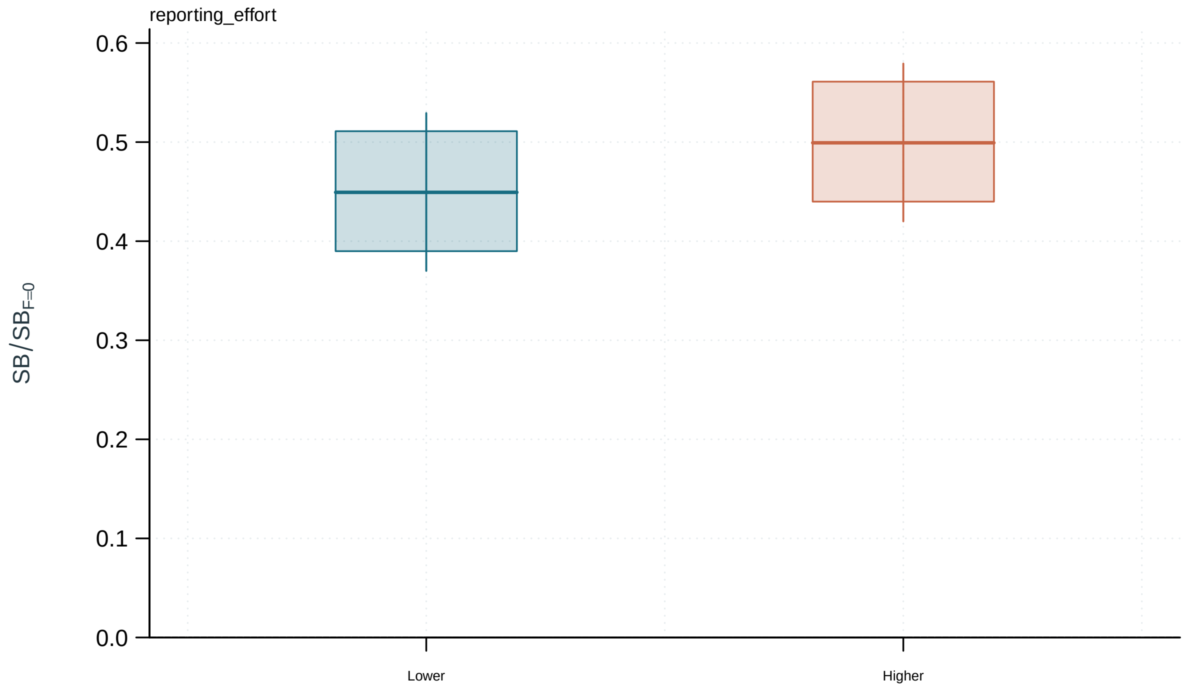

Figure 64: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

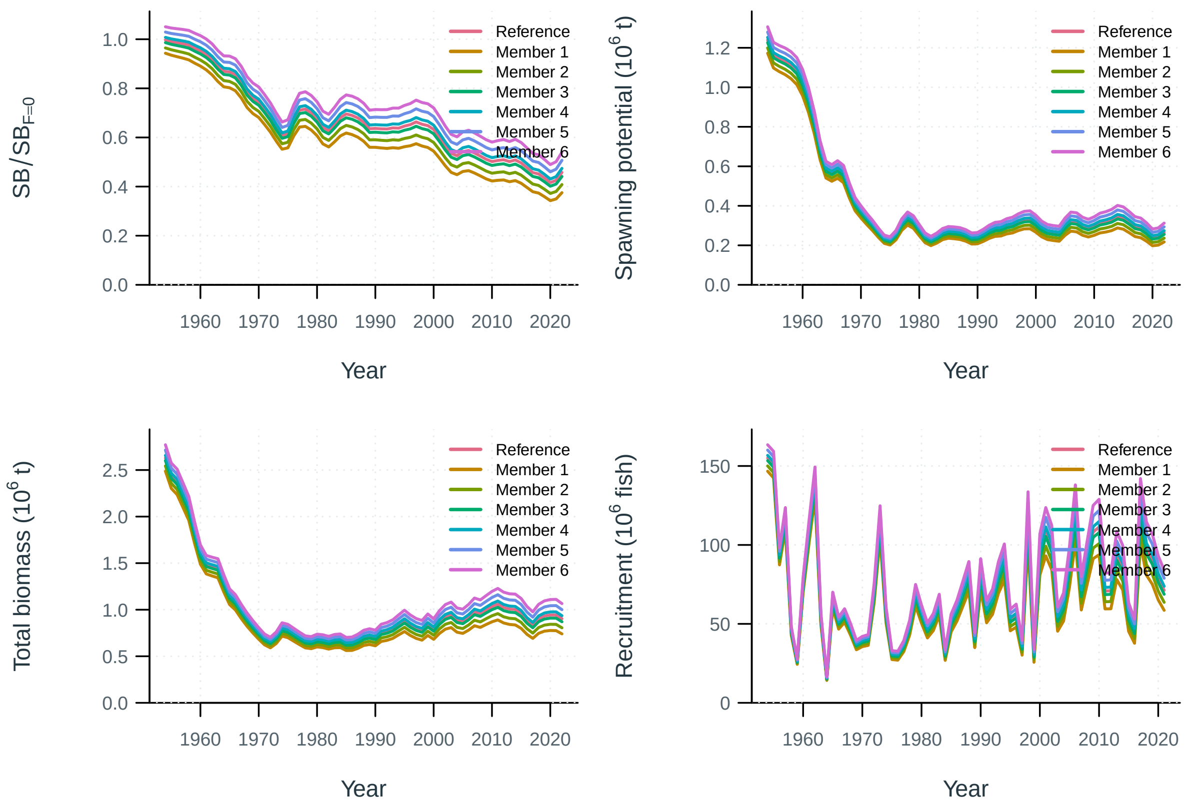

Figure 65: Ensemble model trajectories.

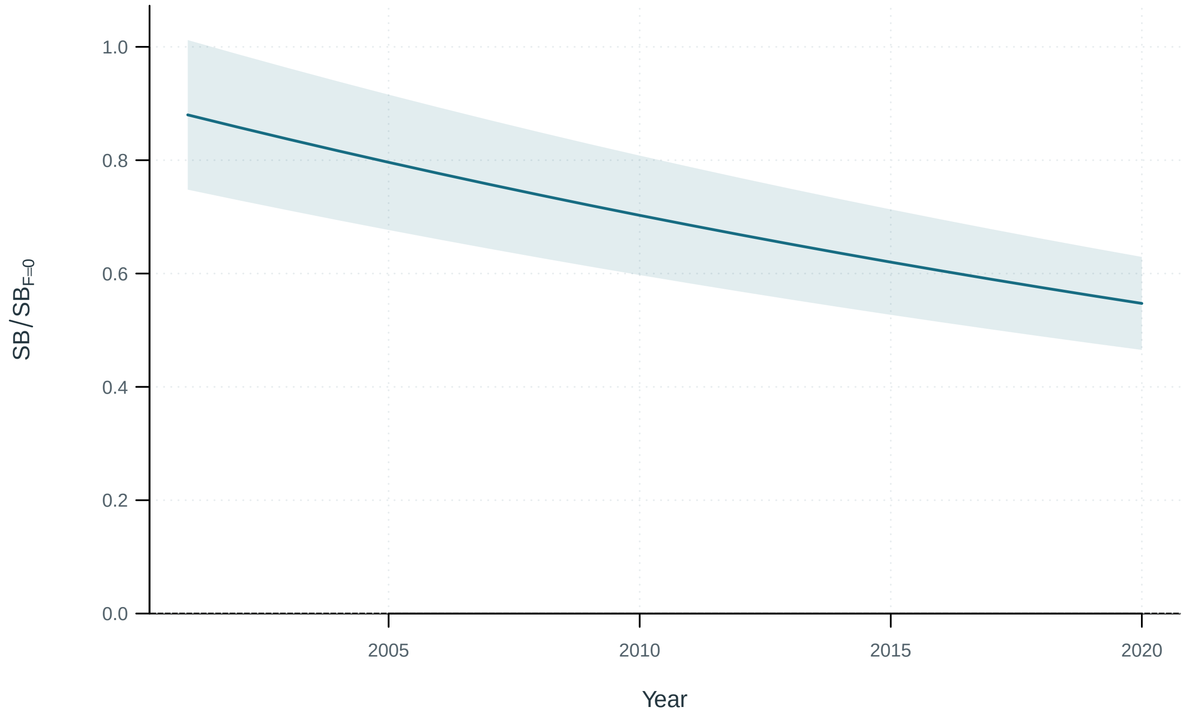

Figure 66: Population trajectories with supplied uncertainty intervals. Central 95% intervals (illustrative supplied quantiles).

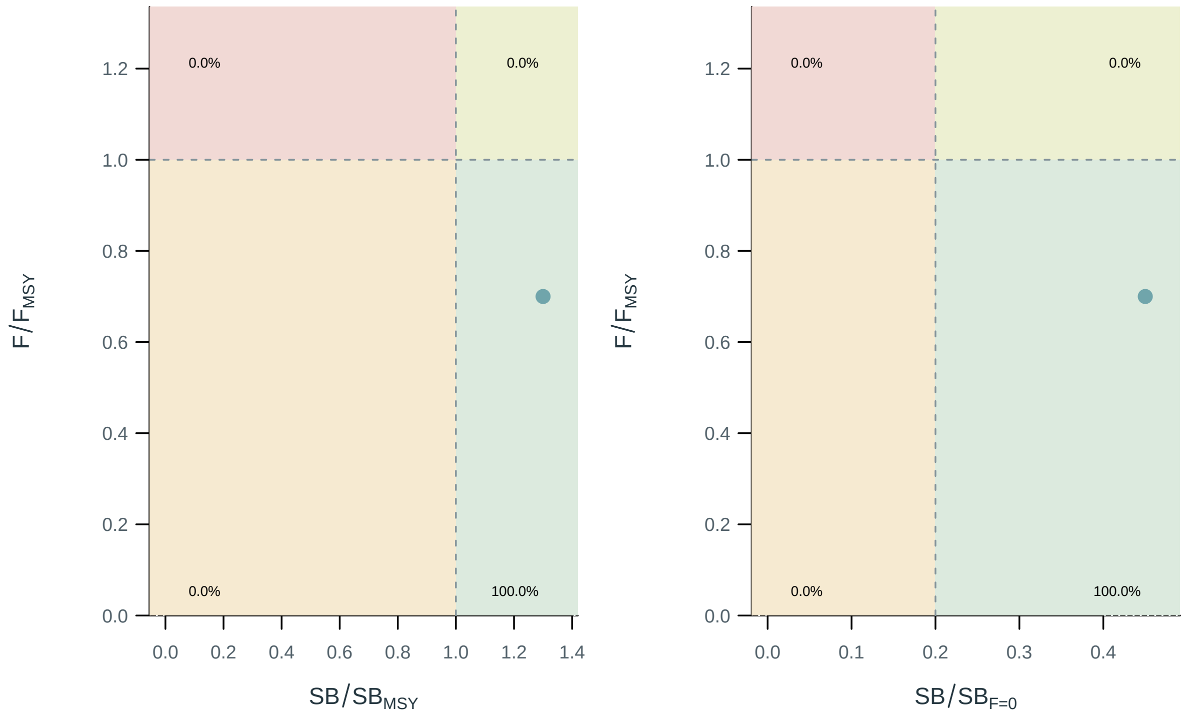

Figure 67: Stock status relative to the supplied spawning and fishing reference points.

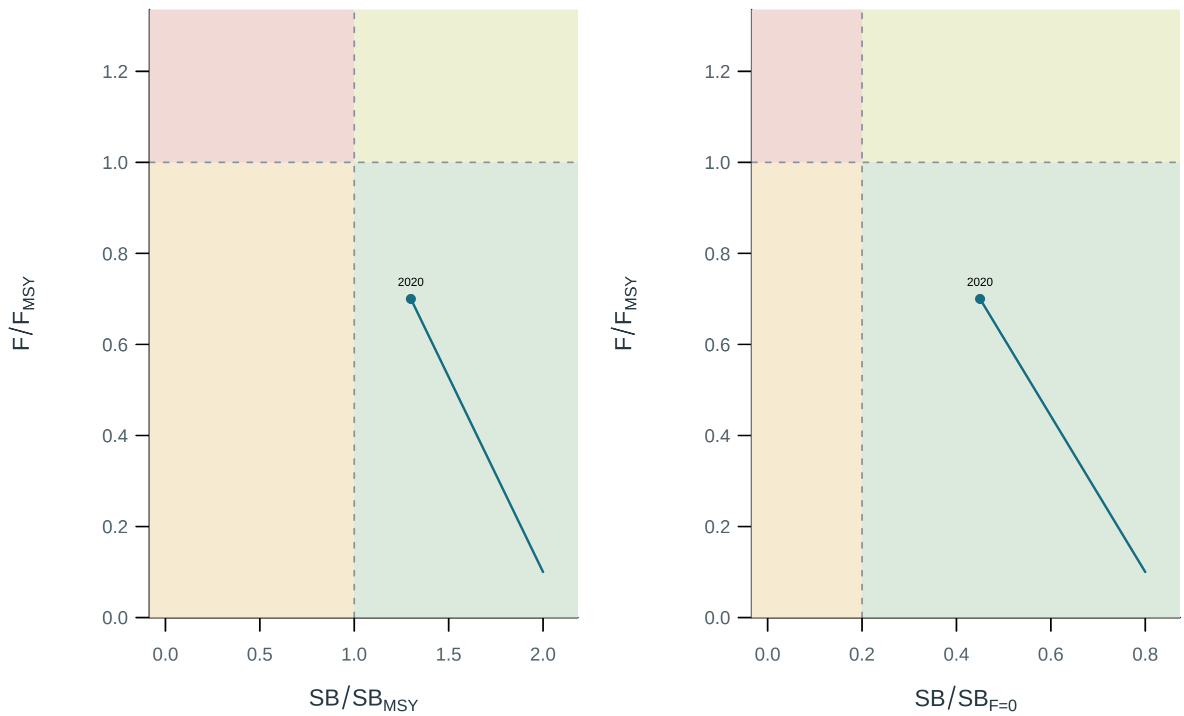

Figure 68: Stock status relative to the supplied spawning and fishing reference points.

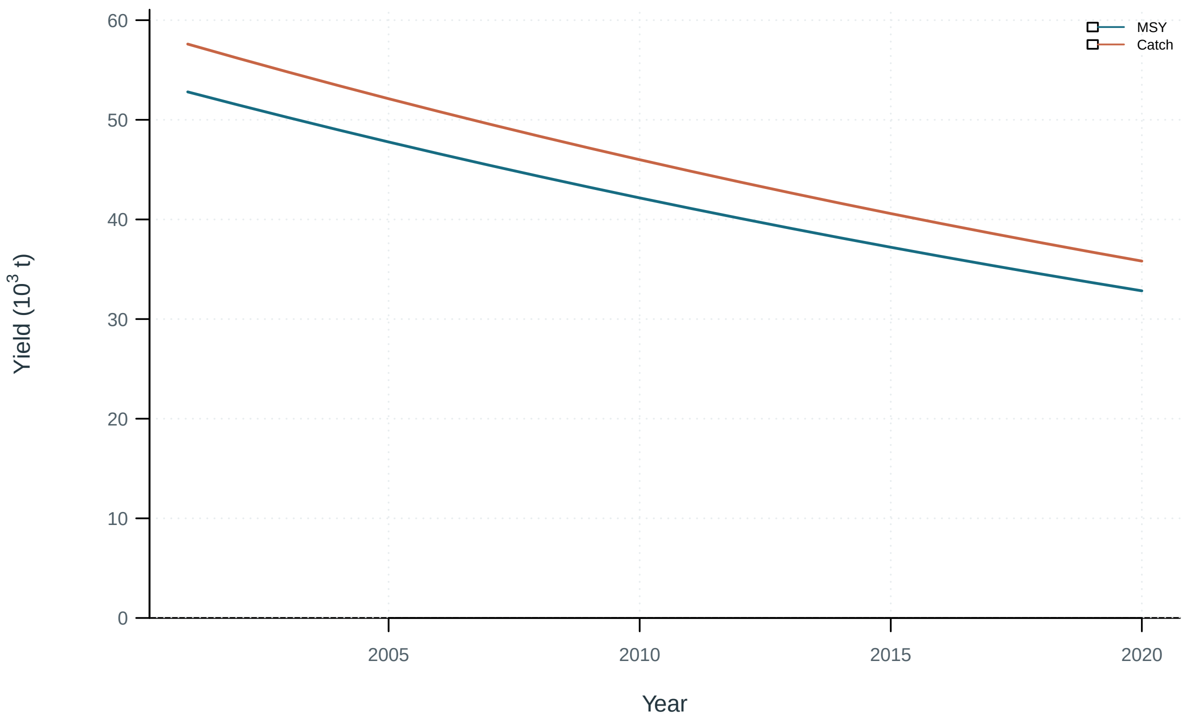

Figure 69: Dynamic maximum sustainable yield and catch.

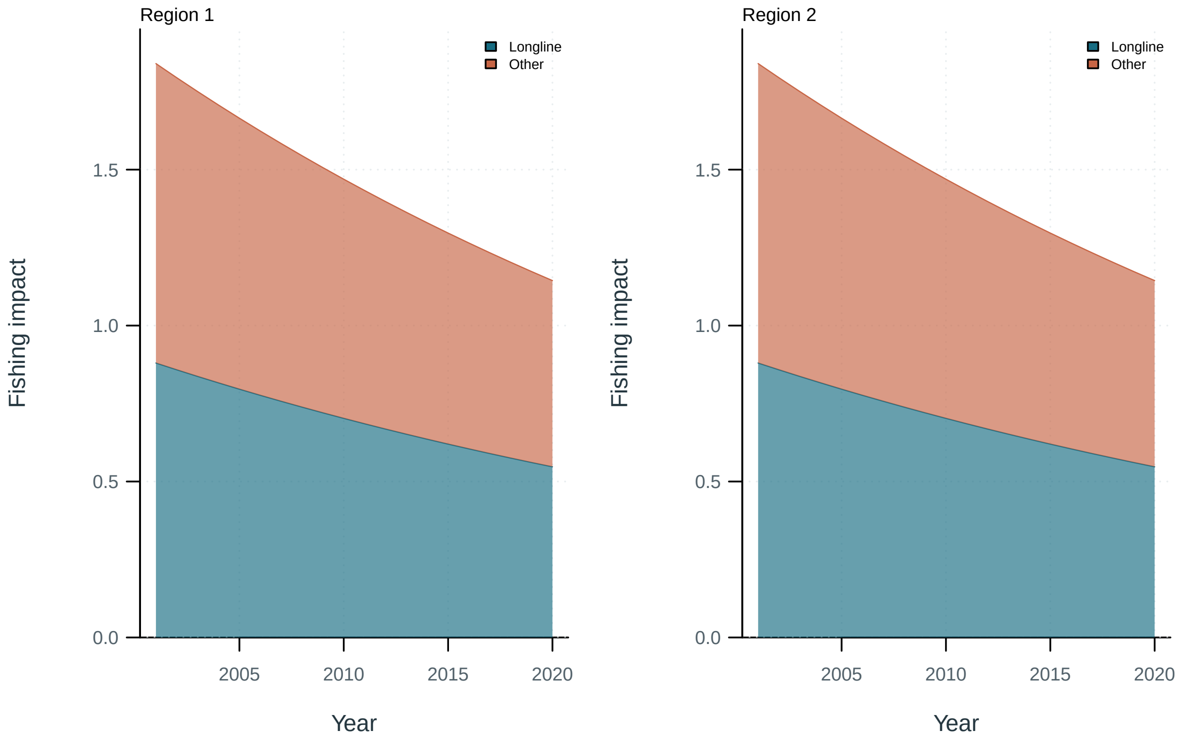

Figure 70: Illustrative output. Contributions of fishing groups to reduced spawning potential.

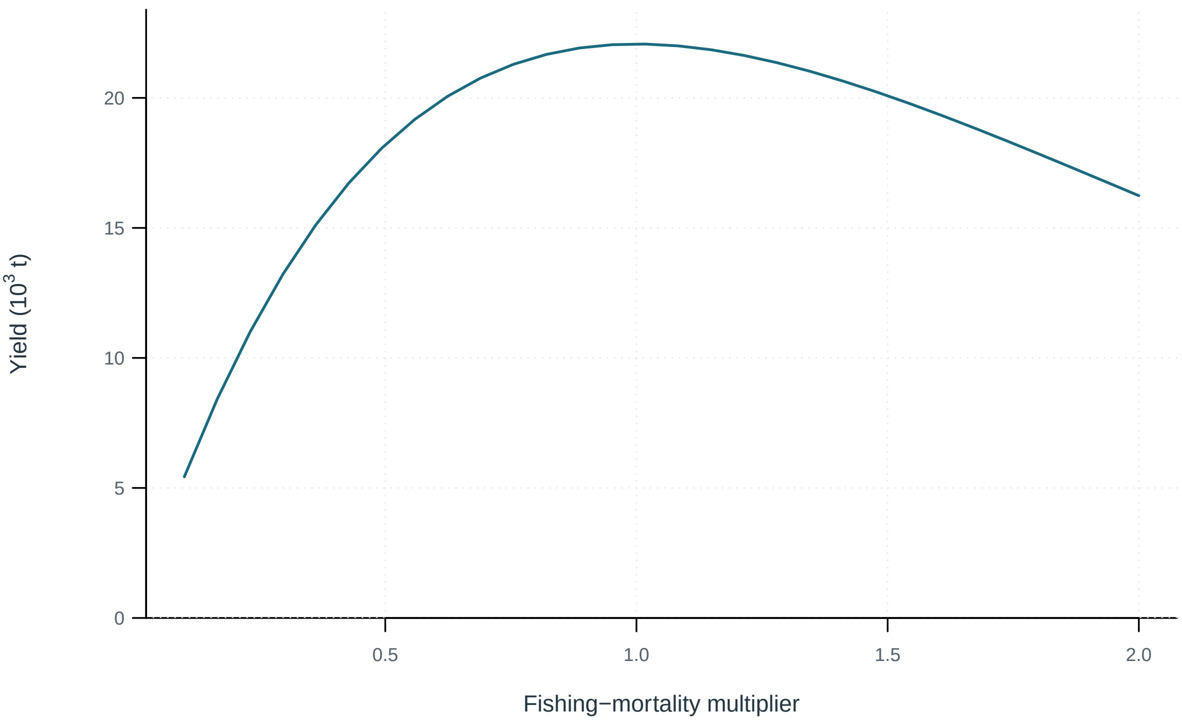

Figure 71: Equilibrium yield across fishing-mortality multipliers.

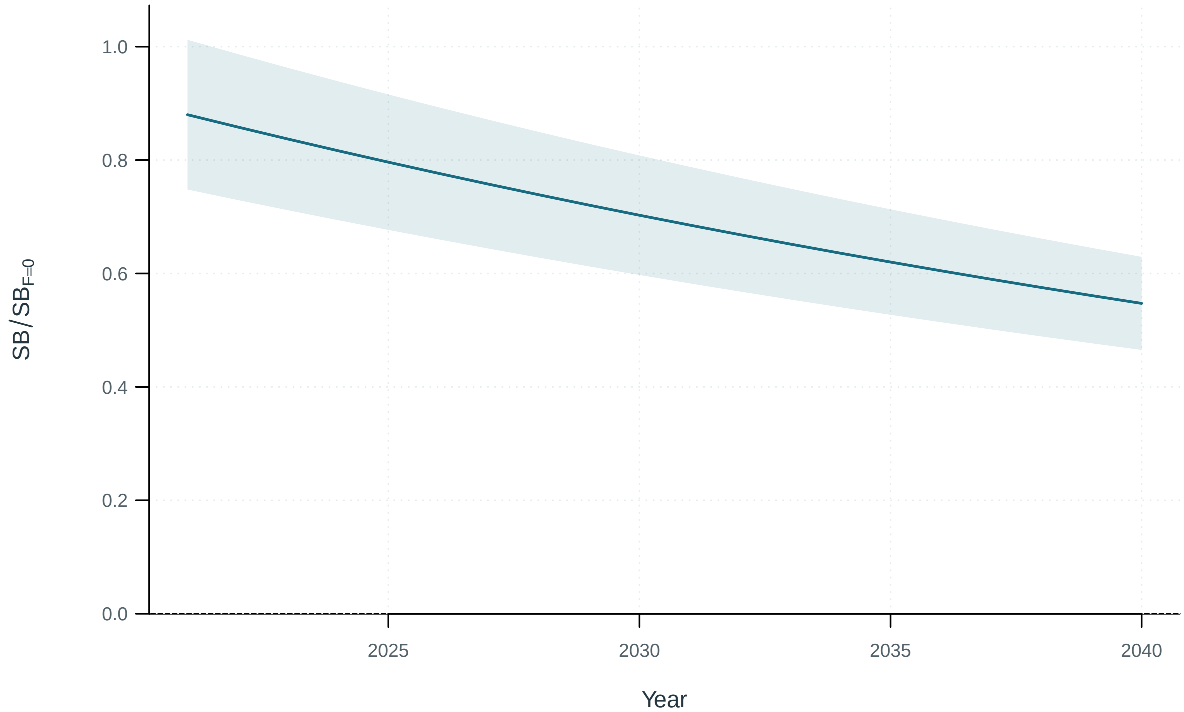

Figure 72: Historical and projected quantities by scenario with supplied uncertainty intervals. Central 95% intervals (illustrative supplied quantiles).

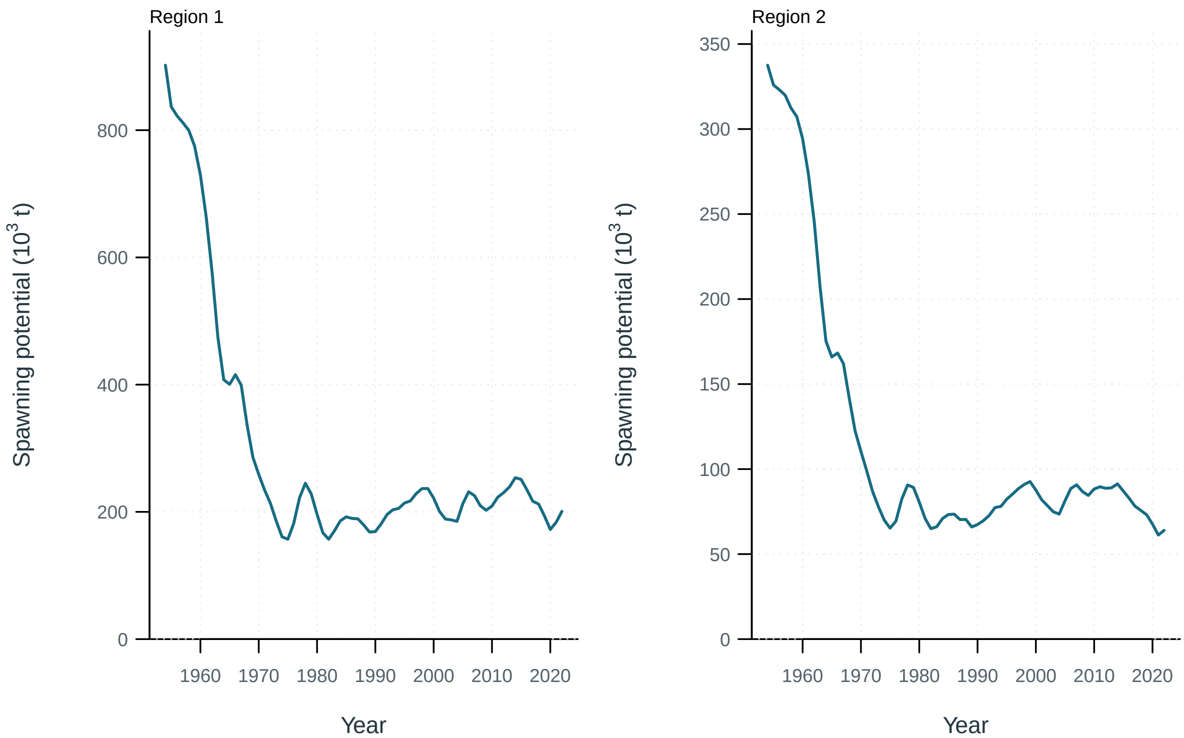

Figure 73: Spawning potential by model region.

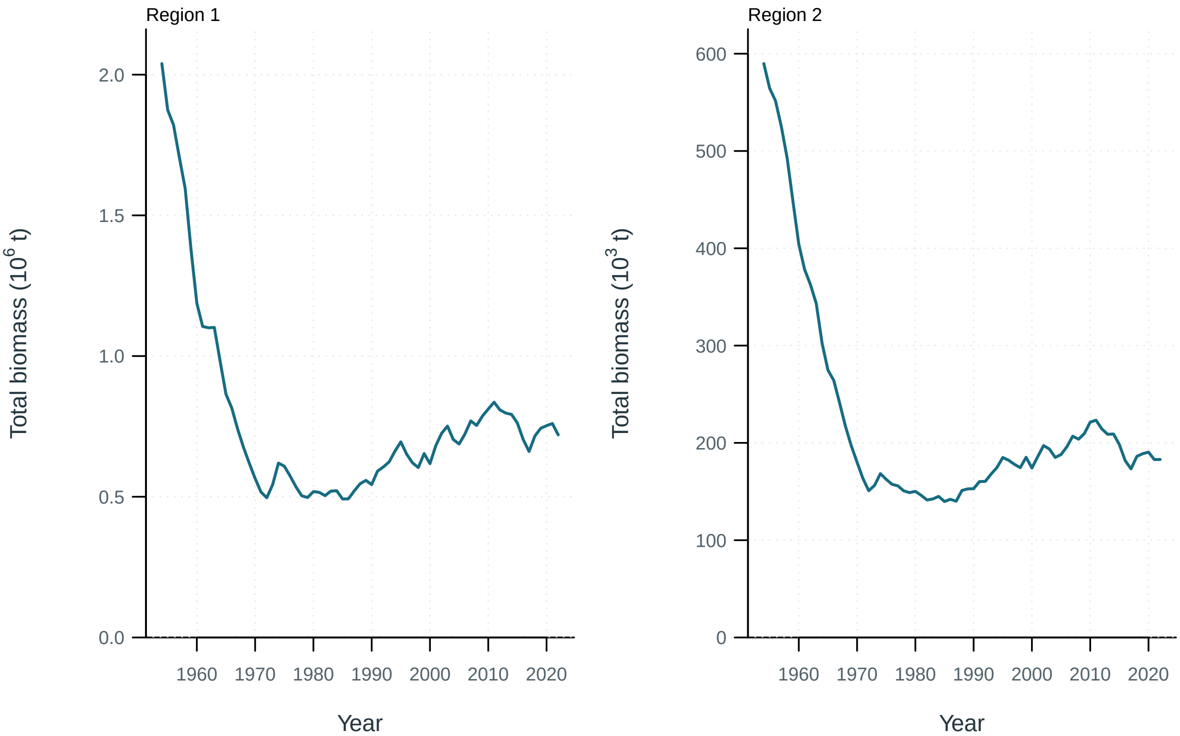

Figure 74: Total biomass by model region.

Figure 75: Dynamic spawning depletion by model region.

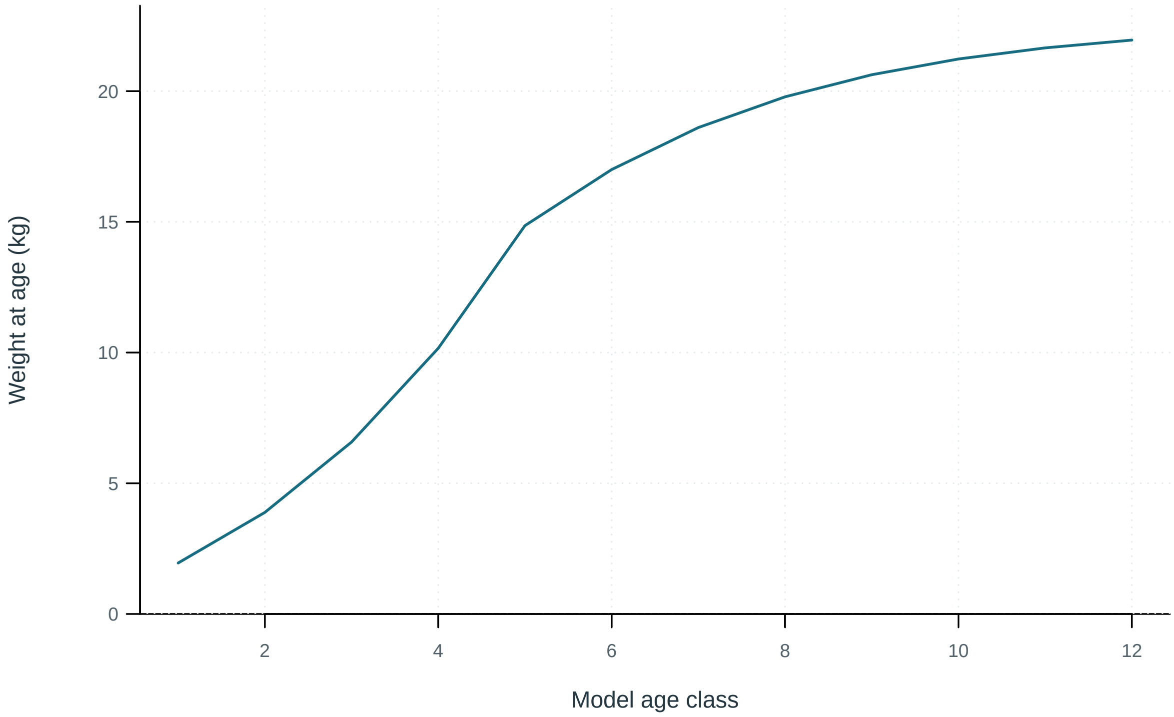

Figure 76: Mean weight at the start of each model age class.

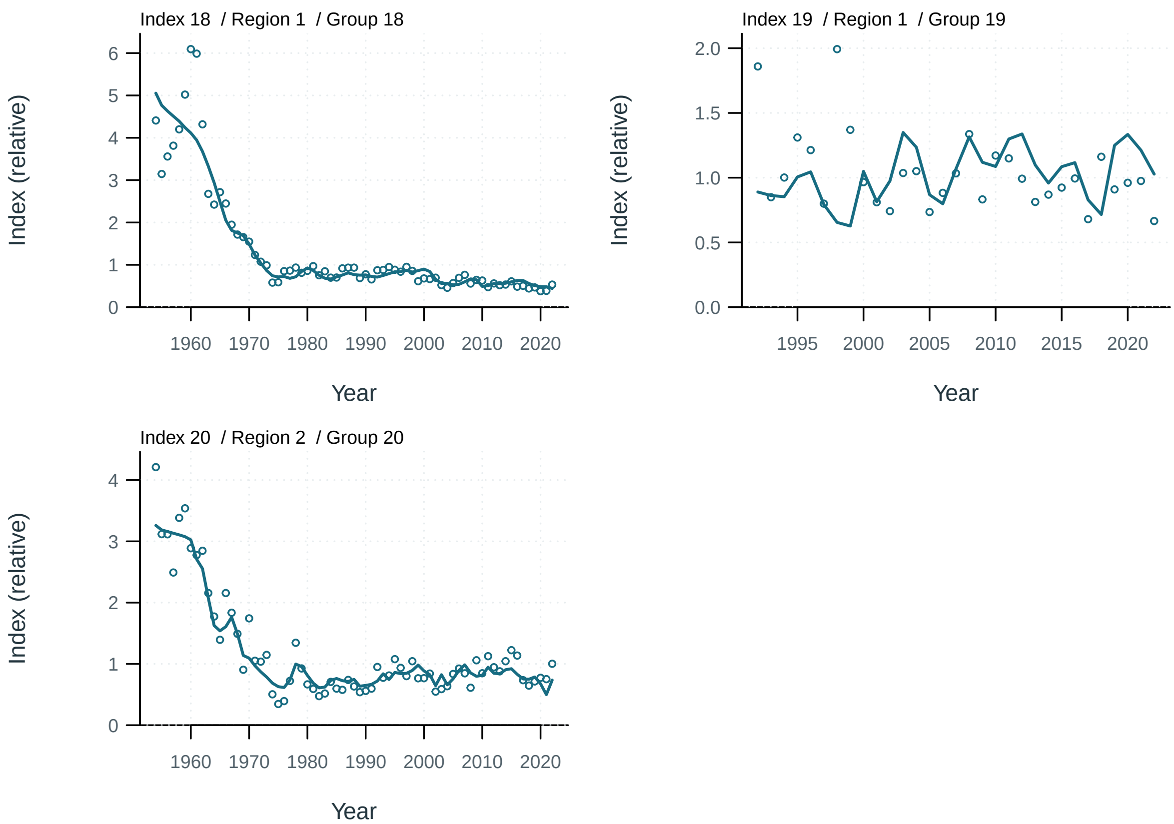

Figure 77: Index observations (points) and predictions (lines) by series.

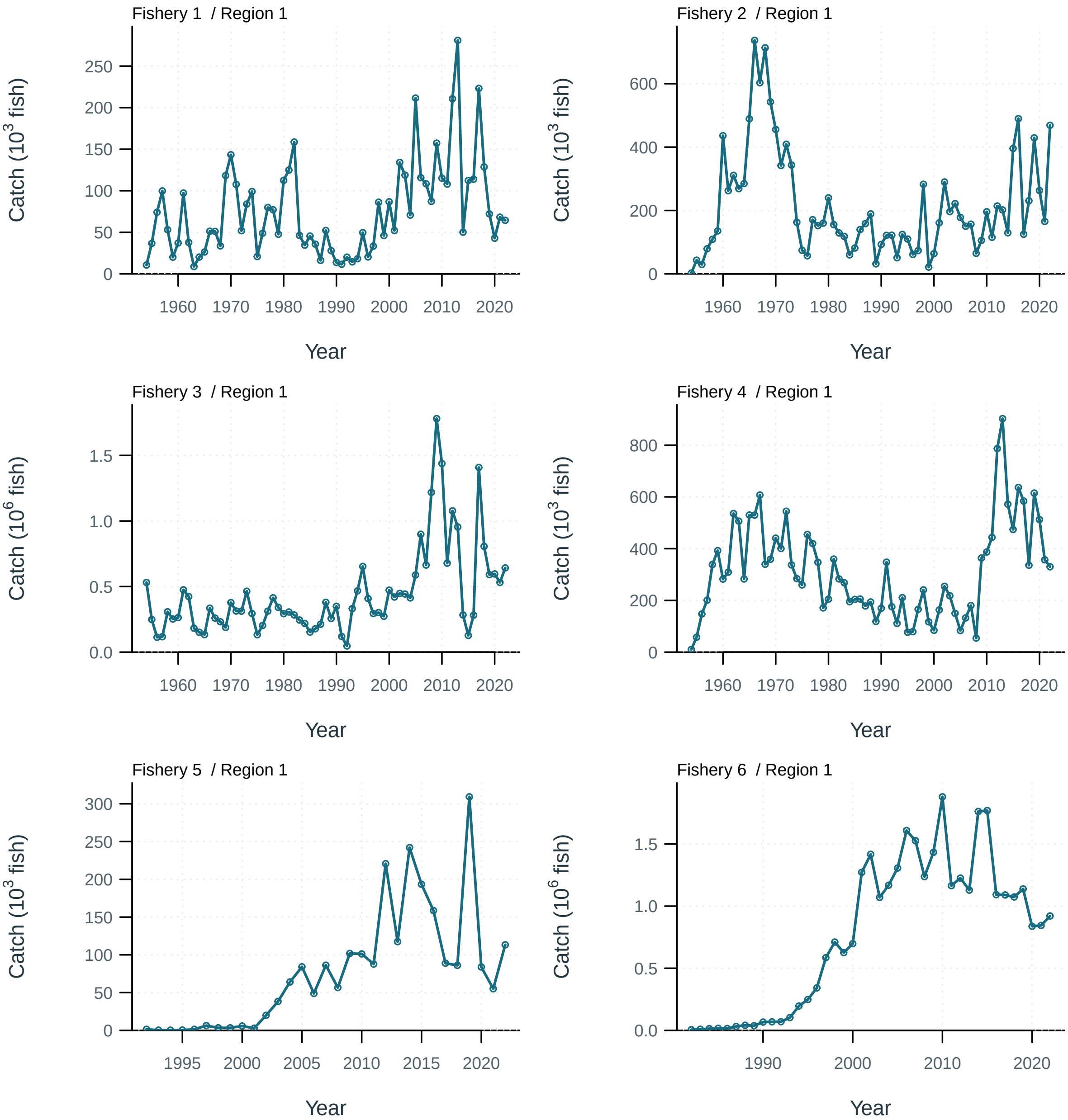

Figure 78: Catch observations (points) and predictions (lines) by series.

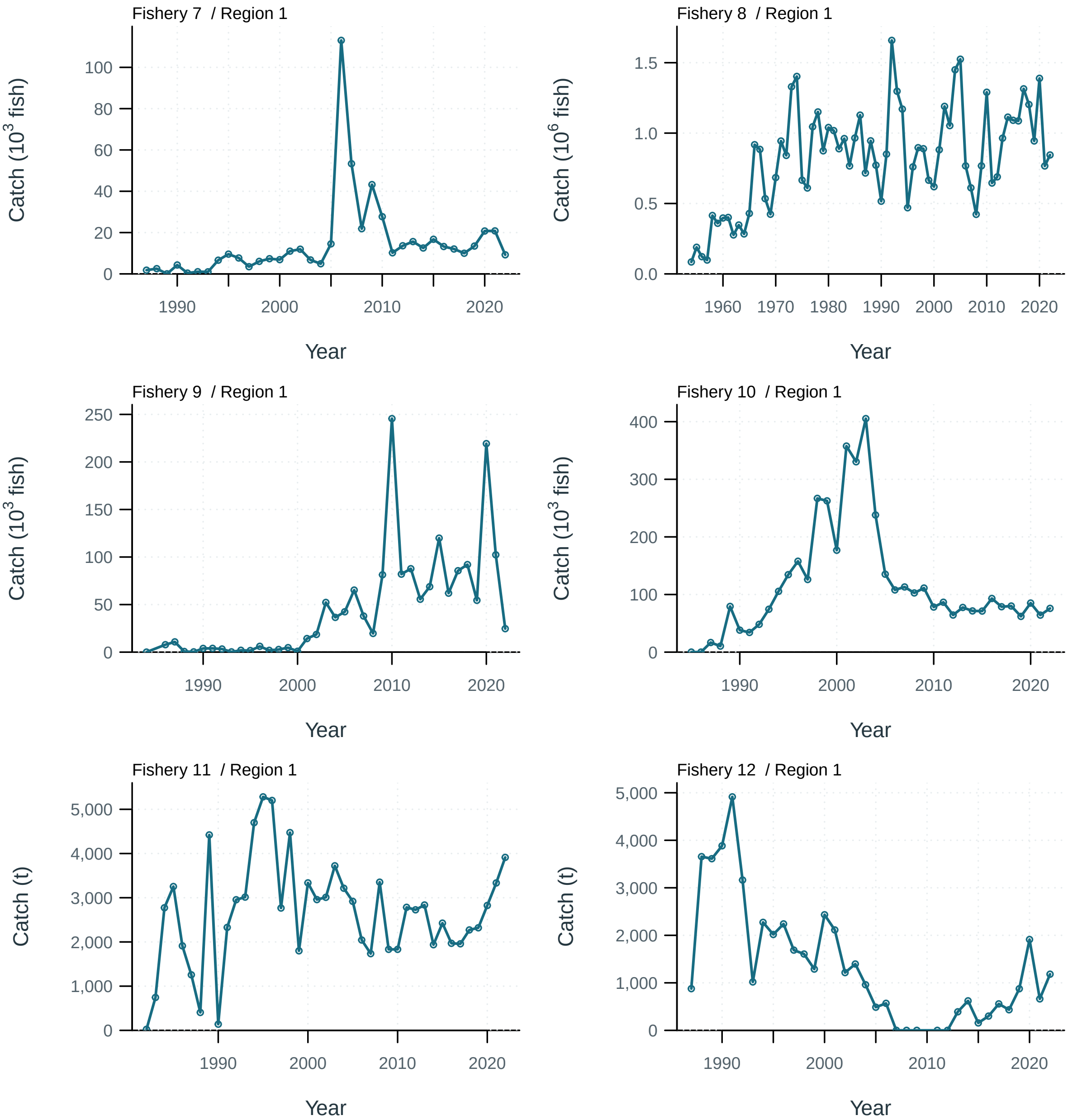

Figure 79: Catch observations (points) and predictions (lines) by series.

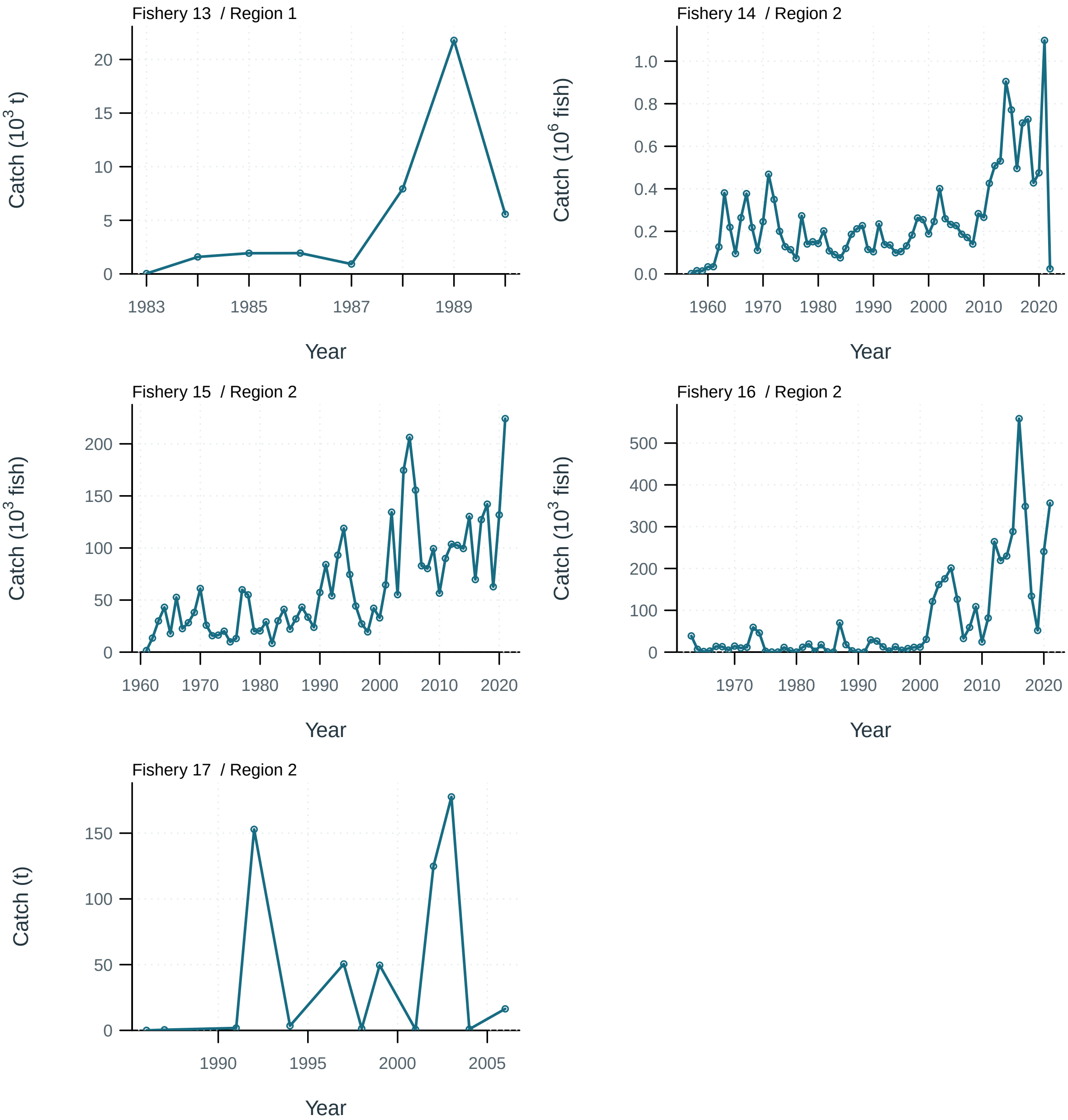

Figure 80: Catch observations (points) and predictions (lines) by series.

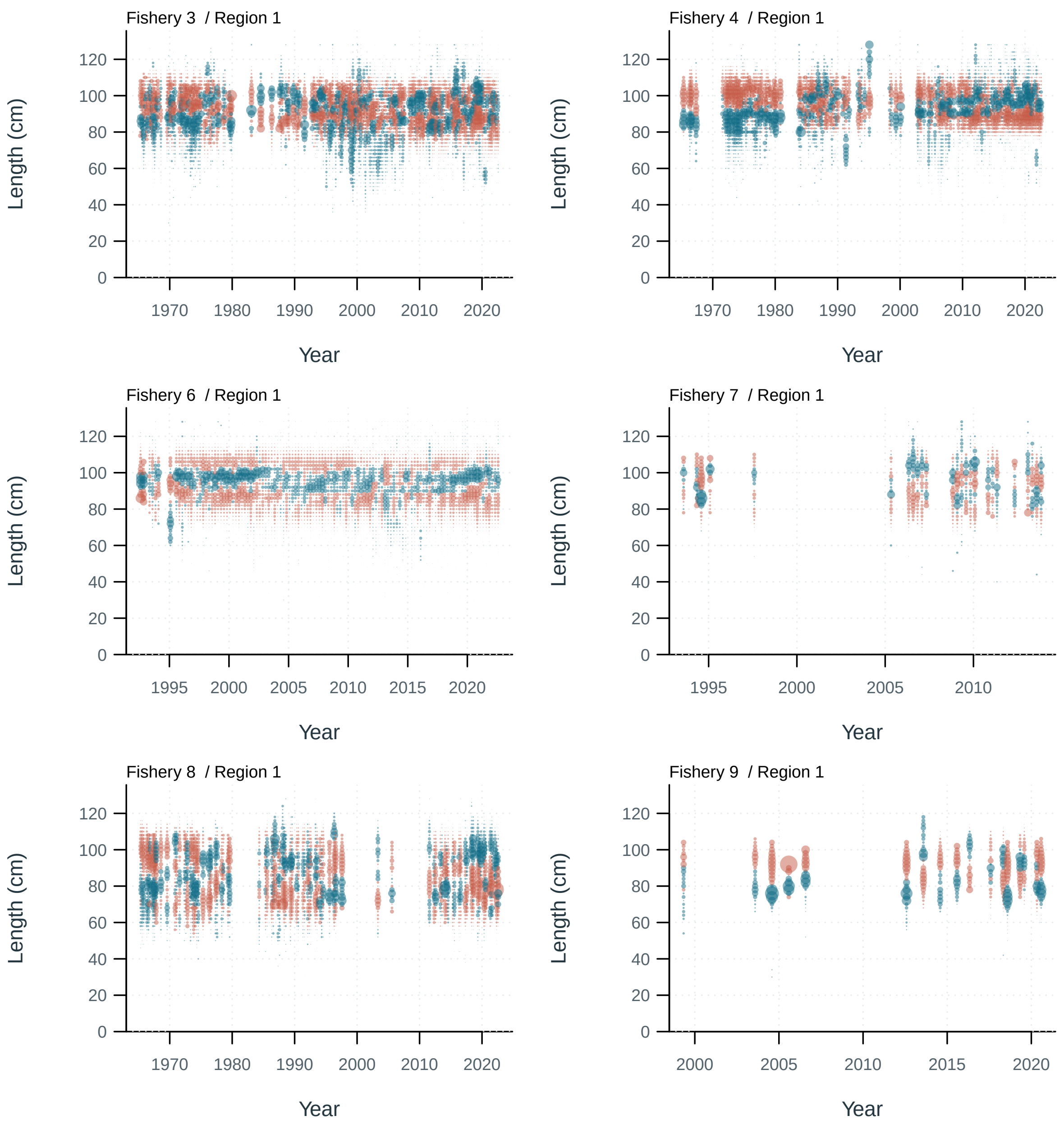

Figure 81: Length composition residuals by fishery. Blue: observed exceeds predicted; red: predicted exceeds observed. Circle area is proportional to absolute residual. Terminal bins include compressed tails.

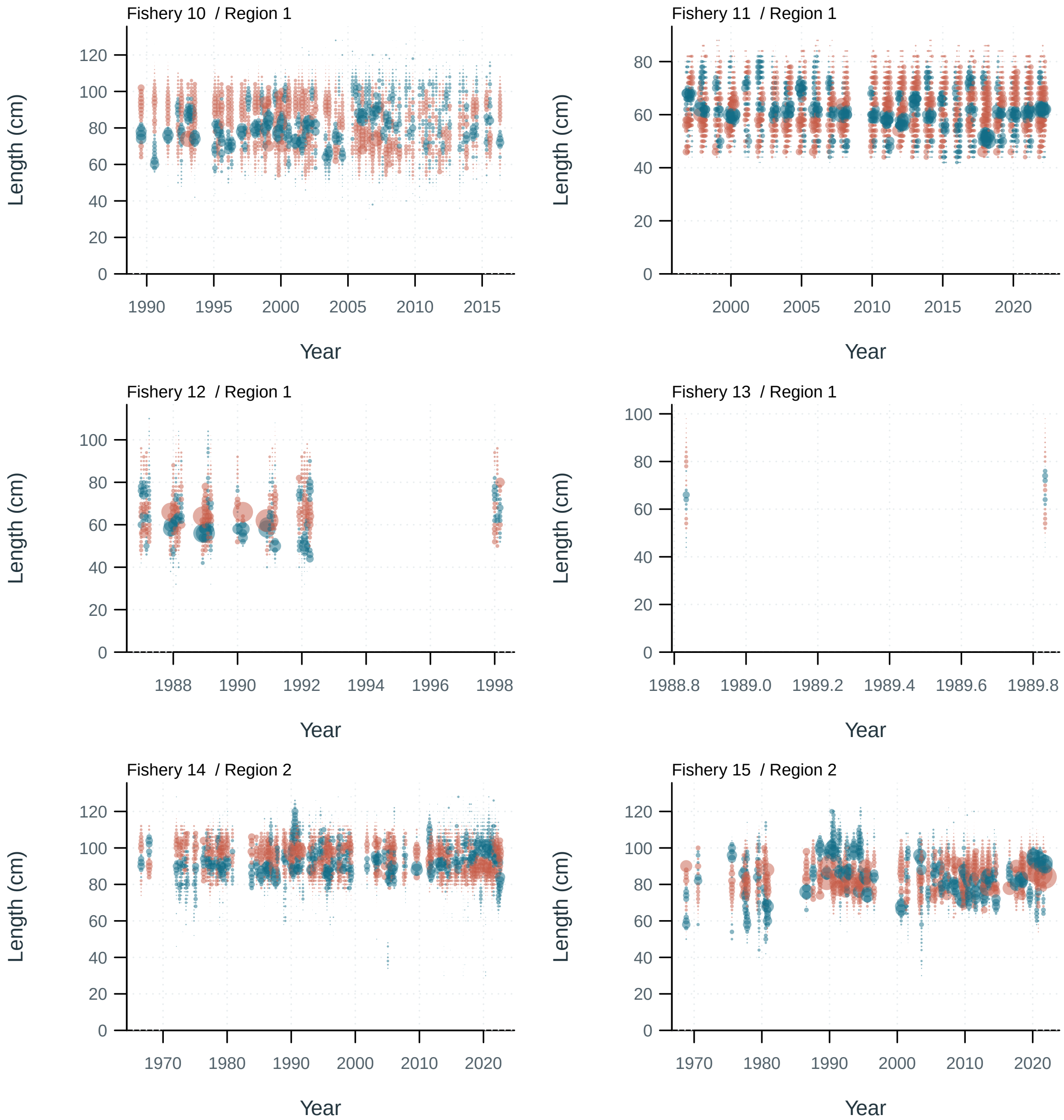

Figure 82: Length composition residuals by fishery. Blue: observed exceeds predicted; red: predicted exceeds observed. Circle area is proportional to absolute residual. Terminal bins include compressed tails.

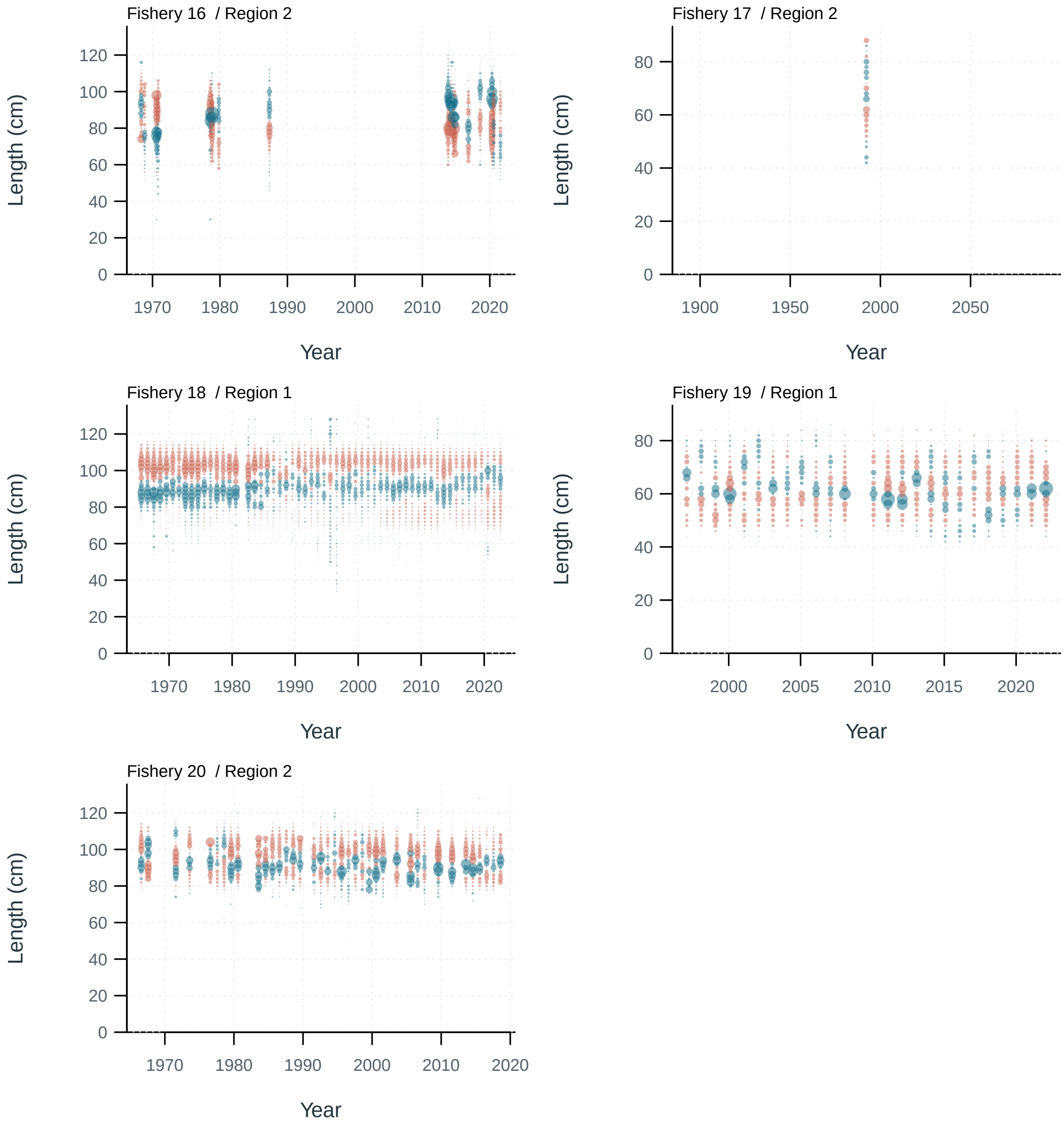

Figure 83: Length composition residuals by fishery. Blue: observed exceeds predicted; red: predicted exceeds observed. Circle area is proportional to absolute residual. Terminal bins include compressed tails.

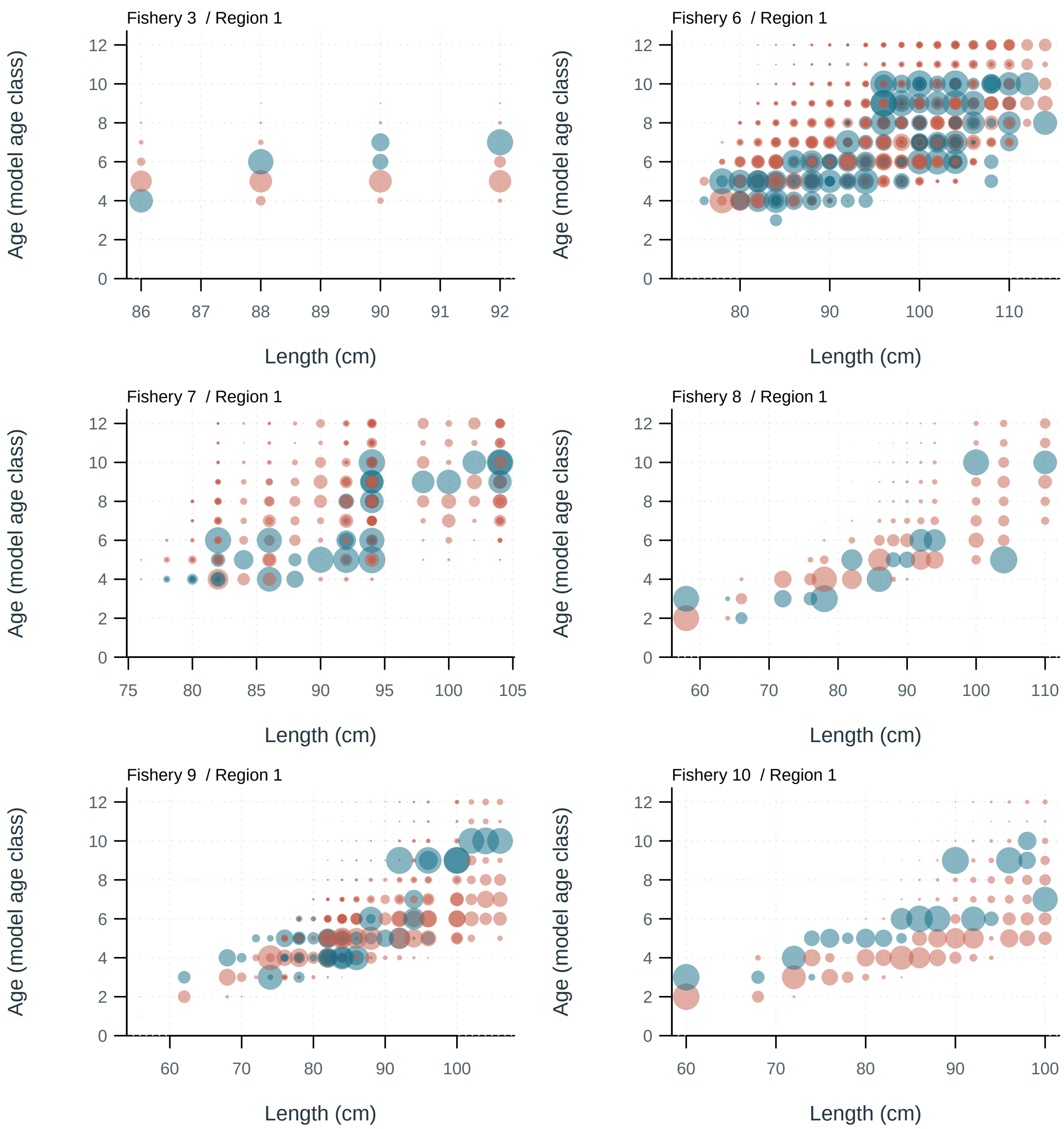

Figure 84: Conditional age-at-length composition residuals by fishery. Blue: observed exceeds predicted; red: predicted exceeds observed. Circle area is proportional to absolute residual.

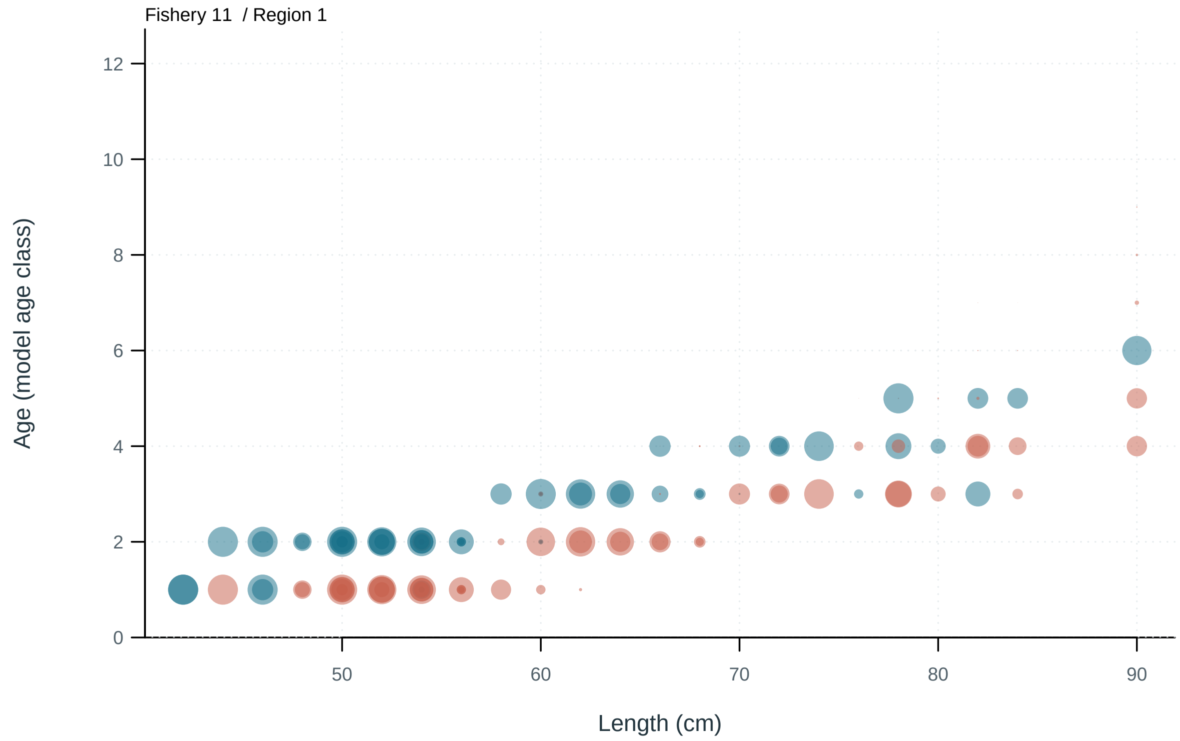

Figure 85: Conditional age-at-length composition residuals by fishery. Blue: observed exceeds predicted; red: predicted exceeds observed. Circle area is proportional to absolute residual.

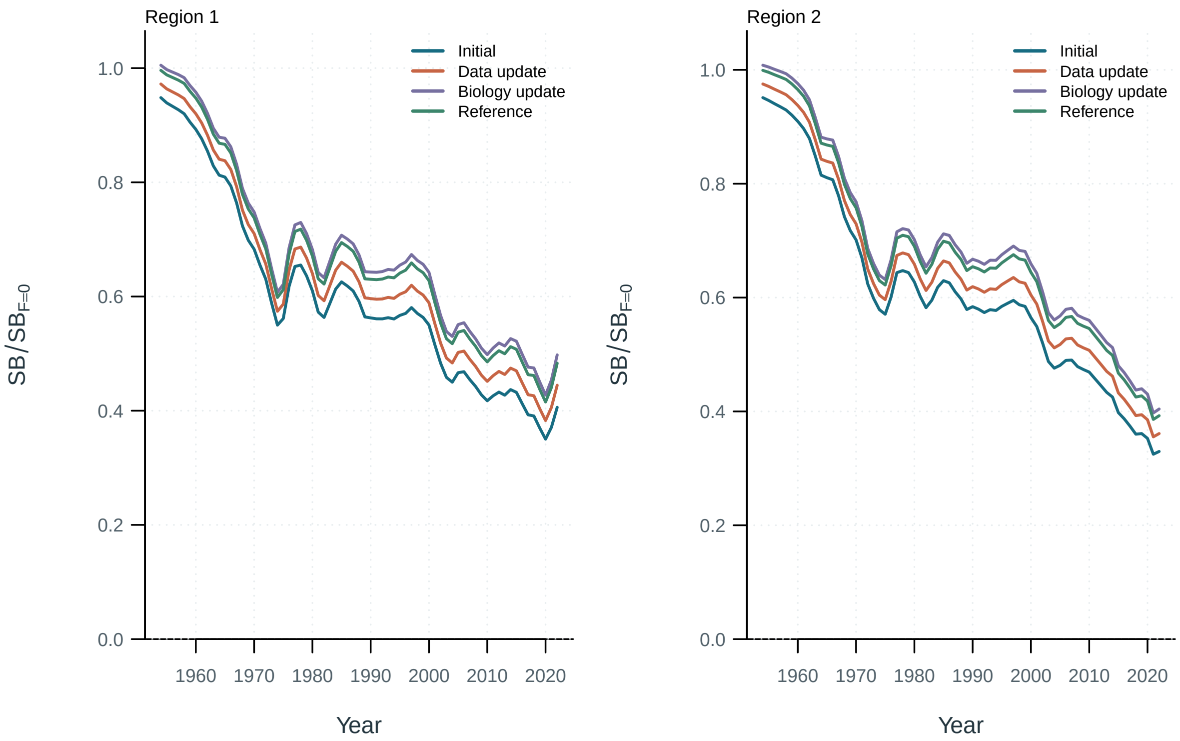

Figure 86: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

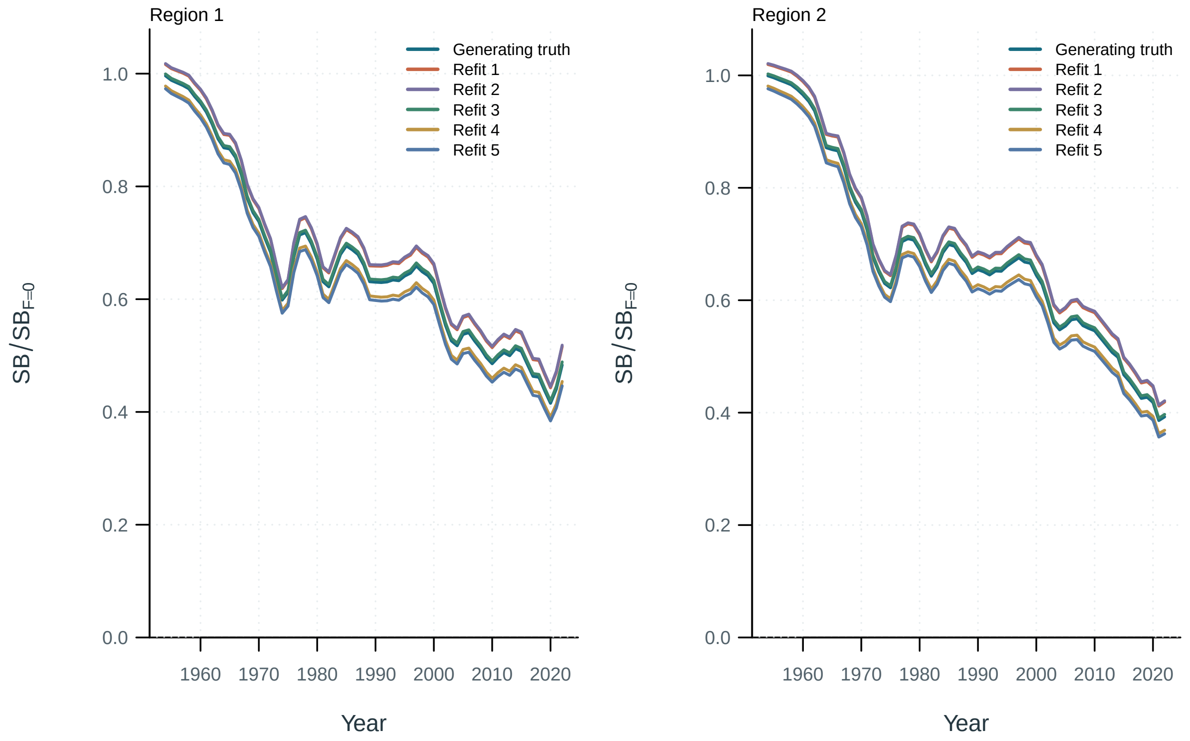

Figure 87: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

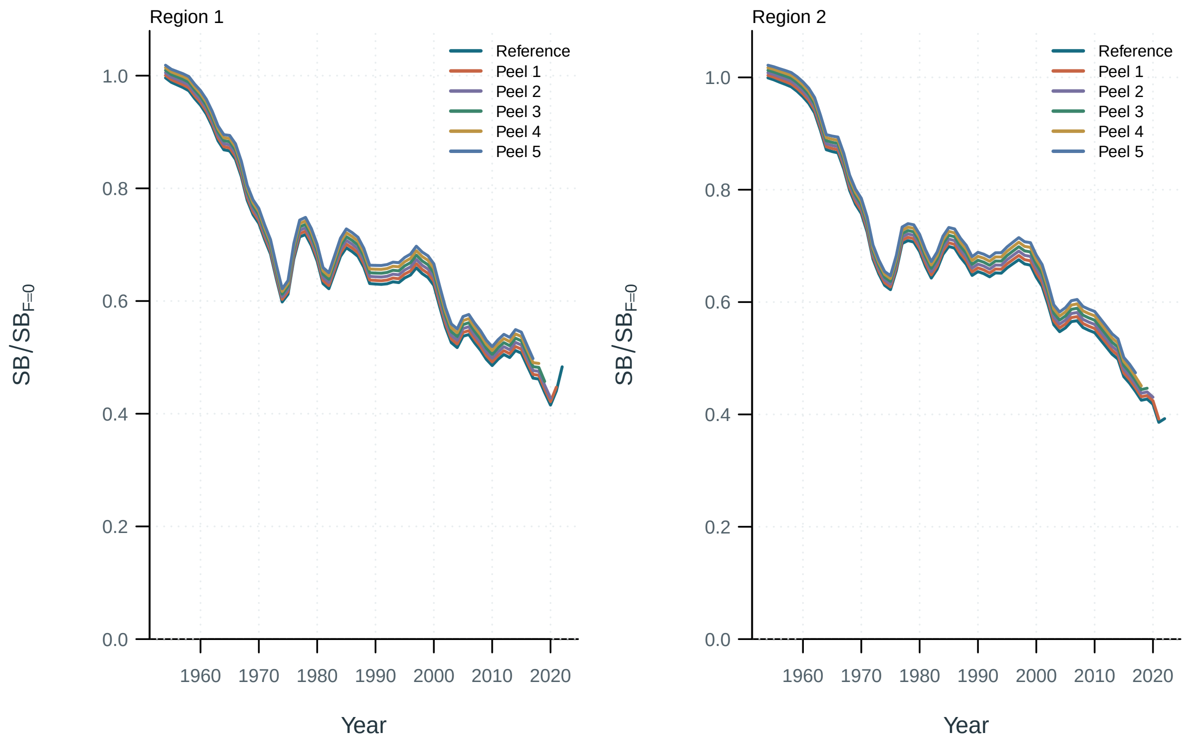

Figure 88: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

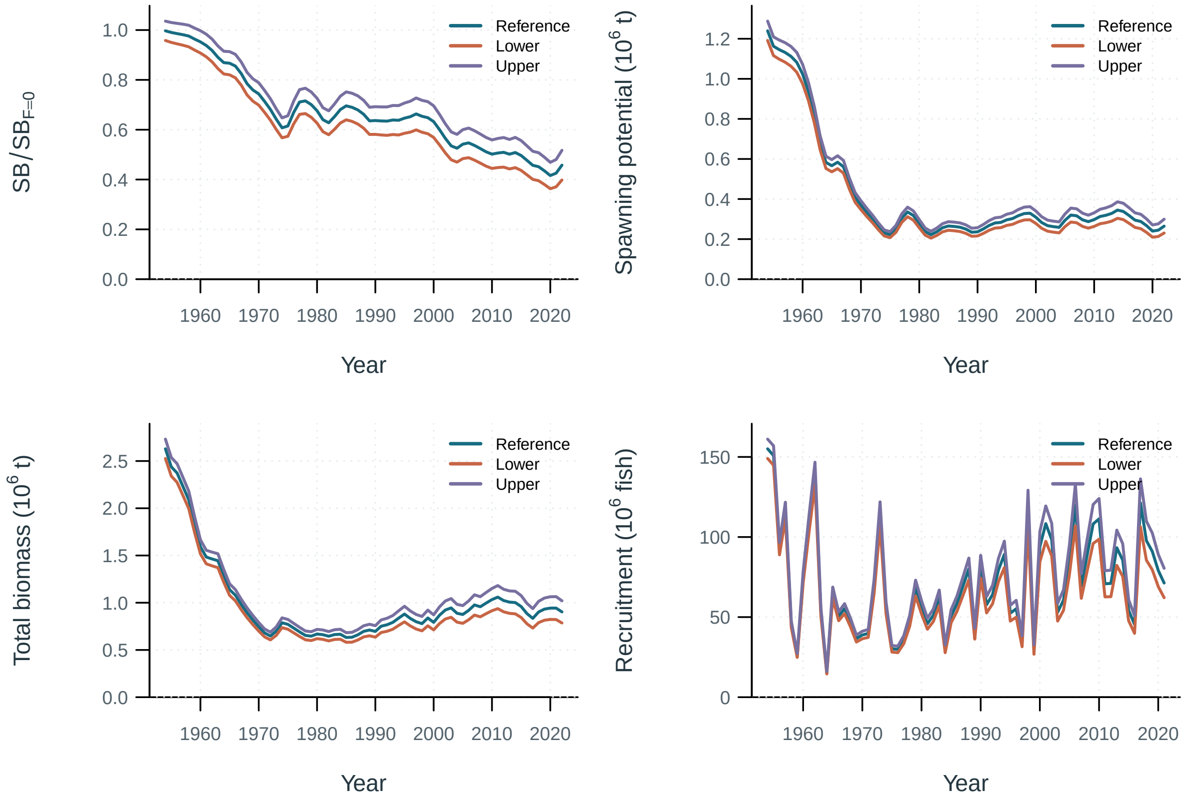

Figure 89: Sensitivity model trajectories.

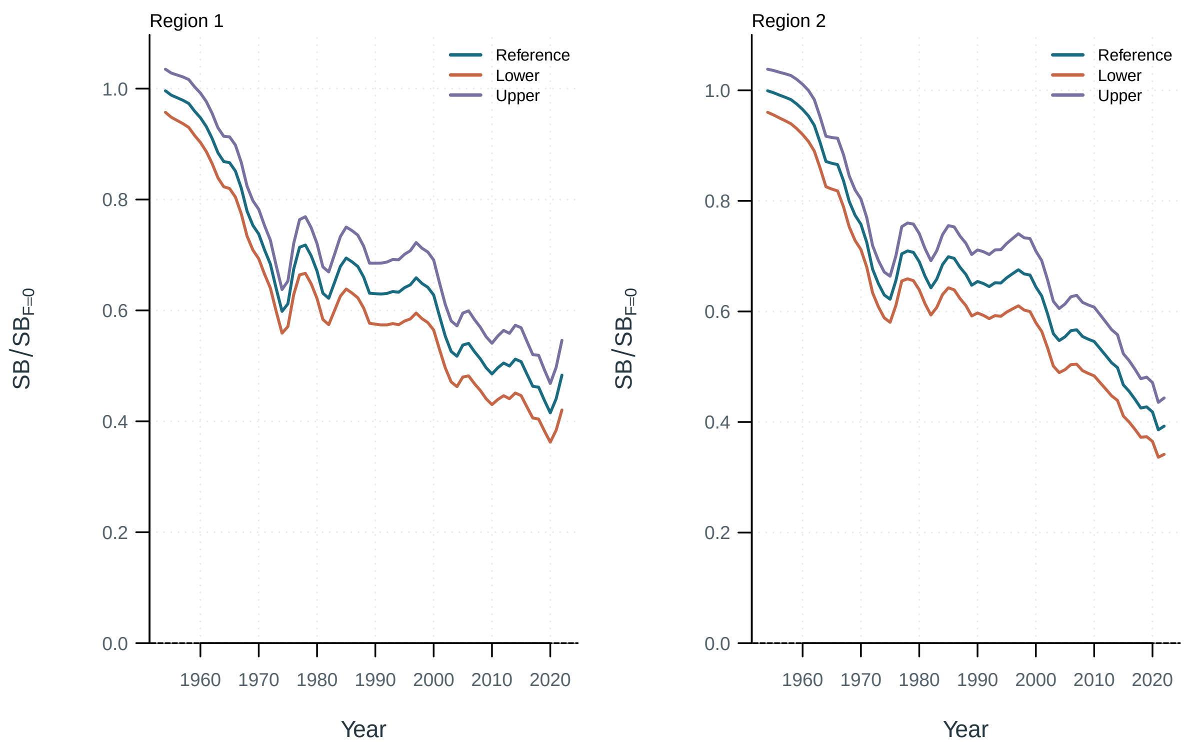

Figure 90: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

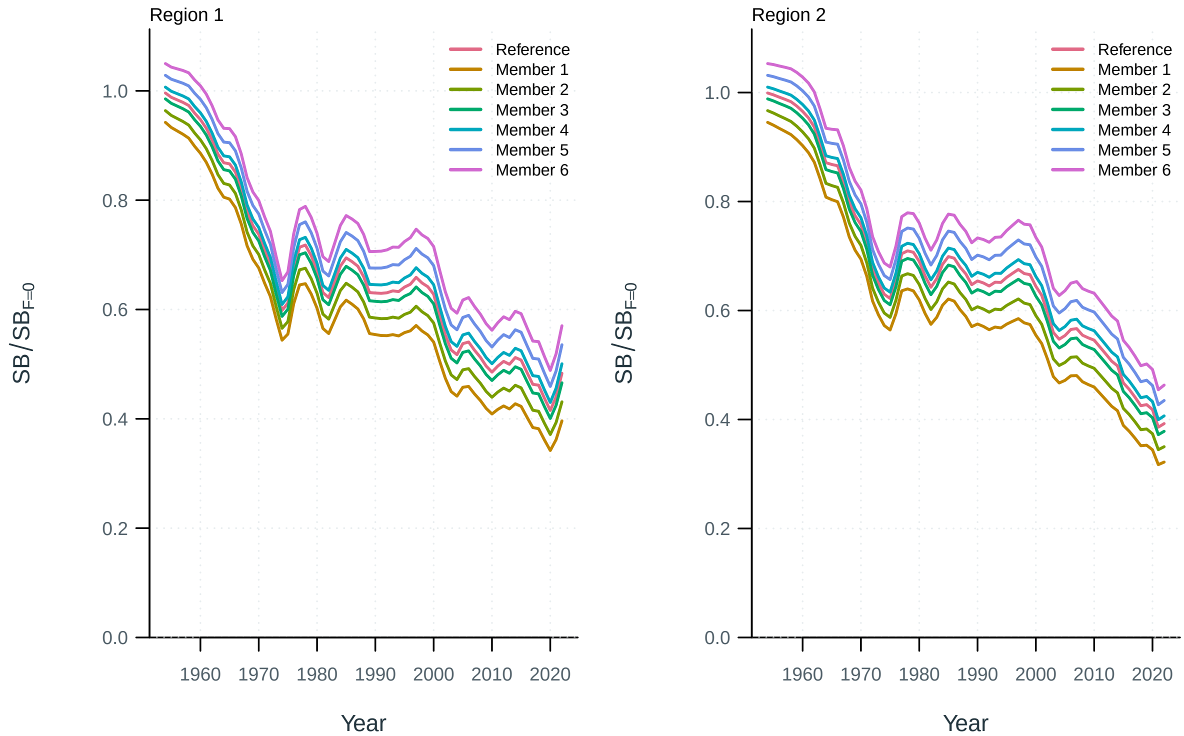

Figure 91: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

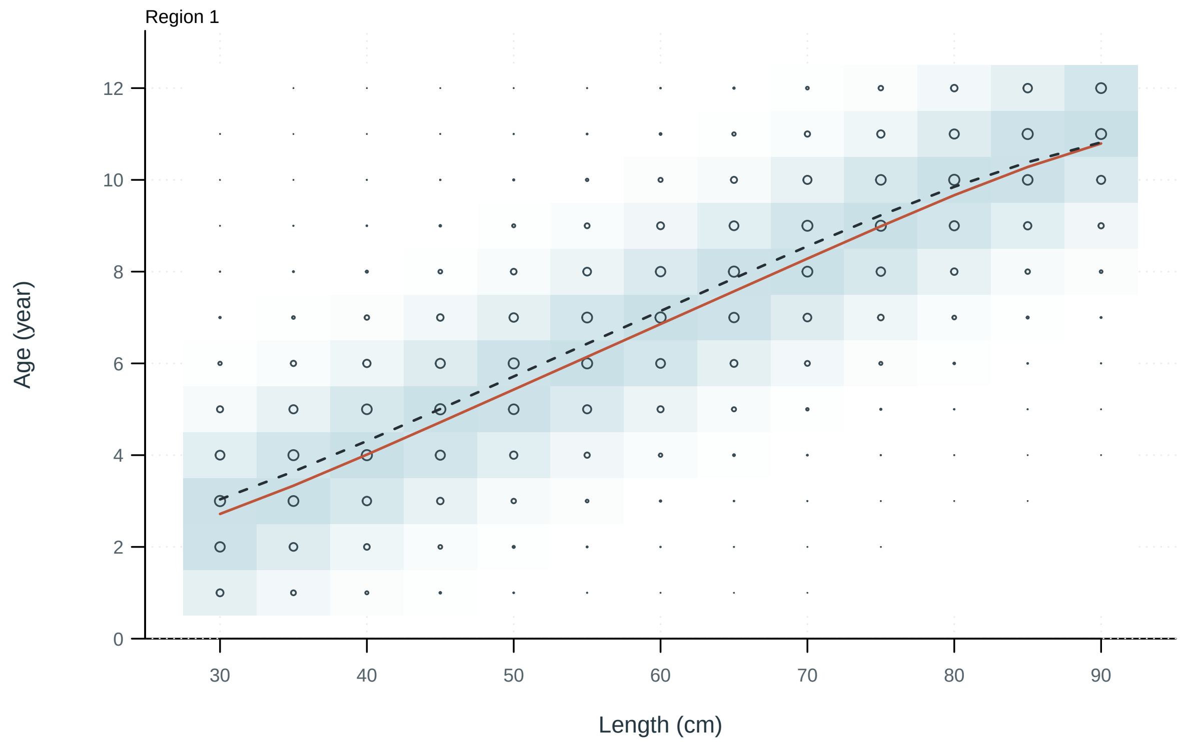

Figure 92: Illustrative output. Observed and predicted age distributions conditional on length.

11 Supplementary outputs

ALB · Diagnostic model interactive comparison

ALB · Diagnostic model figures and tables

ALB · Stepwise development interactive comparison

ALB · Stepwise development figures and tables

ALB · Starting-value trials interactive comparison

ALB · Starting-value trials figures and tables

ALB · Self-test recovery interactive comparison

ALB · Self-test recovery figures and tables

ALB · Retrospective analysis interactive comparison

ALB · Retrospective analysis figures and tables

ALB · Sensitivities interactive comparison

ALB · Sensitivities figures and tables

ALB · Uncertainty ensemble interactive comparison

ALB · Uncertainty ensemble figures and tables

Objective profiles · Display example interactive comparison

Objective profiles · Display example figures and tables

Additional output formats · Display examples interactive comparison

Additional output formats · Display examples figures and tables

12 Appendix: supplementary methods

13 References